

SUPPLEMENT TO NEURAL BELLMAN OPERATORS

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GUIDE TO THE SUPPLEMENT

The supplement collects general approximation arguments, capital-model proofs, the complete original economic applications, implementation details, and independent diagnostics. It uses the main article's notation and cross-references. Supplementary results and equations use separate continuous sequences S.1, S.2, and so forth. Tables retain their appendix-letter prefixes. The final source index identifies the exact preceding numerical record.

APPENDIX A: POLICY EVALUATION, APPROXIMATION, AND COMPLETE ACTION COVERAGE

The distinction changes the population problem. With one action, $f(v) = -v$, $T = 1$, and terminal utility one, the correct solution is $v^*(t) = e^{t-1}$. The perturbation $v_\epsilon = v^* - \epsilon(1 - t)$ preserves the terminal condition. The joint objective under joint differentiation satisfies

$$\Delta(L_A + L_C) = -\frac{\epsilon}{2} + \frac{7\epsilon^2}{3}, \quad \Delta L_C = \frac{7\epsilon^2}{3}. \quad (\text{S.1})$$

Thus the block-specific critic objective has the correct root as its minimizer along this perturbation, whereas the joint objective does not. The same problem is visible in a stationary equation: $-H + \omega H^2$ prefers $H = 1/(2\omega)$, not $H = 0$. No choice of finite positive weight resolves this identification error.

The actor objective is a local policy-improvement surrogate. Under differentiability of the policy value, the derivative of lifetime recursive utility from (t, x) instead involves the adjoint weight

$$D_\phi V^{a_\phi}(t, x) = \mathbb{E}_{t,x} \int_t^\tau e^{\int_t^s f_y(v, S_v, a_v, V^{a_\phi}) dv} D_a \mathcal{H}^{a_\phi}[V^{a_\phi}](s, S_s) D_\phi a_\phi(s, S_s) ds, \quad (\text{S.2})$$

subject to the differentiability and integrability conditions given in the supplement. Arbitrary domain sampling does not reproduce this occupation and recursive-utility weighting. NBO uses (7) for approximate pointwise improvement and verifies the remaining action gap; it does not identify that loss with an exact lifetime policy gradient.

A.0.1. *Conditioning and stopping*

A positive, parameter-independent state weighting can precondition the critic equation without changing its zero set. If that weight depends on learned quantities, its derivative must be included in the stated critic objective. The homothetic laboratory divides the Hamiltonian by its positive coefficient and differentiates that entire normalized critic residual. In the actor step the same coefficient is detached. Error conversion to the original economic equation includes the scale factor.

A zero residual target does not make learning dynamics stationary. For the scalar loss $\ell(z) = a^2 z^2 / 2$, gradient descent is stable only when $0 < \eta a^2 < 2$. Likewise, full differentiation of $\delta^2 / 2$ gives $\delta \nabla \delta$, while a semi-gradient TD update differentiates only the current-value term. The two computations must not be compared using inconsistent signs or discount conventions. If f already includes $-\rho v$, a time discretization must not add a second independent discount to the same continuation equation.

Constant-step Adam is a numerical choice in the calibration laboratory, not an application of a two-timescale theorem. A genuine two-timescale analysis would require nonsummable and square-summable step sizes, an actor-to-critic ratio tending to zero, controlled noise, bounded iterates, and suitable attracting limiting dynamics (Borkar, 2024). A stationary critic gradient need not imply a zero residual, and a stationary multi-agent gradient need not imply Nash optimality. The results below therefore concern verified approximation errors rather than an assumed universal optimizer convergence property.

A.1. *Policy Evaluation and the Recursive Policy Gradient*

Let a_ϕ be differentiable in a scalar parameter direction, and suppose V^{a_ϕ} is sufficiently differentiable for the following calculation. The terminal payoff is inde-

pendent of the policy parameter. Assume differentiation under the expectation is justified, that the differentiated state and utility equations are integrable, and that the boundary conditions of the differentiated problem are zero. Denote the directional derivative of the policy value by w .

Differentiating the fixed-policy equation gives

$$0 = w_t + \mathcal{L}^{a_\phi} w + f_y(t, x, a_\phi, V^{a_\phi}) w + D_a \mathcal{H}^{a_\phi}[V^{a_\phi}] D_\phi a_\phi.$$

The operator derivative includes the dependence of both drift and covariance on the action. The derivative of the generator acting on the changing value is contained in $\mathcal{L}^{a_\phi} w$, while the action derivative of the generator acting on the fixed value is contained in $D_a \mathcal{H}$. No Hessian-action channel is discarded.

Applying the linear Feynman–Kac formula to the differentiated equation gives

$$w(t, x) = \mathbb{E}_{t,x}^{a_\phi} \int_t^\tau \exp \left(\int_t^s f_y(v, S_v, a_v, V^{a_\phi}) dv \right) D_a \mathcal{H}^{a_\phi}[V^{a_\phi}](s, S_s) D_\phi a_\phi(s, S_s) ds.$$

This formula concerns the performance derivative from the stated initial state. A domain-uniform Hamiltonian objective uses a different weighting. Such an objective is useful for approximate pointwise improvement, but it becomes a lifetime policy gradient only with the appropriate occupation and adjoint weights. Trajectories sampled by the current actor are detached in the implemented domain objective. Differentiating through their sampling law would define a different gradient estimator and must be analyzed separately.

The critic’s residual derivative is the derivative of the entire fixed-policy equation. In particular, its level is determined by the recursive evaluation equation together with boundary data. A separate negative-Hamiltonian gradient in the critic is not an innocuous level anchor: in the scalar stationary example it moves the minimizer away from the Bellman root.

A.2. Representation, Boundary Conditions, and Error Measurement

A.2.1. Smooth approximation and what it establishes

A derivative-approximation result concerns a smooth target on a compact domain and a family of networks whose capacity can increase. It does not imply that a fixed

network attains zero residual, that optimization finds its best approximation, or that a nonsmooth target has classical derivatives everywhere. In a smooth problem, increasing the representation class can reduce approximation error in the value and derivatives. In a nonsmooth problem, NBO instead evaluates a consistent positive-weight Bellman equation, or solves an explicitly regularized equation before taking a justified limit.

The hard-terminal architecture is $v(t, x) = G(x) + (T - t)N(t, x)$. It exactly enforces $v(T, x) = G(x)$ independently of training. A lateral Dirichlet face needs an additional compatible lifting and vanishing factor. For a reflected state, the Neumann condition is a different constraint. A loss on only interior points leaves all of these boundary questions unanswered. The regression test evaluates the terminal architecture at randomly chosen states and checks exact equality, not merely a small mean terminal error.

The recursive Epstein–Zin domain is $bV > 0$. An exponential output ensures positivity, while an interval transformation can keep bV between declared positive barriers. If $bG > 0$, one terminal-compatible positive representation is

$$bv(t, x) = \exp [\log(bG(x)) + (T - t)N(t, x)].$$

Its terminal value is exact. It does not by itself impose a lateral exit payoff or a two-sided barrier. These additional conditions must be built into the architecture or measured explicitly. Clipping an invalid utility argument inside a fractional power changes the equation and is not an acceptable hidden numerical repair.

A.2.2. From residual samples to a uniform bound

Let r be K -Lipschitz on a domain in $\mathbb{R}^d$. If every point is within distance h_V of a validation point, the triangle inequality gives $\|r\|_\infty \leq \max_{x \in \mathcal{V}} |r(x)| + Kh_V$. If K is estimated from samples rather than bounded, the resulting expression is itself a diagnostic, not a certified uniform bound.

An integral bound needs more assumptions. Suppose the sampling density is at least $p_0 > 0$, and each ball of radius r_0 or less centered in the domain intersects the domain in volume at least $c_D r^d$. Let $M = \|r\|_\infty$ and assume $M/(2K) \leq r_0$. Near a point approaching the maximum, Lipschitz continuity gives $|r| \geq M/2$ on a ball of

radius $M/(2K)$. Hence

$$\mathbb{E}r^2 \geq \frac{p_0 c_D}{2^{d+2} K^d} M^{d+2}, \quad M \leq \left(\frac{2^{d+2} K^d \mathbb{E}r^2}{p_0 c_D} \right)^{1/(d+2)}.$$

For larger M , use the capped radius r_0 and the corresponding two-case inequality. A concentration argument is additionally needed to replace the population mean by an empirical one. This calculation explains why a dimension-free uniform certificate cannot be read directly from an average training loss.

The action gap is treated similarly. If an action net has covering radius h_A and the Hamiltonian is K_A -Lipschitz in the action, the unobserved supremum exceeds the net maximum by at most $K_A h_A$. Thus the net gap plus this term is an upper bound. A collection of successful deviations without a covering or duality argument is only a lower bound on exploitability.

A.2.3. *Why an integrated differential residual does not select the value*

Consider the exit problem $F(v') = 1 - |v'| = 0$ on $(-1, 1)$ with zero boundary payoff. Its value is $v^*(x) = |x| - 1$. The smooth functions

$$w_\epsilon(x) = \sqrt{1 + \epsilon^2} - \sqrt{x^2 + \epsilon^2}$$

satisfy the same boundary values and have vanishing integrated squared residual, but converge uniformly to $1 - |x|$. At zero the test function one touches the wrong limit from above and gives $F(0) = 1 > 0$, violating the viscosity subsolution inequality. The defect is not removed by exact Hamiltonian maximization.

On a grid with spacing h , the positive-weight exit backup is

$$(Tv)_i = -h + \max(v_{i-1}, v_{i+1}), \quad v_0 = v_N = 0. \quad (\text{S.3})$$

It is an undiscounted shortest-exit problem, not an instance of the stationary contraction theorem. Backward propagation of the boundary values gives $v_i = |x_i| - 1$ exactly. At the peak of the wrong candidate, the *scaled* residual $|w_i - (Tw)_i|/h$ is close to two, rather than close to zero. The laboratory therefore distinguishes correct selection from small integrated differential error.

An explicit positive viscosity term is another possible regularization, provided comparison, boundary convergence, and approximation errors for the regularized equations are controlled before taking the viscosity to zero. Network width or spectral bias is not itself a positive viscosity parameter. Moreover, approximation of a C^2 function and its derivatives does not establish approximation of nonexistent classical derivatives at a kink.

A.2.4. Coverage and independent maximization

A verification distribution is declared independently of the actor's occupation measure. It includes economically relevant counterfactual states and boundary regions, not only states visited by a current policy. If a residual r has a known Lipschitz bound K on a compact domain and a validation set has fill distance d_{fill} , then

$$\|r\|_{\infty} \leq \max_{x \in \mathcal{V}} |r(x)| + Kd_{\text{fill}}. \quad (\text{S.4})$$

The same bound can be applied to an action gap if its modulus is controlled. An empirical mean alone does not give this guarantee. If the sampling density is bounded below and the domain has a uniform interior-volume condition, a Lipschitz residual also admits an L^2 -to-sup estimate with exponent $1/(d+2)$, as derived in the supplement. Its dimension dependence is material.

An independent action solver must provide an *upper bound* on the unobserved maximization error. Trying a few alternative actions supplies only a lower bound on the true gap. Concavity with a duality gap, interval bounds, exhaustive enumeration of a finite action set, or an action net with a known Lipschitz modulus can provide the needed upper bound. An unverified local actor optimizer cannot certify itself.

A.2.5. Differential and discrete error accounts

For the stationary additive problem, write $f = r_0 - \rho v$, with $\rho > 0$, and let R_h restrict a candidate to the nodes of a positive-weight scheme. On interior nodes set

$$T_h^a w = q_h(P_h^a w + hr_0^a), \quad q_h = (1 + \rho h)^{-1}.$$

Stopping nodes instead use their prescribed payoff.

PROPOSITION S.1—Transfer to a Bellman certificate: *Suppose the represented policy is feasible for the discrete action class and*

$$\sup_a \left\| h^{-1}(P_h^a R_h v - R_h v) - R_h \mathcal{L}^a v \right\|_{\infty, \text{int}} \leq \kappa_h.$$

If the differential evaluation residual is at most e and its feasible-action gap at most η , then, on the interior nodes,

$$\|R_h v - T_h^a R_h v\|_{\infty} \leq q_h h(e + \kappa_h), \quad \|T_h R_h v - T_h^a R_h v\|_{\infty} \leq q_h h(\eta + 2\kappa_h).$$

Boundary discrepancies and finite-action coverage errors are additional accounts.

The proof follows by adding and subtracting $R_h \mathcal{L}^a v$; the maximization inequality uses the consistency error twice. The converse is false for arbitrary oscillatory interpolants. For smooth interior functions, local cubature and interpolation estimates can bound κ_h ; they do not justify assigning zero error to a stopping or reflected boundary without checking its realization.

For finite-horizon maps with Lipschitz factors β_t , let δ_t bound evaluation error, ϵ_t bound improvement error, and B_N bound terminal error. Define

$$E_N = B_N, \quad O_N = B_N, \quad R_N = 2B_N, \quad \begin{cases} E_t = \delta_t + \beta_t E_{t+1}, \\ O_t = \delta_t + \epsilon_t + \beta_t O_{t+1}, \\ R_t = 2\delta_t + \epsilon_t + \beta_t R_{t+1}. \end{cases} \quad (\text{S.5})$$

Backward comparison bounds policy-value error by E_t , optimal-value error by O_t , and policy regret by R_t . These arrays are computed, rather than replaced by a training loss, in the NDU experiment. Bounds at later time indices must use their own accounts, not necessarily the account at time zero. Representation and incomplete optimization enter δ_t ; action coverage enters ϵ_t ; the error of replacing the controlled diffusion by a scheme is separate. If a uniform continuous-to-discrete policy-payoff discrepancy is bounded by D_h , transferring a discrete regret bound adds $2D_h$. Without that estimate, the certificate remains a certificate for the stated discrete economy. In particular, this accounting does not establish the optimizer rate required by Theorem 3.

A.2.6. Continuous-action proposals and a budget-safe Bellman enclosure

The learned map and its verifier have different computational roles. In the differential implementation, a feasible actor supplies an action and the critic evaluates that action's generator. In the positive-weight implementation, write

$$(T_n^a w)_i = r_{n,i}(a) + q_n \{P_{n,i}(a)w + b_{n,i}(a)\}, \quad (T_n w)_i = \sup_{a \in \mathcal{A}_i} (T_n^a w)_i. \quad (\text{S.6})$$

Here $P_{n,i}(a)$ has nonnegative entries with sum at most one. The term b contains prescribed stopping payoffs, and $q_n > 0$. Continuous controls need not enter this expression concavely. Reflection, interpolation cells, and stopping decisions can change as the action changes. A supremum, rather than an attained maximizer, suffices below.

Our continuous-action implementation fits the actor by differentiating its own evaluated continuation payoff. It does not construct an all-action target or receive an exact maximizing label. A bounded tanh output enforces the control box. A local improvement step can subsequently compare that proposal with finitely many continuous perturbations. Such a step is recorded as part of a *hybrid* policy; it is not identified with the raw network and is not assumed globally optimal. Critic fitting uses the selected feasible action and the preceding continuation critic. Training ends before any exhaustive reference or action-cover verification begins.

For independent verification, freeze the complete policy $\pi = (\pi_n)$ and cover each action box by closed subboxes. At a state and time, let c_i be an outward lower bound obtained by evaluating any feasible action using a fixed continuation upper array. A box can be discarded only when its verified upper bound does not exceed $c_i + \tau$. Every box not so discarded is either subdivided or retained with its upper bound. Exhausting a time, depth, or box budget therefore enlarges the reported bound; it never converts an unresolved search into a success.

PROPOSITION S.2—Budget-safe action and policy envelopes: *Suppose the terminal arrays satisfy $L_N \leq g \leq U_N$. At each time n , evaluate the fixed-policy map outward to obtain $L_n \leq T_n^{\pi_n} L_{n+1}$. For every state, maintain a complete action cover whose box bounds enclose $T_n^a U_{n+1}$ for all actions in that box. Set $U_{n,i}$ to the maximum of the outward-rounded*

pruning threshold and all unresolved box upper bounds. Then, regardless of which boxes remain unresolved,

$$L_n \leq V_n^\pi \leq V_n^* \leq U_n, \quad 0 \leq V_n^* - V_n^\pi \leq U_n - L_n. \quad (\text{S.7})$$

All inequalities are on the stated nodal economy. No concavity, exact training labels, or convergence of neural optimization is required. If $U_n - T_n U_{n+1} \leq e_n \mathbf{1}$, then its optimal-value excess is at most

$$\left(\prod_{j=n}^{N-1} q_j \right) \|U_N - g\|_\infty + \sum_{k=n}^{N-1} \left(\prod_{j=n}^{k-1} q_j \right) e_k.$$

The proof, including stopping faces, is in the supplement. A stored unresolved bracket supplies an upper bound for e_n ; it is not replaced by the requested tolerance. The policy lower envelope and the Bellman upper envelope are computed separately. The latter may be shared across several frozen policies on the same economy, while each policy requires its own evaluation. This separation prevents an expensive verification oracle from becoming undisclosed actor supervision.

In the preference application, the continuous box has three coordinates and the state grid has two. Bilinear interpolation is enclosed cell by cell. On each rectangle, its extrema occur at cell corners; a second implementation uses cell-slope enclosures. If a box crosses a wealth stopping face, the verifier takes the union of the interior continuation and the prescribed exit payoff. Their numerical values need not coincide at off-grid exit locations. Treating the entire action objective as globally differentiable at this switch would invalidate a derivative-only bound. Reflection is handled by including every crossed fold. Positive consumption and a preference interval strictly above one keep the powers and denominators in their stated domains.

The complete-action bound is not a continuous-state or continuous-time theorem. The verifier interprets stored primitive coefficients and nodal abscissae as exact real inputs and encloses subsequent operations outward. Training and verification agree to numerical precision on the inherited action grid, but the certificate's precise object is this declared nodal economy, not an unspecified sequence of floating-point evaluations. The supplemental transfer result states separately the state interpo-

lation, diffusion approximation, boundary monitoring, and action errors needed to recover the continuous economy. None is inferred from a sampled residual or silently assigned zero.

APPENDIX B: PROOFS

B.0.1. *Proof of Theorem 1*

Extend the aggregator Lipschitz-continuously outside its invariant interval for this comparison argument. The extension leaves all policy utilities and the candidate residual unchanged. For a nonnegative number c , define $d_c(t) = Be^{L(T-t)} + cA_L(T-t)$. Then $d_c(T) = B$, $d_c(t) \geq B$, and $-d'_c = Ld_c + c$. Since a spatially constant shift leaves the generator unchanged and f is L -Lipschitz in utility,

$$F[v + d_c] \geq F[v] - d'_c - Ld_c = F[v] + c, \quad F[v - d_c] \leq F[v] + d'_c + Ld_c = F[v] - c.$$

Because $F[v] = r^a[v] - g^a[v] \in [-e - \eta, e]$, $v + d_{e+\eta}$ is a supersolution and $v - d_{e+\eta}$ is a subsolution. They dominate the required stopping data on every boundary face. A spatially constant shift also preserves an exactly satisfied homogeneous reflected boundary condition. Comparison proves (13).

Apply the same argument to the fixed-policy operator $F^a[v] = -v_t - \mathcal{H}^a[v]$, with $c = e$, to obtain (12). The triangle inequality gives (14), and policy admissibility gives its nonnegative lower bound. The construction uses $L \geq 0$ and $d_c(t) \geq B$ specifically to cover lateral stopping boundaries; it does not incorrectly discount their errors as if every path stopped only at T . For nonsmooth candidates the same argument applies when their residual inequalities are verified in the viscosity sense. The theorem's differential implementation, however, requires the stated smooth candidate. Q.E.D.

B.0.2. *Exact policy improvement*

Suppose $v = V^a$ solves the fixed-policy problem and a feasible $\tilde{a}$ satisfies $\mathcal{H}^{\tilde{a}}[v] \geq \mathcal{H}^a[v]$ pointwise. Then $F^{\tilde{a}}[v] \leq 0$ with the same boundary payoff. Policy comparison gives $V^{\tilde{a}} \geq V^a$. This is a policy-space statement. A single gradient step on a domain-averaged actor loss does not automatically meet its pointwise premise.

The measured action gap and policy-evaluation errors are retained precisely because approximate neural improvement can violate that premise.

B.0.3. *Proof of Theorem 2*

For a fixed state and action, let z and $\tilde{z}$ solve (15) for w and $\tilde{w}$. Strict monotonicity of $z - hf(z)$ gives

$$|z - \tilde{z}| \leq \frac{|P_h^a(w - \tilde{w})|}{1 + \lambda h} \leq q_h \|w - \tilde{w}\|_\infty.$$

The same monotonicity implies order preservation. Taking the maximum over actions preserves both order and the Lipschitz constant. Banach's fixed-point argument supplies the two fixed points on the invariant complete value domain.

Now $\|v - v_h^a\| \leq \|v - T_h^a v\| + q_h \|v - v_h^a\|$, which gives the first bound. Also $\|v - T_h v\| \leq \delta + \epsilon$, giving the second. The triangle inequality gives the third. All norms are over every state of the specified finite model. An omitted action changes T_h and must be accounted for as a separate error, not hidden in δ . Q.E.D.

B.0.4. *Proof of Theorem 3*

First, (16) gives

$$\|v_h - v_h^*\|_\infty \leq \frac{\delta_h + \epsilon_h}{1 - q_h} = \frac{1 + \lambda h}{\lambda} \frac{\delta_h + \epsilon_h}{h} \longrightarrow 0.$$

It remains to identify the limit of the exact monotone scheme. Stability makes the upper and lower half-relaxed limits finite. At a strict local maximum of the upper limit minus a smooth test function, select nearby maximizing grid points. Monotonicity compares the neighboring scheme values to that test function plus the vanishing contact offset. Local consistency then gives the viscosity subsolution inequality. The corresponding minima give the supersolution inequality for the lower limit. Consistent boundary data and the assumed barriers give the required boundary inequalities. Comparison orders the upper limit below the lower limit. The reverse order follows by definition, so the two limits coincide with the unique continuous viscosity solution. The vanishing distance between v_h and v_h^* transfers that convergence to the neural arrays. The discrete policy-value conclusion follows from the third bound. Interpreting a discrete policy as a continuous-time implementation

1 additionally requires convergence of its controlled interpolation and payoff; the 1
 2 theorem does not infer this merely from convergence of its state arrays. This is the 2
 3 usual monotone-scheme selection argument, with the neural approximation error 3
 4 made explicit; see [Barles and Souganidis \(1991\)](#), [Crandall et al. \(1992\)](#). Q.E.D. 4

5 B.0.5. *Finite-horizon Bellman error accumulation* 5

6 Let T_n and T_n^a be the time- n Bellman and fixed-policy maps, with Lipschitz 6
 7 constants q_n (which may exceed one). For arrays v_n , define $\delta_n = \|v_n - T_n^a v_{n+1}\|$ and 7
 8 $\epsilon_n = \|T_n v_{n+1} - T_n^a v_{n+1}\|$. If the terminal error is B_N , backward induction gives 8
 9

$$\|v_n - v_n^*\| \leq \left(\prod_{j=n}^{N-1} q_j \right) B_N + \sum_{k=n}^{N-1} \left(\prod_{j=n}^{k-1} q_j \right) (\delta_k + \epsilon_k).$$

10 The policy-evaluation bound replaces $\delta_k + \epsilon_k$ by δ_k . Their sum bounds regret. Empty 10
 11 products equal one. For the implicit Lipschitz recursive step, $q_n \leq (1 - hL)^{-1}$; for 11
 12 additive exponential discount, $q_n = e^{-\rho h}$. Thus a per-step error $o(h)$ and consis- 12
 13 tent terminal data suffice on a fixed finite horizon, without asserting a stationary 13
 14 contraction in an inapplicable utility regime. 14
 15 15
 16 16
 17 17

18 B.0.6. *Derivation of the temporal-self condition* 18

19 A self at (t, x) replaces a^* by a fixed feasible local action a for duration ϵ and then 19
 20 returns to a^* . For each exponential component, the deviation value is 20
 21

$$\mathbb{E} \left[\int_t^{t+\epsilon} e^{-\rho_j(s-t)} u(a) ds + e^{-\rho_j \epsilon} W_j(t + \epsilon, S_{t+\epsilon}^a) \right].$$

22 Itô expansion gives its first-order increment relative to $W_j(t, x)$ as $\epsilon[u(a) + (W_j)_t +$ 22
 23 $\mathcal{L}^a W_j - \rho_j W_j] + o(\epsilon)$. Weighting the two components and subtracting their evalua- 23
 24 tion equations under a^* leaves 24
 25 25
 26 26
 27 27

$$\epsilon \{u(a) - u(a^*) + (\mathcal{L}^a - \mathcal{L}^{a^*})V\} + o(\epsilon).$$

28 Nonpositive first-order gain for every feasible action is exactly [\(S.71\)](#). This deriva- 28
 29 tion assumes the smoothness and local integrability needed for the expansion; it is 29
 30 an equilibrium deviation condition, not an assertion of optimal commitment over a 30
 31 31
 32 32

finite interval. The homothetic derivation follows by integrating $e^{-[\rho_j - b(r-m)]s}$ and applying the consumption first-order condition. *Q.E.D.*

B.1. *Proofs for the Guarded and Signed Map*

B.1.1. *Proof of Proposition S.7*

At a replaced node the output equals the computed target. At any other node its distance from that target is at most τ_h . The triangle inequality gives the one-step conclusion. For a frozen policy sequence, let e_n be the sup-norm difference between the guarded and exact policy values at date n . Positivity of the transition weights and their mass bound imply $e_n \leq \tau_h + \eta_h + q_h e_{n+1}$, with $e_N = 0$ for exact terminal data. Backward induction gives the geometric sum. As $N = T/h$ and $q_h \leq 1$, the bound is at most $T(0.1h + \eta_h/h)$ under the stated schedule. No property of the optimizer was used. The assertion concerns the guarded nodal values. An off-node neural interpolant, an improved policy, and a diffusion approximation each require their separate error account. In fixed-precision arithmetic, $\eta_h = o(h)$ is a maintained refinement condition, not a fact that can be inferred as h tends to zero indefinitely. $\square$

B.1.2. *Proof of Theorem S.8*

On a regular box, the payoff restricted to a line segment is continuous and piecewise continuously differentiable, with finitely many interpolation and reflection knots, or is the limit of such segments. Integration of its almost-everywhere derivative yields

$$Q_w(a) - Q_w(c) = \int_0^1 \nabla Q_w(c + s(a - c)) \cdot (a - c) ds.$$

Every partial derivative is included in its interval hull. Maximizing the resulting interval products proves (S.82). Taking the minimum with another valid upper enclosure preserves validity. If a partial derivative is strictly positive throughout the box, integration along that coordinate shows that replacing it by its upper endpoint cannot lower the payoff; the negative case is analogous. Repeating the argument proves face domination and the dimension statement.

A subdivision replaces a parent by children whose union contains it. A face reduction replaces a set by a subset with the same supremum. Removal occurs only after that supremum has a bound no larger than the incumbent plus tolerance. Feasible point evaluations can only increase the incumbent, so an earlier removed box stays bounded by the final incumbent plus tolerance. The remaining boxes keep their upper bounds without any success inference from an interrupted search. Induction over all operations proves (S.83). At a stopping switch the line-segment continuity premise can fail; the implementation therefore uses the natural union enclosure and does not invoke the derivative argument there.

For the local quadratic observation, a K -Lipschitz gradient with a zero-gradient point in the box has norm at most Ks throughout a box of diameter s . Integration over a segment of length at most s bounds the variation by Ks^2 . This conditional observation does not extend to an arbitrary switching box, nor does it assert that all nodes possess such a box. $\square$

B.1.3. *Derivative hulls for the preference economy*

Write $y = \log X$, and let F be the continuation interpolant evaluated at a cubature transition. On an interior leg, its action derivatives are

$$\partial_c F = -hF_y, \quad \partial_p F = F_y \{h(0.06 - 0.04p) + s_z\}, \quad \partial_\theta F = hF_u r_u,$$

where s_z is the stored portfolio shock coefficient and $r_u \in \{-1, 1\}$ is the reflection slope; both slopes enter the hull at a fold. The flow derivatives are

$$h \exp\{(1-u)y - u \log c\}, \quad 0, \quad -hk\theta.$$

Outside a wealth stopping face, clipped wealth is constant and F is the exact stopping payoff. Its y derivative with respect to the transition is zero, while

$$F_u = \frac{\exp\{(1-u)y\} \{1 - (1-u)y\}}{(1-u)^2}.$$

Every crossed bilinear cell contributes its one-sided derivative range, restricted by the other coordinate's interval. Interval evaluation includes these ranges and the flow terms. A leg whose wealth interval meets both sides of a stopping switch

disables the signed bound for that box. Point and interval evaluations share the stated nodal primitives but are computed independently of the actor's derivative graph.

B.1.4. *Proof of Proposition S.9*

Let $\bar{m} = d^{-1} \sum_i m_i$. Jensen's inequality gives $d^{-1} \sum_i \log m_i \leq \log \bar{m}$. Moreover, $d^{-1} \sum_i \tanh((BY)_i) \leq 1$. The mean-state drift is consequently no larger than $b_+ - \bar{m}$. The dispersion penalty at the terminal date is nonnegative and can be dropped for an upper bound. Fubini's theorem applied to the integrated expected mean state and terminal mean state yields the kernel $w(t)$. The Brownian terms have zero mean; bounded controls and bounded nonlinear drift ensure the required integrability. Thus every admissible policy has expected payoff at most

$$\int_0^T e^{-\rho t} \max_{m \in [\underline{m}, \bar{m}]} \{ \log m - \zeta m^2 / 2 + w(t)(b_+ - m) \} dt.$$

The integrand is strictly concave in m . Its derivative is $1/m - \zeta m - w(t)$, whose positive root gives $m_0(t)$ before clipping. This proves the upper bound.

For the feasible common schedule, the expected mean-state drift is at least $b_- - m_0(t)$. Define the normalized dispersion norm $\|Z\|_{2,d}^2 = \mathbb{E}[d^{-1} \sum_i Z_i^2]$. Centred common controls and common noise vanish from the dispersion equation. For numbers in $[-1, 1]$, their cross-sectional variance is at most one. Hence Minkowski's inequality bounds the norm of the integrated centred nonlinear drift by κT . The centred idiosyncratic Brownian part has norm $\sigma_I \sqrt{(1 - 1/d)T}$. A second application of Minkowski and squaring give the upper bound D_d on expected terminal dispersion. Subtracting $\chi e^{-\rho T} D_d$ from the lower mean-payoff calculation proves $J(m_0) \geq L_d$. Feasibility gives $J(m_0) \leq V^*$. The derivation never restricts the dense matrix B beyond the stated bounded-tanh form. $\square$

B.1.5. *Compensation and arithmetic scope*

For each realized state and positive consumption, CRRA felicity is increasing in consumption, including when $u > 1$ and its level is negative. The financed supplement leaves the state law and every other payoff term unchanged. Its expected

payoff is therefore nondecreasing in e . Positivity of the nodal transition operator propagates a lower interval payoff at each date. If its lower endpoint exceeds the optimal upper envelope at every active starting node and date, the corresponding financed supplement compensates the loss throughout that nodal economy. Bisection searches for a sufficient supplement; only the final verified inequality provides the stated upper bound.

All implemented interval conclusions use the inherited binary64 execution contract, with outward elementary operations and separately bounded transcendental evaluations. They concern exact real interpretations of stored primitive coefficients and grid nodes. They are not a formal verification of the interpreter, hardware, or all library internals. In the capital bound, square-root seeds are accepted only after interval squaring verifies both endpoint inequalities, and the time integral is bounded by interval rectangles rather than by an adaptive quadrature tolerance. Random stress tests are regression evidence, not substitutes for these inclusion arguments.

B.2. Additional Proofs and Numerical Contracts

B.2.1. Costate error and action quality

Let $\hat{m}$ maximize $H_{\hat{q}}$ on the stated convex feasible set, and let $e = q - \hat{q}$. Strong concavity and the variational first-order condition at $\hat{m}$ imply

$$H_{\hat{q}}(\hat{m}) - H_{\hat{q}}(m) \geq \frac{\mu}{2} \|m - \hat{m}\|_d^2$$

for every feasible m , including boundary maximizers. Therefore $\|\tilde{m} - \hat{m}\|_d \leq \sqrt{2\delta/\mu}$. Decomposing the true Hamiltonian difference yields

$$\begin{aligned} H_q(m_q) - H_q(\tilde{m}) &\leq \delta - \frac{\mu}{2} \|m_q - \hat{m}\|_d^2 + \langle e, \tilde{m} - m_q \rangle_d \\ &\leq \delta + \|e\|_d \sqrt{2\delta/\mu} + \frac{\|e\|_d^2}{2\mu}. \end{aligned}$$

The last step completes the square in $\|m_q - \hat{m}\|_d$. This is (28). The Hessian of H_q is the negative diagonal matrix with entries $1/(dm_i^2)$ minus the positive semidefinite

mean-penalty matrix. It is bounded above by $-\mu I/d$, proving the strong-concavity premise.

For (29), write $Z - g(X) = [Z - \mathbb{E}(Z | X)] + [\mathbb{E}(Z | X) - g(X)]$. The cross term has zero expectation by conditioning on X . Integrability justifies this orthogonal decomposition. If the conditional Euler costate differs from the continuous policy costate by $b(X)$, that term remains an additional bias; it is not removed by the projection identity. Neither the sampled costate loss nor a successful optimizer return is a uniform upper bound on this bias.

B.2.2. *Cancellation, clipping and finite-population inference*

For each candidate, the exact payoff identity used in Proposition S.5 consists of a policy production term, its consumption term, its terminal dispersion, and the corresponding reference terms. When two candidates use the identical reference path, subtraction cancels every exact reference term. Comparing the two policy terms with their numerical approximations therefore requires the two policy transfer errors and the statistic-roundoff accounts, but no reference-state transfer error. The implementation retains conservative original statistic budgets even where their reference quadrature components also cancel.

For any real x, y and positive thresholds b_a, b_b ,

$$(|x - y| - b_a - b_b)_+ \leq (|x| - b_a)_+ + (|y| - b_b)_+.$$

Taking expectations proves the summed clipping-tail allowance without requiring independence between policies. Applying the empirical Bernstein inequality to the clipped difference and then to its negative gives the two one-sided bounds. A union bound covers all declared tests. Our allocation counts both directions and every method, profile design and auxiliary evaluation; it does not spend new confidence mass for deterministic transformations such as (39).

For the finite population, draw I with the fixed probabilities p_j , independently of each future innovation path. The pair (I, W) is then independent across paths.

Conditional transfer gives

$$\left| \mathbb{E}[X_{a,h}] - \sum_j p_j \{J_{y_j}(a) - J_{y_j}(m_0)\} \right| \leq \sum_j p_j E_{a,h}(y_j).$$

The clipping threshold is the maximum of the profile-specific thresholds; its expectation-tail allowance is bounded by the maximum of their allowances. The prescribed implementation uses equal probabilities over nine fixed profiles. Both averaging and the final subtraction are interval-enclosed. Random empirical profile frequencies are not substituted for the population weights when calculating deterministic error budgets. They are part of the sampling variation of the final statistic.

For the paired method difference, the same profile is assigned to both policies on each path. Common random numbers across different methods or training seeds do not invalidate a finite-family union bound. They also do not create independent replications of the optimizer. The ten fixed training streams are reported individually and descriptively; no confidence assertion about an unspecified distribution of future hyperparameter choices is inferred.

B.2.3. *Observed-history implementation*

Under the observation assumptions, M is a continuous martingale with bracket $\langle M \rangle_t = Qt$. Let $A = \Sigma^\top Q^{-1}$. The matrix $A\Sigma$ is the orthogonal projector onto the row space of Σ , and $nn^\top$ is the projector onto its orthogonal complement. Independence of the private Brownian motion gives

$$\langle AM + nB^0 \rangle_t = (AQA^\top + nn^\top)t = I_{d+1}t.$$

The multivariate Lévy characterization proves the Brownian assertion. Also $\Sigma A = I_d$ and $\Sigma n = 0$, so $\Sigma \tilde{W} = M$. On each decision cell, the action is determined at its left endpoint by the stored internal state and past recovered increments. With that held action, the physical state equation has a unique strong solution because the production drift is globally Lipschitz and bounded. Induction over the cells establishes uniqueness in law for the joint physical and internal process. Driven by the Brownian motion just constructed, it therefore has the same law as the controller originally

defined using directly supplied innovations. Applying the specified componentwise clipping to the recovered vector preserves this distributional equivalence.

The proposition uses only private randomization and observed state history; it does not reveal the original null-space economic driver. Randomized controls are admissible in the enlarged filtration. Since the original deterministic performance envelope holds against all such adapted controls and independent randomization cannot improve an unconstrained expected-payoff supremum beyond the best deterministic randomization realization, the comparison with the original optimum is preserved. The result is not an implementability theorem for discrete noisy state measurements. The exact drift integral is stated explicitly, and errors in approximating it must be treated separately.

B.2.4. *Independent scalar HJB approximation*

For $d = 1$, the terminal dispersion penalty vanishes but production remains nonlinear in log capital. The reference solves the same primitive action-box problem, not a Riccati or homothetic reduction. Put $f(y) = \kappa \tanh(By)$ and $\nu = (\sigma_L^2 + \sigma_C^2)/2$. On a uniform grid of width Δy , define the positive rates

$$\ell_i(m) = \frac{\nu}{\Delta y^2} + \frac{(m - c - f(y_i))_+}{\Delta y}, \quad r_i(m) = \frac{\nu}{\Delta y^2} + \frac{(c + f(y_i) - m)_+}{\Delta y}.$$

The implicit Bellman step solves

$$\frac{v_i - v_i^+}{h} + \rho v_i = \max_m \{ \log m + y_i - \zeta m^2/2 + \ell_i(m)(v_{i-1} - v_i) + r_i(m)(v_{i+1} - v_i) \}.$$

For a fixed action vector the coefficient matrix is strictly diagonally dominant with positive diagonal and nonpositive off-diagonal entries. Its inverse is nonnegative. Howard updates maximize over each drift-sign interval; on either interval the positive root of $1/m - \zeta m = p$ is clipped to its feasible endpoints, and the two candidates are compared. This is an independent monotone matrix computation, not a trained critic used as its own reference.

At the two outer boundaries, the prescribed affine asymptotes replace production by its limiting saturation value. If $w(t)$ is the coefficient on initial log capital and $m_0(t)$ the common schedule, their intercept is the numerical integral of the

corresponding deterministic discounted flow. The reference reports the grid, time step, boundary distance, linear-system residual, policy-iteration convergence and domain sensitivity. Fitted actors are evaluated as Markov feedbacks on this finite grid. These comparisons do not conflate that controller with the innovation-driven implementation. Grid differences and asymptotic boundary values are not asserted to be rigorous continuous-domain error enclosures.

B.2.5. From evaluation error to economic improvement

Proposition 5 controls a decision at a given state. The distribution of states at which that decision is used matters for the economic conclusion. We make this link explicit. In this subsection, π is a fixed Markov policy with a sufficiently regular policy value V^π to justify Itô's formula and the expectations below. Specifically, $V^\pi \in C^{1,2}([0, T] \times \mathbb{R}^d)$ is continuous at the terminal date with $V^\pi(T, \cdot) = g$. Along each candidate considered, the drift and terminal terms in the discounted Itô identity are integrable and its stochastic integral is a zero-mean martingale. The condition

$$\mathbb{E} \int_0^T e^{-2\rho t} \|\Sigma^\top D_y V^\pi(t, Y_t^a)\|_2^2 dt < \infty$$

is sufficient for the latter. The parameters of π are frozen. Let a be any admissible candidate, including a history-dependent candidate, and let Y^a be its physical state. Set $q^\pi = dD_y V^\pi$ and define

$$g^\pi(t, y) = \max_{m \in \mathcal{A}} H_{q^\pi(t, y)}(m) - H_{q^\pi(t, y)}(\pi(t, y)) \geq 0,$$

where $\mathcal{A}$ is the same convex action set for both maximizations and both policies. One may always take the primitive action box. For a costate predictor $\hat{q}$, let δ_t bound the $H_{\hat{q}(t, Y_t^a)}$ maximization gap of a_t over that set. Write $\bar{q}_h(t, y)$ for the conditional mean of the frozen-policy rollout costate at mesh h . Define, under the candidate's state law,

$$\mathcal{G}_a = \mathbb{E} \int_0^T e^{-\rho t} g^\pi(t, Y_t^a) dt,$$

$$\mathcal{D}_a = \mathbb{E} \int_0^T e^{-\rho t} \delta_t dt,$$

$$\mathcal{E}_{a,h}^2 = \mathbb{E} \int_0^T e^{-\rho t} \|\widehat{q}(t, Y_t^a) - \bar{q}_h(t, Y_t^a)\|_d^2 dt,$$

$$\mathcal{B}_{a,h}^2 = \mathbb{E} \int_0^T e^{-\rho t} \|\bar{q}_h(t, Y_t^a) - q^\pi(t, Y_t^a)\|_d^2 dt.$$

The quantity $\mathcal{E}_{a,h}$ concerns regression accuracy; $\mathcal{B}_{a,h}$ retains the rollout discretization bias.

PROPOSITION S.3—Evaluation, action quality, and payoff: *Suppose the displayed quantities are finite and the consumption Hamiltonian is μ -strongly concave on $\mathcal{A}$. Then*

$$J_y(a) - J_y(\pi) \geq \mathcal{G}_a - \left\{ \sqrt{\mathcal{D}_a} + \frac{\mathcal{E}_{a,h} + \mathcal{B}_{a,h}}{\sqrt{2\mu}} \right\}^2. \quad (\text{S.8})$$

In particular, an upper bound on the action-loss term and a lower bound on $\mathcal{G}_a$ establish a sufficient condition for policy improvement. The proposition applies to any costate estimator; a method comparison still requires the independently evaluated payoffs and their associated work.

PROOF: The fixed-policy equation for V^π , evaluated along Y^a , gives the performance-difference identity

$$J_y(a) - J_y(\pi) = \mathbb{E} \int_0^T e^{-\rho t} \{H_{q^\pi(t, Y_t^a)}(a_t) - H_{q^\pi(t, Y_t^a)}(\pi(t, Y_t^a))\} dt. \quad (\text{S.9})$$

The diffusion and the state-dependent part of felicity are common to the two actions, so their differences cancel. The terminal terms agree. Localization followed by the assumed integrability justifies the martingale expectation. Subtract the true greedy Hamiltonian inside the integrand and use Proposition 5. Minkowski's inequality for the measure $e^{-\rho t} dt d\Pr$, first for $\sqrt{\delta_t}$ and the costate norm, and then for the regression and bias components of the costate, gives

$$\mathbb{E} \int_0^T e^{-\rho t} \left(\sqrt{\delta_t} + \frac{\|q^\pi - \widehat{q}\|_d}{\sqrt{2\mu}} \right)^2 dt \leq \left\{ \sqrt{\mathcal{D}_a} + \frac{\mathcal{E}_{a,h} + \mathcal{B}_{a,h}}{\sqrt{2\mu}} \right\}^2.$$

Combining these statements proves the result.

Q.E.D.

Independent costate diagnostics. Condition on a fitted policy, its critic, and a held-out state X . Let $\bar{Z}_1, \bar{Z}_2$ be means from independent rollout banks with the same conditional mean $\bar{q}_h(X)$. Conditional independence and zero conditional mean of the two errors imply

$$\mathbb{E}[\langle \hat{q}(X) - \bar{Z}_1, \hat{q}(X) - \bar{Z}_2 \rangle_d \mid X] = \|\hat{q}(X) - \bar{q}_h(X)\|_d^2.$$

With r independent replicates per bank and conditional single-rollout variance $v_h(X) = \mathbb{E}[\|Z - \bar{q}_h(X)\|_d^2 \mid X]$,

$$\frac{1}{2} \mathbb{E}[\|\bar{Z}_1 - \bar{Z}_2\|_d^2 \mid X] = \frac{v_h(X)}{r}.$$

The first sample average is not constrained to be positive. Both identities refer to the actual held-out state distribution. They estimate the quantities in (S.8) only when that distribution is the candidate's discounted state occupation law, or when a justified change-of-measure account is supplied. Nested-grid differences diagnose time-step sensitivity; without a remainder bound they do not bound $\mathcal{B}_{a,h}$.

A continuous-time costate error account. There is also a direct sufficient condition for controlling the true costate, independent of a rollout bias estimate. For a smooth candidate value v , define its fixed-policy residual and terminal error by

$$r^\pi[v] = -v_t - \mathcal{L}^\pi v - u(\cdot, \pi) + \rho v, \quad b_v = v(T, \cdot) - g.$$

Suppose $b^\pi = c1 + f - \pi$ is continuously differentiable with $\|D_y b^\pi\|_2 \leq L_b$, and suppose differentiating the following Feynman–Kac expectations is justified. If $\|dD_y b_v\|_d \leq C_T$ and $\|dD_y r^\pi[v](s, \cdot)\|_d \leq C_r(s)$ everywhere, then

$$\|dD_y v(t, y) - q^\pi(t, y)\|_d \leq e^{(L_b - \rho)(T-t)} C_T + \int_t^T e^{(L_b - \rho)(s-t)} C_r(s) ds. \quad (\text{S.10})$$

To see this, write $v - V^\pi$ as the expected discounted terminal error plus the expected discounted residual integral. The state Jacobian $J_{t,s}$ satisfies $\dot{J}_{t,s} = D_y b^\pi(s, Y_s) J_{t,s}$, $J_{t,t} = I$, and hence $\|J_{t,s}\|_2 \leq e^{L_b(s-t)}$. Differentiate the expectation and apply that bound. A hard terminal condition gives $C_T = 0$. Equation (S.10) is a sufficient mathematical account for a smooth continuous policy. A sampled residual, a rollout

grid difference, or the derivatives of a different deployed implementation do not supply its global hypotheses.

APPENDIX C: CAPITAL ERROR BOUNDS AND NUMERICAL ACCOUNTS

This appendix supplies the complete supporting constants and proofs for the capital model and policy-specific theorem in the main article. The economic state, objective, action box, benchmark schedule, and implemented controller are those in (18)–(20) and (23). The analytical anchor, diffusion transfer, and clipping account concern different error terms; none is a training convergence assertion.

C.1. The analytical anchor and tube bound

Let $\beta \geq \kappa \|B\|_2$, let s_0 bound the initial cross-sectional standard deviation, and set

$$Q_d = s_0 + \kappa T + \sigma_I \sqrt{(1 - 1/d)T}, \quad (\text{S.11})$$

$$K(s) = \beta \int_s^T e^{-\rho t} w(t) e^{\beta(t-s)} dt + 2\chi e^{-\rho T} Q_d e^{\beta(T-s)}, \quad (\text{S.12})$$

$$\mu(t) = \frac{2\{\log[m_0(t)/M] + [M - m_0(t)]/m_0(t)\}}{[M - m_0(t)]^2}. \quad (\text{S.13})$$

The quantity $\mu(t)$ is the smallest tangent-remainder curvature of log utility on the entire primitive action interval, relative to $m_0(t)$. It is not a Hessian bound estimated from training points. In particular, it can be substantially sharper than the uniform lower curvature M^{-2} .

THEOREM S.4—Curvature-controlled policy guarantee: Suppose $\rho > 0$, $\zeta, \chi, \kappa \geq 0$, the schedule in (20) is interior, and $\|\Pi y\|_d \leq s_0$. Let $V^*(y) = \sup_m J_y(m)$, where the supremum is over the full original admissible class. Then

$$0 \leq V^*(y) - J_y(m_0) \leq G_d := \frac{1}{2} \int_0^T e^{\rho s} \frac{K(s)^2}{\mu(s)} ds. \quad (\text{S.14})$$

For $\epsilon \geq 0$ satisfying $[m_0(t) - \epsilon, m_0(t) + \epsilon] \subset [\underline{m}, M]$, every admissible policy m^ϕ with $|m_i^\phi(t, Y) - m_0(t)| \leq \epsilon$ for all coordinates, times, and states satisfies

$$0 \leq V^*(y) - J_y(m^\phi) \leq G_d + D_{d,\epsilon}, \quad (\text{S.15})$$

where, with $A_\beta(T) = \int_0^T e^{\beta t} dt$,

$$D_{d,\epsilon} = \int_0^T \left[\epsilon K(s) + \frac{\epsilon^2 e^{-\rho s}}{2} \{[m_0(s) - \epsilon]^{-2} + \zeta\} \right] ds + \chi e^{-\rho T} \epsilon^2 A_\beta(T)^2. \quad (\text{S.16})$$

These bounds hold for every initial mean, every initial state with the stated dispersion bound, and arbitrary adapted competitors in the original action box. They require no neural training convergence or discretization error estimate.

The proof uses synchronous state comparison, an exact discounted mean-state identity, and completion of a quadratic bound on an arbitrary policy deviation. The complete proof appears in Section C.2. The result is a model-specific numerical principle: curvature limits the benefit from changing consumption, whereas a spectral bound limits the propagation of that change through dense nonlinear production. Their balance yields a scalar integral certificate for a high-dimensional policy. Bounded nonlinearity, the additive diffusion structure, and the chosen felicity are essential assumptions, rather than generic properties of all HJB equations.

C.1.1. Computable scope and cost

An SVD is used only to propose a scalar λ . Positive pivots in an outward interval $LDL^\top$ calculation for $\lambda I - B^\top B$ certify the spectral majorant. A failed pivot triggers enlargement, not acceptance of the SVD output. The remaining integrals use outward rectangle sums, polynomial exponential and logarithm bounds, and square-root endpoints checked by interval squaring. For a fixed majorant with verified pivots, this certificate requires $O(d^3)$ arithmetic operations and $O(d^2)$ storage for the matrix check, followed by $O(N)$ scalar quadrature work. This is not a dimension-independent training theorem or a worst-case bound on the number of attempted majorants.

The guarantee holds already at initialization. Training is intended to improve the feasible policy inside the tube, not to manufacture its certificate. The anchor itself has the tighter bound G_d and requires no training. A comparison omitting it would

overstate the role of the actor. Neither a small $G_d + D_{d,\epsilon}$ nor faster actor evaluation proves a net economic advantage over that benchmark.

C.2. Proof of the Continuous-Capital Guarantee

All norms in this proof use the normalization $d^{-1} \sum_i z_i^2$. This normalization is important: the spectral norm of B remains the ordinary Euclidean operator norm, while the stochastic moment of cross-sectional dispersion is of order one rather than order d .

C.2.1. Logarithmic tangent remainder

For $a > 0$ and $x > -a$, twice integrating the second derivative gives

$$\frac{x}{a} - \log(1 + x/a) = x^2 \int_0^1 \frac{1-r}{(a+rx)^2} dr.$$

For $x \geq 0$ the integral decreases as x increases. For $x < 0$ it is at least $1/(2a^2)$, its value at zero. Consequently, for every $m \in [\underline{m}, M]$ and interior $a < M$,

$$\log(m/a) - \frac{m-a}{a} \leq -\frac{\mu(a)}{2}(m-a)^2, \quad \mu(a) = \frac{2\{\log(a/M) + (M-a)/a\}}{(M-a)^2}. \quad (\text{S.17})$$

On the smaller interval $|m-a| \leq \epsilon < a$, the same integral also yields

$$0 \leq \frac{m-a}{a} - \log(m/a) \leq \frac{(m-a)^2}{2(a-\epsilon)^2}. \quad (\text{S.18})$$

These are global inequalities on the specified intervals, including their endpoints; no empirical concavity check is used.

C.2.2. Synchronous comparison and baseline moments

Construct Y^m and Y^0 on the same probability space with identical initial state and Brownian drivers, where the second process uses m_0 . With $\Delta Y = Y^m - Y^0$ and $\Delta m = m - m_0 \mathbf{1}$, additive noise cancels. Since componentwise hyperbolic tangent is one-Lipschitz,

$$\|\Delta Y_t\|_d \leq \int_0^t \{\beta \|\Delta Y_s\|_d + \|\Delta m_s\|_d\} ds.$$

Gronwall's inequality followed by Minkowski's integral inequality implies

$$(\mathbb{E}\|\Delta Y_t\|_d^2)^{1/2} \leq \int_0^t e^{\beta(t-s)} r(s) ds, \quad r(s) = (\mathbb{E}\|\Delta m_s\|_d^2)^{1/2}. \quad (\text{S.19})$$

All terms are integrable because controls and the nonlinear production term are bounded. For the baseline, common consumption, common drift constants, and common noise vanish under Π . Furthermore, $\|\Pi \tanh(BY)\|_d \leq 1$ and $\mathbb{E}\|\Pi W_T\|_d^2 = (1 - 1/d)T$. Minkowski's inequality therefore gives

$$(\mathbb{E}\|\Pi Y_T^0\|_d^2)^{1/2} \leq \|\Pi y\|_d + \kappa T + \sigma_I \sqrt{(1 - 1/d)T} \leq Q_d. \quad (\text{S.20})$$

This estimate is uniform in the initial mean. It does not suppose that the optimal policy is symmetric or that the dense economy separates.

C.2.3. An exact payoff-difference identity

Put $h(Y) = \overline{\tanh(BY)}$. Integrating the mean-state equation inside the discounted flow and terminal mean, and interchanging integrable expectations and time integrals, yields

$$\begin{aligned} J_y(m) - J_y(m_0) &= \int_0^T e^{-\rho t} \mathbb{E} \left[\overline{\log(m_t/m_0(t))} - w(t) \overline{\Delta m_t} - \frac{\zeta}{2} \{\bar{m}_t^2 - m_0(t)^2\} \right] dt \\ &\quad + \kappa \int_0^T e^{-\rho t} w(t) \mathbb{E}[h(Y_t^m) - h(Y_t^0)] dt \\ &\quad - \chi e^{-\rho T} \mathbb{E} \left[2 \langle \Pi Y_T^0, \Pi \Delta Y_T \rangle_d + \|\Pi \Delta Y_T\|_d^2 \right]. \end{aligned} \quad (\text{S.21})$$

The martingale mean terms have zero expectation. The schedule satisfies $m_0(t)^{-1} - w(t) - \zeta m_0(t) = 0$. Thus all terms linear in $\overline{\Delta m_t}$ cancel. Equation (S.17) bounds the first integral above by $-\frac{1}{2} \int e^{-\rho t} \mu(t) r(t)^2 dt$; the remaining mean-adjustment square is nonpositive and may be dropped in an upper bound.

Next, $|h(Y^m) - h(Y^0)| \leq \|B\|_2 \|\Delta Y\|_d$. Equations (S.19) and (S.20), Cauchy-Schwarz, and a change in integration order bound the potentially positive production and terminal-dispersion terms by $\int_0^T K(s) r(s) ds$. Dropping the nonpositive terminal square, we obtain

$$J_y(m) - J_y(m_0) \leq \int_0^T \left[K(s) r(s) - \frac{1}{2} e^{-\rho s} \mu(s) r(s)^2 \right] ds \leq \frac{1}{2} \int_0^T e^{\rho s} \frac{K(s)^2}{\mu(s)} ds.$$

Taking the supremum over the original adapted controls proves (S.14). Attainment of the supremum is not needed.

For a tube policy, $r(s) \leq \epsilon$. Reverse the payoff difference in (S.21). Equation (S.18) and $\mathbb{E}(\overline{\Delta m})^2 \leq \mathbb{E}\|\Delta m\|_d^2$ bound the consumption and adjustment loss by

$$\frac{\epsilon^2}{2} \int_0^T e^{-\rho s} \{[m_0(s) - \epsilon]^{-2} + \zeta\} ds.$$

The production and terminal cross terms contribute at most $\epsilon \int K(s) ds$. The terminal square now has positive sign and is bounded by $\chi e^{-\rho T} \epsilon^2 A_\beta(T)^2$ using (S.19). This proves $J_y(m_0) - J_y(m^\phi) \leq D_{d,\epsilon}$. Adding (S.14) proves (S.15). Feasibility supplies the nonnegative lower bound. *Q.E.D.*

C.2.4. Interval spectral and deployment checks

For the stored exact-real matrix B , outward products and sums enclose each entry of $B^\top B$. The Schur complement recurrence for an $LDL^\top$ factorization is enclosed entry by entry. If every pivot interval has strictly positive lower endpoint, every exact pivot is positive, so $\lambda I - B^\top B$ is positive definite. Hence $\|B\|_2 < \sqrt{\lambda}$. The proposed numerical singular value is not part of this proof; an unsuccessful interval factorization supplies no spectral bound.

If $\ell \leq m_0(t) \leq u$, then $\text{up}(u - \epsilon) \geq m_0(t) - \epsilon$ and $\text{down}(\ell + \epsilon) \leq m_0(t) + \epsilon$. Every finite output after projection onto (22) therefore meets the theorem's tube condition under the declared arithmetic contract, regardless of errors in the neural proposal. The proof does not verify the interpreter or processor. The arithmetic assumptions, overflow checks, saved weights, and input-clock convention remain explicit components of the computational claim.

C.3. A network-independent diffusion-transfer bound

Here the word independent concerns the transfer constants, not the economic statistic. Let

$$q_i = \sigma_I^2 \|B_{i\cdot}\|_2^2 + \sigma_C^2 \left(\sum_j B_{ij} \right)^2,$$

$$H = \kappa \left(d^{-1} \sum_i q_i \right)^{1/2}, \quad Q = \frac{\kappa}{2} \left(d^{-1} \sum_i q_i^2 \right)^{1/2}.$$

The letter Q here denotes a generator bound, not the dispersion bound Q_d in (S.11). For a policy in a radius- ϵ interval at sampling dates, put

$$M_\epsilon = \max_{m \in [m_0(0) - \epsilon, m_0(T) + \epsilon]} |c - m| + \kappa, \quad K_\epsilon = \beta M_\epsilon + Q.$$

The terminal dispersion majorant is

$$S_\epsilon = s_0 + (\kappa + \epsilon)T + \sigma_I \sqrt{(1 - 1/d)T}.$$

The formula uses the monotonicity of this economy's m_0 . For more general schedules their extrema replace the endpoints. Define

$$u_\epsilon(t) = h e^{\beta t} \{ K_\epsilon t/2 + H \sqrt{t/3} \}, \quad (\text{S.22})$$

$$\begin{aligned} E_\epsilon(h) &= \beta \sum_k B_k u_\epsilon(t_k) + K_\epsilon \sum_k \int_{t_k}^{t_{k+1}} W(t)(t - t_k) dt \\ &\quad + \chi e^{-\rho T} u_\epsilon(T) \{ 2S_\epsilon + u_\epsilon(T) \}. \end{aligned} \quad (\text{S.23})$$

PROPOSITION S.5—Transfer of paired policy values: *For exact arithmetic and untruncated internal innovations, the preceding construction satisfies*

$$|\Delta_\phi - \mathbb{E} X_{\phi,h}| \leq E_\epsilon(h) + E_0(h).$$

With clipped internal innovations, forward rounding, and enclosed scalar integration, replace each u_ϵ by its stated enlarged bound and add the statistic's quadrature and arithmetic error. Denote the resulting saved upper bound by $E_{\phi,h}^{\text{all}}$. It is $O(h)$ apart from these explicit numerical terms and does not involve a network derivative bound. The economic process remains the original continuous diffusion.

The complete proof appears in Section C.5. Its key step is to retain the martingale cancellation in the accumulated local drift error. Bounding each local stochastic error separately by its absolute magnitude would yield only an $O(\sqrt{h})$ account and would obscure the small policy differences studied here. This proposition

concerns the sampled implementation (23); it does not prove this rate for arbitrary state-observed neural feedback.

C.4. The deterministic clipping and tail account

Paired statistics are not uniformly bounded, because the terminal dispersion difference contains Gaussian shocks. This is dealt with explicitly, rather than ignored in a bounded concentration inequality. For each declared initial state and radius, the supplement constructs constants C_ϕ, A_ϕ such that

$$\Pr\{|X_{\phi,h}| > C_\phi + A_\phi(\sqrt{d-1} + v)\} \leq e^{-v^2/2}, \quad v \geq 0.$$

They are obtained from synchronous state separation and the norm of the projected reference innovations. Neither an empirical output range nor an empirical maximum enters their construction. With $v_* = 8$, put $b_\phi = C_\phi + A_\phi(\sqrt{d-1} + v_*)$ and $\tau_\phi = A_\phi e^{-v_*^2/2}/v_*$. Then clipping $X_{\phi,h}$ to $[-b_\phi, b_\phi]$ changes its expectation by at most τ_ϕ .

C.5. Proofs for Policy-Specific Verification

This section proves the continuous-time transfer, clipping, and statistical statements used in the main text. Norms are normalized by dimension throughout. The internal controller and the economic state are distinct. This distinction is maintained in every comparison below.

C.5.1. Generator and stochastic local-error bounds

Let $f_i(y) = \kappa \tanh(B_i \cdot y)$, $b(y, m) = c\mathbf{1} + f(y) - m$, and let Σ have idiosyncratic columns σ_{Ie_i} and common column $\sigma_C\mathbf{1}$. For the ordinary spectral majorant $\beta \geq \kappa\|B\|_2$,

$$\|f(y) - f(z)\|_d \leq \beta\|y - z\|_d, \quad \|b(y, m)\|_d \leq M_\epsilon.$$

Each row of $Df(y)\Sigma$ has squared norm at most $\kappa^2 q_i$, so its normalized squared Hilbert–Schmidt norm is at most H^2 . Also

$$\left| \frac{1}{2} \text{tr}\{\Sigma \Sigma^\top D^2 f_i(y)\} \right| \leq \frac{\kappa}{2} q_i,$$

because $|\tanh''| \leq 1$. The generator of f under any held admissible action therefore satisfies

$$\|\mathcal{A}^m f(y)\|_d \leq \beta M_\epsilon + Q = K_\epsilon. \quad (\text{S.24})$$

The controls used during the comparison are adapted, but fixed over each cell. The same bound applies to the reference's deterministic, time-varying common control, with $\epsilon = 0$ and the time-dependent generator.

First take exact arithmetic and untruncated internal innovations. On cell k , Itô's formula and stochastic Fubini give

$$\begin{aligned} \int_{t_k}^{t_{k+1}} \{f(Y_s) - f(Y_{t_k})\} ds &= \int_{t_k}^{t_{k+1}} (t_{k+1} - v) \mathcal{A}^{a_k} f(Y_v) dv \\ &\quad + \int_{t_k}^{t_{k+1}} (t_{k+1} - v) Df(Y_v) \Sigma dW_v =: D_k + M_k. \end{aligned}$$

The deterministic term has L^2 norm at most $K_\epsilon h^2/2$. The stochastic terms are martingale increments over disjoint cells. Itô's isometry, not a sum of their absolute norms, gives

$$\left(\mathbb{E} \left\| \sum_{j < k} M_j \right\|_d^2 \right)^{1/2} \leq Hh \sqrt{t_k/3}.$$

Set $e_k = Y_{t_k} - Z_k$. The chosen actions cancel from the recursions. The triangle inequality yields

$$(\mathbb{E} \|e_k\|_d^2)^{1/2} \leq \beta h \sum_{j < k} (\mathbb{E} \|e_j\|_d^2)^{1/2} + h \{K_\epsilon t_k/2 + H \sqrt{t_k/3}\}.$$

The inhomogeneous term is nondecreasing in t_k . Discrete Gronwall and $(1 + \beta h)^k \leq e^{\beta t_k}$ prove (S.22). Importantly, no derivative of a_k with respect to its internal state appears in this recursion. Replacing a_k by a policy evaluated separately at Y_{t_k} and Z_k would destroy this cancellation and would be a different proposition.

For $t_k \leq t < t_{k+1}$, conditioning Itô's formula gives

$$\|\mathbb{E}\{f(Y_t) - f(Y_{t_k})\}\|_d \leq K_\epsilon(t - t_k).$$

Combining this inequality with the grid-point strong error bounds the absolute bias in the weighted mean-production sum by

$$\beta \sum_k B_k u_\epsilon(t_k) + K_\epsilon \sum_k \int_{t_k}^{t_{k+1}} W(t)(t - t_k) dt.$$

The common components of drift, action, and Brownian noise vanish after applying Π . The policy's coordinate deviations from its common schedule at grid dates are at most ϵ . Therefore

$$(\mathbb{E} \|\Pi Y_T\|_d^2)^{1/2} \leq s_0 + (\kappa + \epsilon)T + \sigma_I \sqrt{(1 - 1/d)T} = S_\epsilon.$$

For any coupled Y, Z , Cauchy–Schwarz gives

$$\left| \mathbb{E} \{ \|\Pi Y\|_d^2 - \|\Pi Z\|_d^2 \} \right| \leq (\mathbb{E} \|Y - Z\|_d^2)^{1/2} \{ 2(\mathbb{E} \|\Pi Y\|_d^2)^{1/2} + (\mathbb{E} \|Y - Z\|_d^2)^{1/2} \}.$$

This proves the terminal term in (S.23). Applying the calculation once to the policy and once to the reference proves Proposition S.5 in exact arithmetic.

C.5.2. *Clipped innovations and numerical integration*

Let g be a standard normal variable and $\text{clip}_z(g)$ its projection onto $[-z, z]$. Symmetry gives zero mean for $g - \text{clip}_z(g)$, and

$$\begin{aligned} \mathbb{E} \{ g - \text{clip}_z(g) \}^2 &\leq \mathbb{E} [g^2 \mathbf{1}\{|g| > z\}] = 2\{z\varphi(z) + \Pr(g > z)\} \\ &\leq 2\varphi(z)(z + z^{-1}) =: v_z. \end{aligned}$$

Thus the accumulated internal-noise discrepancy has normalized L^2 norm at most $\sqrt{\sigma_I^2 + \sigma_C^2} \sqrt{T v_z}$. It is added to the same Gronwall recurrence, giving the conservative enlargement $e^{\beta T} \sqrt{\sigma_I^2 + \sigma_C^2} \sqrt{T v_z}$ at every grid point. This bounds a change in the controller's internal update, not a replacement of the economic innovation law.

The scalar weights A_k, B_k are integrated from their exponential expressions. C_k , the reference action integral, and the within-cell production weight are enclosed on a dyadic subdivision with 64 panels per cell. Their midpoint errors are explicitly accumulated. If δ_M bounds the sum of the absolute errors in the integrated reference actions, $e^{\beta T} \delta_M$ enlarges the reference state error. The errors in A_k, B_k, C_k are multiplied by global bounds on consumption, its logarithm, and bounded production,

and are added to the statistic error. These terms are not inferred from a difference between two grids.

C.5.3. Arithmetic contract and forward-error budget

All stored binary64 model coefficients denote exact real inputs to the economic problem. The computation assumes round-to-nearest elementary arithmetic, the usual dot-product forward-error bound, and correctly rounded input conversion. It does not assume correct rounding of a platform exponential, logarithm, or hyperbolic tangent when proving a payoff endpoint. Network library functions produce proposals only; the interval schedule and inward projection enforce feasibility independently of proposal accuracy. The simulation drift and log utility use the polynomial kernels described next. The Python interpreter, numerical libraries, compiler, hardware, and random-number generator are not machine-formally verified.

For $|x| \leq 20$, write $r = x/256$ and approximate e^r by its degree-14 Taylor polynomial in Horner form. Nine squarings approximate e^{2x} . The Taylor remainder is bounded by $e^{|r|}|r|^{15}/15!$. The coefficient representation and Horner operations satisfy the standard γ_{30} forward bound, with $\gamma_j = ju/(1 - ju)$ and $u = 2^{-52}$, a conservative choice relative to binary64 unit roundoff. Propagating relative error through nine squarings bounds the exponential relative error by less than 2^{-35} on this range. The derivative of $1 - 2/(1 + z)$ with respect to $\log z$ is $2z/(1 + z)^2 \leq 1/2$. Including the final arithmetic and saturation outside $[-20, 20]$ therefore bounds absolute tanh error by 2^{-34} . The saturation error itself is below $2e^{-40}$.

For positive consumption, binary exponent extraction writes $x = 2^{e-1}(2m)$, with $m \in [1/2, 1)$. Then

$$\log x = 2 \sum_{j=0}^{\infty} \frac{q^{2j+1}}{2j+1} + (e-1) \log 2, \quad q = \frac{2m-1}{2m+1} \in [0, 1/3].$$

The implementation retains 18 terms and evaluates $\log 2$ with the same series at $q = 1/3$. The geometric remainder and Horner forward error, including the bounded exponents on the primitive consumption interval, are below 2^{-34} . No empirical comparison with the platform logarithm serves as this error bound. The optional

compiled kernels use identical binary64 coefficients, operation order, and disabled fast-math transformations; the array implementation is retained as a fallback.

Clipped innovations yield a deterministic internal state cap. A conservative cap for the implemented ranges is

$$S_{\text{cap}} = 2\{\|y\|_{\infty} + M_{\epsilon}T + (\sigma_I + \sigma_C)10T/\sqrt{h} + 1\}.$$

Using $\gamma_{d+2}S_{\text{cap}} \max_i \sum_j |B_{ij}|$ for the matrix-product error, the code propagates drift, innovation-combination, and state-addition errors through the same discrete Gronwall bound. It verifies the bootstrap condition that this enlargement is less than one. This condition, together with the doubled cap, bounds the entire numerical path. Separately, it accumulates the errors of all statistic sums, log evaluations, and terminal squares. The exact formulas and intermediate constants are returned by `constants`; each final record saves the resulting state cap, forward-roundoff bound, scalar-integration bound, and total bias endpoint. Scalar transcendental evaluations used only to assemble conservative budgets receive additional 10^{-12} bias and 10^{-10} clipping cushions on the declared bounded parameter range; they do not replace the enclosed spectral or integral calculations.

The sampling contract is ideal independent standard-normal innovations rounded to binary64 before the specified clipping, with the forward-error allowance for their conversion. Pseudorandom draws are a reproducible implementation of this contract, not a proof of ideal independence. The resulting certificate is conditional numerical mathematics and conditional statistical inference. It must not be described as a machine-formal guarantee of a particular hardware random stream.

C.5.4. *A global clipping envelope*

Let Z^0 denote the internal reference trajectory. In exact arithmetic, synchronous coupling gives

$$\|Z_k^{\phi} - Z_k^0\|_d \leq D_k, \quad D_k = \epsilon_{\text{eff}}\{e^{\beta t_k} - 1\}/\beta,$$

where ϵ_{eff} also covers schedule movement within one cell, integrated-action error, and both forward-rounding increments. For the retained parameters, $|m'_0| \leq 1$,

so $\epsilon + h$ suffices before these numerical enlargements. Define D_T analogously. Production differences have absolute sum at most $\beta \sum_k B_k D_k$. The consumption deficit is at most

$$\frac{\epsilon_{\text{eff}}^2}{2} \{[m_0(0) - \epsilon_{\text{eff}}]^{-2} + \zeta\} \int_0^T e^{-\rho t} dt.$$

The code verifies the strictly positive denominator. Terminal differences are bounded by $\chi e^{-\rho T} \{D_T^2 + 2D_T \|\Pi Z_T^0\|_d\}$.

The projected reference state satisfies

$$\|\Pi Z_T^0\|_d \leq s_0 + \kappa T + e_{\text{round}} + \frac{\sigma_I \sqrt{T}}{\sqrt{d}} H_0,$$

where H_0 is the Euclidean norm of the projected sum of the standardized clipped idiosyncratic innovations, normalized to unit total variance. As a function of all underlying independent standard Gaussians, H_0 is one-Lipschitz. Independence, symmetry, and the variance reduction under clipping give $\mathbb{E} H_0 \leq \sqrt{d-1}$. Gaussian concentration therefore gives $\Pr\{H_0 > \sqrt{d-1} + v\} \leq e^{-v^2/2}$. Collecting the deterministic terms defines C_ϕ and setting $A_\phi = 2\chi e^{-\rho T} D_T \sigma_I \sqrt{T/d}$ gives the claimed tail envelope. Integrating that envelope beyond $v_* > 0$ gives

$$\mathbb{E}[(|X_{\phi,h}| - b_\phi)_+] \leq A_\phi \int_{v_*}^{\infty} e^{-v^2/2} dv \leq A_\phi e^{-v_*^2/2} / v_*.$$

The construction does not require the learned network to be Lipschitz as a function of shocks; only its feasible action range is used in the deterministic separation bound.

For completeness, the Gaussian concentration fact can be obtained from the Gaussian logarithmic Sobolev inequality. If P_t is the Ornstein–Uhlenbeck semigroup, entropy differentiation and the identity $\nabla P_t = e^{-t} P_t \nabla$ imply $\text{Ent}(e^g) \leq \frac{1}{2} \mathbb{E}[e^g \|\nabla g\|^2]$ by Cauchy–Schwarz and integration of e^{-2t} . For $g = \lambda H_0$, the one-Lipschitz bound gives $\log \mathbb{E} e^{\lambda(H_0 - \mathbb{E} H_0)} \leq \lambda^2/2$ by integrating the entropy inequality in λ . Chernoff optimization yields the stated one-sided tail. Smooth approximation and truncation justify the calculation for the norm and clipping functions.

C.5.5. *Finite families, selection, and interpretation*

Conditioning on the fitted weights and their validation selection makes each final-test statistic a fixed function of one fresh path. Each row is therefore an independent sample across paths, even when its Brownian bank is shared with other policies. Applying [Maurer and Pontil \(2009, Theorem 4\)](#) to both signs and taking a union bound proves the simultaneous statement. The implementation allocates 2,000 one-sided statements, so at most 1,000 two-sided rows may be reported under this allocation. The audit rejects an overrun. It also checks source weights, path counts, raw-array hashes, every recomputed mean and variance, and all displayed table inputs.

A certificate may fail to establish improvement. That is distinct from an exception in the numerical code. Both outcomes are retained. A zero correction, a deliberately saturated poor correction, and trained corrections are compared under the same accounting. Small training loss or successful completion is never substituted for a positive lower improvement endpoint. If a candidate is chosen after inspecting several final rows, the simultaneous guarantee continues to hold only within the predeclared finite family; an undeclared search requires a new allocation and independent data.

APPENDIX D: PROOFS AND COMPUTABLE ACCOUNTS FOR THE BELLMAN BRIDGE

Throughout this section, $\langle u, v \rangle_d = d^{-1} u^\top v$ and $\|u\|_d^2 = \langle u, u \rangle_d$. Conditioning on fitted parameters and selection is understood. The initial distribution has finite support and maximal dispersion s_0 . The same arguments apply under any initial distribution with the stated moment and tail bounds.

D.1. *Finite performance difference and the action bridge*

Bounded drift, bounded controls, clipped innovations, and a finite horizon give finite state moments of every order. For the deterministic reference π_k , the finite composition of smooth transition maps has derivatives bounded by a polynomial in the initial state and innovations. The running state reward is bounded and

smooth, and the terminal reward is quadratic. Dominated differentiation therefore establishes (32) and its first two derivatives. This argument concerns the finite continuation, so no continuous-time PDE regularity or derivative-discretization estimate is needed.

Let $Q_k^\pi(z, b) = \ell_k(z, b) + S_k^\pi\{x_b(z)\}$. The reference recursion is $V_k^\pi(z) = Q_k^\pi(z, \pi_k)$. For a possibly history-dependent candidate, condition on its history at k , then telescope $V_{k+1}^\pi(Z_{k+1}) - V_k^\pi(Z_k)$. The two terminal functions agree and the initial law is common, giving

$$\mathcal{R}_h(a) - \mathcal{R}_h(\pi) = \sum_k \mathbb{E}\{Q_k^\pi(Z_k, a_k) - Q_k^\pi(Z_k, \pi_k)\}. \quad (\text{S.25})$$

Put $e_k = \hat{q}_k - q_k^\pi$ and $v_k = a_k - \pi_k \mathbf{1}$. Since $x_{a_k} - x_{\pi_k} = -hv_k$, the fundamental theorem of calculus implies, pathwise,

$$\begin{aligned} [\hat{Q}_k(Z_k, a_k) - \hat{Q}_k(Z_k, \pi_k)] - [Q_k^\pi(Z_k, a_k) - Q_k^\pi(Z_k, \pi_k)] \\ = -h \int_0^1 \langle e_k(X_{k,u}), v_k \rangle_d du. \end{aligned}$$

Cauchy–Schwarz with respect to h times counting measure on the grid, the candidate probability law, and uniform measure on $[0, 1]$ proves Proposition 7. The sign of the line integral is immaterial to its absolute bound. Independence of future reference innovations from the candidate history is used in the recursion.

D.2. A computable action gap

The normalized action Hessian of the approximate continuation problem is

$$dD_b^2 \hat{Q}_k(z, b) = -A_k \text{diag}(b_i^{-2}) - \frac{\zeta A_k}{d} \mathbf{1} \mathbf{1}^\top + h^2 d D^2 \hat{S}_k\{x_b(z)\} \preceq \Gamma_k I_d.$$

The second-order integral remainder on the convex action box proves (35). Maximizing its scalar quadratic in each coordinate gives the stated endpoint or projection rule, including $\Gamma_k = 0$. The value is nonnegative because $b = a_k$ is allowed.

For the quadratic conclusion, let b_k maximize $\hat{Q}_k$. When $-\Gamma_k \geq \mu_0 h$, strong concavity and the constrained first-order inequality at b_k give

$$\|a_k - b_k\|_d^2 \leq \frac{2\delta_k}{\mu_0 h}, \quad \hat{Q}_k(Z_k, b_k) - \hat{Q}_k(Z_k, \pi_k) \geq \frac{\mu_0 h}{2} \|b_k - \pi_k \mathbf{1}\|_d^2.$$

Use $\|b_k - \pi_k \mathbf{1}\|_d^2 \geq \|a_k - \pi_k \mathbf{1}\|_d^2 / 2 - \|a_k - b_k\|_d^2$ and subtract the actor's gap. Summation yields $\mathcal{M}_h \geq \mu_0 \mathcal{A}_h / 4 - 2\mathcal{D}_h$. Consequently

$$\sqrt{\mathcal{A}_h \mathcal{E}_h} \leq 2\sqrt{(\mathcal{M}_h + 2\mathcal{D}_h) \mathcal{E}_h / \mu_0} \leq \mathcal{M}_h / 2 + \mathcal{D}_h + 2\mathcal{E}_h / \mu_0.$$

Together with (34), this proves (36). If $a_k = b_k$, the stronger relation $\mathcal{M}_h \geq \mu_0 \mathcal{A}_h / 2$ proves the stated improvement. No claim of global concavity of the original stochastic control functional enters this argument.

D.3. Continuation targets on one global grid

The following recursion computes an unbiased normalized derivative target without a matrix-valued state Jacobian. Start at the supplied postdecision mean x , add innovation k , and then simulate the reference states at $j = k + 1, \dots, N$ with the same global grid and the disclosed π_j . With $\gamma_T = 2\chi e^{-\rho T}$, set

$$\begin{aligned} p_N &= -\gamma_T \Pi Z_N, \\ p_j &= B_j Df(Z_j)^\top \mathbf{1} + \{I + hDf(Z_j)\}^\top p_{j+1}, \quad j = N - 1, \dots, k + 1. \end{aligned} \tag{S.26}$$

The output is p_{k+1} ; for $k = N - 1$ it is p_N . There is no derivative of the reference action because the reference depends only on the date. Every running consumption term is likewise constant in the starting state. The chain rule proves (S.26), and the dominated-differentiation argument above shows its expectation is $q_k^\pi(x)$.

The implemented scalar critic has the explicit decomposition, at $s = t_{k+1}$,

$$\hat{S}_k(x) = -\chi e^{-\rho T} \{\|\Pi x\|_d^2 + \sigma_I^2(1 - 1/d)(T - s + h)\} + (T - s)\{F_0(s) + F_1(s, x)/d\}. \tag{S.27}$$

The two networks have distinct parameters. Thus

$$\hat{q}_k(x) = -\gamma_T \Pi x + (T - s) D_x F_1(s, x) :$$

fitting the time level cannot change the spatial costate. Using one instead of the exact clipped-normal second moment in the known state-independent noise term changes neither an action difference nor the costate. The terminal costate is exact. Fitting on a 64-cell grid and applying this scalar predictor on a 2,048-cell grid does not make their continuation problems equal; their difference remains in the assessed $\mathcal{E}_h$.

For two hidden tanh layers, write $F_1(s, x) = w_3^\top \tanh\{W_2 \tanh(W_1(s, x) + b_1) + b_2\} + b_3$. Let W_{1x} contain the state columns of W_1 . Any verified spectral upper bounds $l_1 \geq \|W_{1x}\|_2$, $l_2 \geq \|W_2\|_2$, $l_3 \geq \|w_3\|_2$ give

$$\begin{aligned} \|\hat{q}_k(x) + \gamma_T \Pi x\|_d &\leq C_{\theta,k} := (T - s)l_1 l_2 l_3 / \sqrt{d}, \\ dD^2 \hat{S}_k(x) &\preceq \Lambda_k I_d, \quad \Lambda_k = (T - s)l_3 l_1^2 (l_2^2 + l_2). \end{aligned} \quad (\text{S.28})$$

The Hessian bound uses $|\tanh'| \leq 1$ and $|\tanh''| \leq 1$. The negative quadratic lift may be omitted in this upper bound; an absolute normalized Hessian bound is $\gamma_T + \Lambda_k$. These are computable bounds from the saved parameters, not fitted derivative maxima.

D.4. Antithetic cancellation and nonasymptotic ranges

We give explicit ranges for the statistics used in (37). All displayed constants are upper endpoints when implemented. The raw costate comparison uses precisely one bank with r independent antithetic pairs. The executed choice is $r = 2$, or four paths per bank. Set

$$\beta \geq \kappa \|B\|_2, \quad n_k = N - k - 1, \quad F_k = (1 + h\beta)^{n_k}, \quad \tau_k = (N - k)h,$$

$$P_k = \beta \sum_{j=k+1}^{N-1} B_j (1 + h\beta)^{j-k-1}.$$

A larger enclosed bound $e^{\beta n_k h}$ may replace F_k . A smaller proved bound on $\|Df(z)^\top \mathbf{1}\|_d$ may replace the leading β in P_k . Suppose $\|\Pi(a_k - m_0(t_k)\mathbf{1})\|_d \leq \epsilon$. Define $s_d = \sqrt{d-1}$ and

$$C_k = C_{\theta,k} + P_k + \gamma_T F_k \kappa n_k h$$

$$\begin{aligned}
& + \gamma_T(F_k - 1) \left\{ s_0 + (\kappa + \epsilon)T + \frac{\sigma_I}{\sqrt{d}}(\sqrt{t_k} + \sqrt{\tau_k})s_d \right\}, \\
a_k^{\text{tail}} &= \gamma_T(F_k - 1) \frac{\sigma_I}{\sqrt{d}}(\sqrt{t_k} + \sqrt{\tau_k/r}), \quad C_E = \max_k C_k, \quad A_E = \max_k a_k^{\text{tail}}. \quad (\text{S.29})
\end{aligned}$$

Then the ideal statistic E satisfies

$$\Pr\{|E| > T(C_E + A_E v)^2\} \leq 3e^{-v^2/2}, \quad v \geq 0. \quad (\text{S.30})$$

For the executed choice $v_* = 8$, a clipping threshold and an absolute expectation allowance are

$$b_E = T(C_E + A_E v_*)^2, \quad \tau_E = 6TA_E e^{-v_*^2/2}(C_E/v_* + A_E). \quad (\text{S.31})$$

The same proof bounds E^{raw} by $2T(C_Z + A_Z v)^2$, where C_Z, A_Z omit $C_{\theta,k}$ from (S.29). Its expectation allowance is twice the corresponding value of (S.31). For $E - E^{\text{raw}}$, one can add these two ranges and allowances; equivalently the direct bound $T(C_E + A_E v)^2 + 2T(C_Z + A_Z v)^2$ uses the same three-event union. Numerical errors are enclosed separately.

To prove these claims, let J be the derivative of a reference terminal state with respect to x . Then $\|J\|_2 \leq F_k$ and $\|J - I\|_2 \leq F_k - 1$. Writing the terminal state as its initial value, the accumulated production, common deterministic drift, and total innovation gives, after subtracting $q_0(x) = -\gamma_T \Pi x$, the production-adjoint term plus

$$-\gamma_T \{(J^\top - I)\Pi x + J^\top \sum_{j>k} h \Pi f(Z_j) + (J^\top - I)\Pi \xi + \Pi \xi\}.$$

The average of the last term is exactly zero within an antithetic bank: the clipping map is odd. The production norm is at most $\kappa n_k h$. The first term is bounded by P_k . Common noise disappears from $\Pi \xi$, although it still affects the states and their Jacobians; the bounds on those terms are global.

For the candidate prefix, let $H_0 = \|\Pi \sum_{j<k} \text{clip}_z(G_{j,I})\|_2 / \sqrt{k}$ when $k > 0$, with zero contribution at $k = 0$. The map defining H_0 is 1-Lipschitz on its standard Gaussian input and $\mathbb{E}H_0 \leq s_d$. The average of the analogous future norms from r independent antithetic base paths has mean at most s_d and Lipschitz constant $r^{-1/2}$. Gaussian concentration therefore bounds the failure of $H_0 \leq s_d + v$ and of the two

future bounds $H_l \leq s_d + v/\sqrt{r}$ by $3e^{-v^2/2}$ in total. On their intersection,

$$\|\Pi X_{k,U}\|_d \leq s_0 + (\kappa + \epsilon)T + \sigma_I \sqrt{t_k/d}(s_d + v),$$

and each bank error has norm at most $C_E + A_E v$. Cauchy–Schwarz proves (S.30). Integrating its tail, using $\int_{v_*}^{\infty} e^{-v^2/2} dv \leq e^{-v_*^2/2}/v_*$, proves (S.31). Subtracting two bank residuals proves the raw and paired-error bounds.

For completeness, the predicted improvement has a substantially smaller range than the sum of two global value bounds. Let $\Delta \geq \sup_k \|a_k - \pi_k \mathbf{1}\|_d$ and let l_k, u_k enclose the candidate, reference, and line segments between them. Put

$$L_\ell = \max_k \max_{m \in \{l_k, u_k\}} |(A_k/h)(m^{-1} - \zeta m) - B_k/h|, \quad C_\theta = \max_k C_{\theta,k},$$

and

$$C_M = T\Delta \left[L_\ell + C_\theta + \gamma_T \{s_0 + (\kappa + \epsilon)T + \sigma_I \sqrt{T/d} s_d\} \right],$$

$$A_M = T\Delta \gamma_T \sigma_I \sqrt{T/d}.$$

The mean value theorem and the prefix event above imply $\Pr\{|M| > C_M + A_M v\} \leq e^{-v^2/2}$. Thus $b_M = C_M + A_M v_*$ and $\tau_M = A_M e^{-v_*^2/2}/v_*$ are valid. Also $0 \leq A \leq T\Delta^2$. If w_A bounds the normalized diameter of the action boxes and $\Gamma_+ = \max_k \max(0, \Gamma_k/h)$, then

$$D \leq T w_A \{L_\ell + C_\theta + \gamma_T \|\Pi x_{a_K}(Z_K)\|_d\} + T \Gamma_+ w_A^2/2.$$

This gives b_D, τ_D from the same affine Gaussian-tail calculation. All ranges remain valid under mixtures over the disclosed initial profiles and, using the maximum constants, over fitted streams.

D.5. Statistics, arithmetic, and payoff transfer

Conditional independence of the two continuation banks gives

$$\mathbb{E}[E \mid K, Z_K, U] = T \|\hat{q}_K(X_{K,U}) - q_K^\pi(X_{K,U})\|_d^2,$$

and the usual two-replicate identity gives

$$\mathbb{E}[E^{\text{raw}} \mid K, Z_K, U] = T\mathbb{E}[\|Z_1 - q_K^\pi(X_{K,U})\|_d^2 \mid K, Z_K, U].$$

Uniform sampling of K proves all the mean identities in (37). Equal independent samples from each declared stream identify the finite uniform mean. Independence across streams and the nonidentical-variable version of the empirical Bernstein inequality, Maurer and Pontil (2009, Theorem 11), justify pooling. Dependence among M, A, E, D on one path does not invalidate a union bound across their confidence statements. Clipping changes each expectation by at most its proved tail allowance.

The finite-grid identities above concern exact real transition and reward maps with the disclosed coefficients. A protected numerical audit must additionally enclose its scalar weights, forward-state error, neural values and derivatives, adjoint recurrence, and statistic arithmetic. The controller's rounded actions are held fixed when its rounded state is compared with the exact internal state, so this state account does not require a derivative of the actor. For example, if the action bridge has normalized representation error at most e_x and the normalized critic and true continuation costates have Lipschitz bounds $L_{\hat{q}}, L_q$ on the enclosing domain, Minkowski's inequality gives

$$\sqrt{\mathcal{E}_h^{\text{exact}}} \leq \sqrt{\mathcal{E}_h^{\text{represented}}} + \sqrt{T}(L_{\hat{q}} + L_q)e_x.$$

A deterministic clipped-noise state enclosure may be used to multiply these small arithmetic errors. It is not used as the statistical range in place of (S.29). The audit saves the resulting cushions and evaluates the endpoint with outward rounding. An ordinary platform derivative without such an account remains a finite-grid diagnostic, rather than a protected confidence endpoint.

The executed account makes these numerical terms explicit. Let $C_{\max}$ enclose every coordinate of a candidate prefix followed by a reference continuation, including its numerical representatives. A sufficient choice is

$$C_{\max} = 4 \left\{ \max_y \|y\|_\infty + M_\epsilon T + (\sigma_I + \sigma_C)zT/\sqrt{h} + 1 \right\}.$$

The maximum is over the disclosed initial profiles. This deterministic bound is used for numerical error only. Put $u = 2^{-52}$, $\gamma_n = nu/(1 - nu)$, and $\varepsilon_{\tanh} = 2^{-34}$. The inherited polynomial kernels have uniform tanh and consumption-log error at most $\varepsilon_{\tanh}$. The row dot-product error is bounded by $\gamma_{d+2}C_{\max} \max_i \sum_j |B_{ij}|$. Multiplication by κ , addition of the drift constants and action, and the protected state update give the saved one-step bound ξ_h . The source evaluates all its positive terms outward, with γ_{32} for the scalar update and an additional $16u(\sigma_I + \sigma_C)z\sqrt{h}$ for represented innovations. Thus

$$e_Z = Ne^{\beta T} \xi_h, \quad e_x = (1 + h\beta)e_Z + 2\xi_h + \gamma_{16}(C_{\max} + hM_\epsilon)$$

are sufficient prefix and bridge representation bounds. The last term also covers the line interpolation and a coupling of the binary uniform representative to a continuous uniform variable within 2^{-53} .

Let $b_r = \max_i \|B_{i\cdot}\|_2$ and $b_1 = \max_i \sum_j |B_{ij}|$. A sufficient operator error for the drift derivative used in the adjoint is

$$\varepsilon_{Df} = \beta \left\{ b_r \sqrt{d} e_Z + \gamma_{d+2} C_{\max} b_1 + 2\varepsilon_{\tanh} + \gamma_5 (1 + \varepsilon_{\tanh})^2 \right\}. \quad (\text{S.32})$$

The first term uses $|\tanh''| \leq 1$ to propagate the state error; the last two enclose the computed $1 - \tanh^2$. If $\tilde{p}_{j+1}$ is a computed adjoint and e_{j+1} encloses its normalized error, a valid next error is

$$e_j = (1 + h\beta)e_{j+1} + (h\|\tilde{p}_{j+1}\|_d + B_j^+) \varepsilon_{Df} + \beta r_{B_j} + \omega_j, \quad (\text{S.33})$$

$$\omega_j = \gamma_{d+24} [\|\tilde{p}_{j+1}\|_d + \kappa \|B\|_2 \{h\|\tilde{p}_{j+1}\|_d + |\bar{B}_j|\} (1 + 4\varepsilon_{\tanh})] + 10^{-24}.$$

Here B_j^+ and r_{B_j} are its scalar upper endpoint and midpoint radius. The terminal error combines an interval evaluation of $-\gamma_T \Pi \tilde{Z}_N$ with $\gamma_T e_Z$. The coefficient γ_{d+24} covers the products, row dot product, and addition in $\tilde{p} + \{(h\tilde{p} + \bar{B}_j)\kappa(1 - \widetilde{\tanh^2})\}B$. The relative dot-product bound uses $\|B\|_2$, bounded outward by its Frobenius norm. The small absolute term dominates gradual-underflow errors in this fixed $d \leq 50$ update: the primitive coefficients and step are bounded, $|B_{ij}| \leq 1$, and even a sum of all its elementary underflow errors is less than 10^{-24} . The positive recurrence (S.33) itself is evaluated by interval operations, so no additional empirical error or

terminal correction is inferred from a comparison of grids. Averaging the four path errors and enclosing the bank's summation gives its recorded error.

An explicit costate Lipschitz bound, used only to multiply e_x , is obtained by putting $H_f = \kappa \|B\|_2^2$ and $\mathcal{B} = \sum_j B_j$. The Euclidean bilinear second derivative of the reference state flow is at most $H_f T e^{2\beta T}$, while the Hessian of $\bar{f}$ has norm at most H_f/d . The chain rule therefore gives

$$L_q = H_f e^{2\beta T} \mathcal{B} (1 + \beta \sqrt{d} T) + \gamma_T e^{2\beta T} (1 + \sqrt{d} C_{\max} H_f T). \quad (\text{S.34})$$

For the neural costate one may use $L_{\hat{q}} = \gamma_T + T l_3 l_1^2 (l_2^2 + l_2)$. The error in the predicted advantage caused by the prefix replacement is at most $T \Delta L_{\hat{q}} (1 + h\beta) e_Z$. This uses the difference of the two continuation gradients before taking an absolute bound. The corresponding actor-gap error is at most $T w_{\mathcal{A}} L_{\hat{q}} (1 + h\beta) e_Z$. The squared-error account uses the previously displayed Minkowski bound with $\sqrt{T} (L_q + L_{\hat{q}}) e_x$. The source saves all these constants and the verified positive-curvature indicator.

Finally, numerical enclosures of sample values can be used more accurately than a uniform arithmetic allowance. Suppose the exact clipped sample lies in $[L_i, U_i]$, let m_i be the disclosed binary midpoint, and let r_i enclose $\max(m_i - L_i, U_i - m_i)$. Projection onto the subspace orthogonal to the constant vector is nonexpansive. Consequently the exact sample mean differs from $\bar{m}$ by at most $\bar{r}$, and its sample standard deviation is at most

$$s_m + \sqrt{\frac{n}{n-1}} \left(\frac{1}{n} \sum_i r_i^2 \right)^{1/2}.$$

With $\ell = \log(4/\alpha_{\text{event}})$, it suffices to add

$$\bar{r} + \sqrt{\frac{2\ell}{n-1}} \left(\frac{1}{n} \sum_i r_i^2 \right)^{1/2}$$

to each side of the midpoint empirical Bernstein interval. This construction encloses the unknown exact sample moments. Its data-dependent numerical radii do not replace the independently proved population clipping ranges. For the costate product, for example, bank and critic norm errors η_1, η_2 give the sample radius $T\{\eta_1 \|\tilde{e}_2\|_d + \eta_2 \|\tilde{e}_1\|_d + \eta_1 \eta_2\}$, in addition to the interval dot product. All mean

and variance corrections are evaluated outward. The raw costate comparison uses the identical represented queries; the welfare endpoint additionally includes their exact-state representation account.

Finally, subtraction of the reference transformed reward gives exactly

$$\mathbb{E}X_{a,h} = \mathcal{R}_h(a) - \mathcal{R}_h(\pi) - D_{\text{hold}}$$

at the level of expectations, where $X_{a,h}$ is the paired statistic in Proposition S.5. The reference internal drift uses precisely $h\pi_k = \bar{M}_k$; the difference from the continuous schedule is already included in that proposition's scalar quadrature and reference-state account. Pointwise optimality of $m_0(t)$ in the consumption-only part implies $D_{\text{hold}} \geq 0$. Applying Proposition 7, the simultaneous endpoints, and the paired payoff-transfer allowance proves Theorem 9. Applying Proposition 8 on the same event proves its optional quadratic bound. The nonnegative parts of upper risk and action endpoints are valid because their true expectations are nonnegative. The maximum of two valid lower bounds is again valid on that event.

APPENDIX E: A PAIRED TRANSFER TO CONTINUOUS-TIME WELFARE

The direct comparison of two executed policies permits a sharper transfer than the sum of their separate discretization bounds. This section establishes that transfer for the original capital diffusion and the original held-action implementation. It changes neither the simulated observations nor the confidence event. The construction uses only the economic primitives, the action tube, and the assessment grid.

E.1. *The compared policies and the exact finite statistic*

Condition on all training and selection information. Let a_k, b_k be the two implemented actions on $[t_k, t_{k+1})$, where $t_k = kh$ and $Nh = T$. They may depend on the entire innovation history through t_k , including the rounded internal states used by their implementations. Both satisfy

$$\|a_k - m_0(t_k)\mathbf{1}\|_\infty \leq \varepsilon, \quad \|b_k - m_0(t_k)\mathbf{1}\|_\infty \leq \varepsilon. \quad (\text{S.35})$$

The inward guard in the executed verifier enforces these inequalities. As in the capital implementation, this tube is contained in the primitive positive consumption box. The two policies receive the same initial profile and Brownian innovations. Their physical states satisfy

$$dY_t^v = \{c\mathbf{1} + f(Y_t^v) - v_k\} dt + \Sigma dW_t, \quad f(y) = \kappa \tanh(By), \quad v \in \{a, b\}.$$

Here $\Sigma = (\sigma_I I_d, \sigma_C \mathbf{1})$ and $c = \text{productivity} - (\sigma_I^2 + \sigma_C^2)/2$. The actions in these equations are the actions actually returned by the original implementation. In particular, they are not recomputed at a different state or on a different grid.

For the proof, introduce an auxiliary Euler state with exact arithmetic and unclipped Brownian increments,

$$Z_{k+1}^v = Z_k^v + h\{c\mathbf{1} + f(Z_k^v) - v_k\} + \Sigma(W_{t_{k+1}} - W_{t_k}), \quad Z_0^v = Y_0^v. \quad (\text{S.36})$$

This is an analytical coupling. It is not a second executed policy. Its actions remain those generated by the original clipped, rounded controller. Consequently, no Lipschitz or differentiability assumption on the actor is needed below.

Use $\langle x, y \rangle_d = d^{-1} x^\top y$, $\|x\|_d^2 = \langle x, x \rangle_d$, and $\Pi = I_d - d^{-1} \mathbf{1} \mathbf{1}^\top$. Put

$$\psi(y) = \overline{f(y)}, \quad w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)}, \quad \omega(t) = e^{-\rho t} w(t),$$

and define the exact scalar weights

$$A_k = \int_{t_k}^{t_{k+1}} e^{-\rho t} dt, \quad B_k = \int_{t_k}^{t_{k+1}} \omega(t) dt, \quad J_k = \int_{t_k}^{t_{k+1}} \omega(t)(t - t_k) dt. \quad (\text{S.37})$$

Let $X_h^{a,b}$ be the following exact-real random variable:

$$\begin{aligned} X_h^{a,b} = & \sum_{k=0}^{N-1} \left[B_k \{ \psi(Z_k^a) - \psi(Z_k^b) \} + A_k \{ \overline{\log a_k} - \overline{\log b_k} \} - B_k (\bar{a}_k - \bar{b}_k) \right] \\ & - \frac{\zeta}{2} \sum_{k=0}^{N-1} A_k (\bar{a}_k^2 - \bar{b}_k^2) - \chi e^{-\rho T} \{ \|\Pi Z_N^a\|_d^2 - \|\Pi Z_N^b\|_d^2 \}. \end{aligned} \quad (\text{S.38})$$

Integration by parts applied to $e^{-\rho t} w(t) \overline{Y_t^v}$ shows that $J(a) - J(b)$ equals the expectation of (S.38), with the production sum replaced by $\int_0^T \omega(t) \{ \psi(Y_t^a) - \psi(Y_t^b) \} dt$

and the terminal states replaced by their physical counterparts. The consumption terms in (S.38) are already integrated exactly because the implemented actions are held on each cell. The common initial and exogenous-drift terms cancel.

E.2. Computable global coefficients

In this section $\|\cdot\|_2$ for a matrix is its Euclidean operator norm. Choose a verified bound $b_* \geq \|B\|_2$ and set

$$\beta = \kappa b_*, \quad L_2 = \frac{4}{3\sqrt{3}}, \quad q_i = \sigma_I^2 \|B_i\|_2^2 + \sigma_C^2 (B_i \mathbf{1})^2, \quad r_B = \max_i \|B_i\|_2.$$

B_i denotes row i of B ; the q_i in this section are covariance coefficients, not costates.

Let

$$C_0 = \max_{0 \leq k < N} |c - m_0(t_k)|, \quad M_\varepsilon = C_0 + \kappa + \varepsilon, \quad R_i = C_0 |B_i \mathbf{1}| + (\kappa + \varepsilon) \sum_j |B_{ij}|.$$

Any upper bounds for these quantities may be substituted. In the declared economy, m_0 is increasing and $c < m_0(0)$, so the calculation bounds C_0 by $m_0(T) - c$ using interval endpoints.

The scalar production map admits the Lipschitz bound

$$|\psi(x) - \psi(y)| \leq \beta_\psi \|x - y\|_d, \quad \beta_\psi = \min \left\{ \beta, \kappa \left[\frac{1}{d} \sum_{i,j} \max\{(BB^\top)_{ij}, 0\} \right]^{1/2} \right\}. \quad (\text{S.39})$$

Indeed, $dD\psi = \kappa B^\top u$ with $0 \leq u_i \leq 1$. Bounding $u^\top BB^\top u$ either by $b_*^2 d$ or by the sum of the positive entries proves (S.39). The implementation replaces every displayed quantity by an outward upper bound before taking the minimum.

Define

$$\begin{aligned} H_m &= \kappa L_2 b_*^2, & H_x &= \kappa L_2 b_* r_B \sqrt{d}, \\ G &= \kappa \left(\frac{1}{d} \sum_i q_i \right)^{1/2}, & H_\Sigma &= \kappa L_2 \|\text{diag}(\sqrt{q}) B\|_2, \\ K &= \kappa \{C_0 \|B \mathbf{1}\|_d + b_*(\kappa + \varepsilon)\} + \frac{\kappa L_2}{2} \|q\|_d, \end{aligned}$$

$$\begin{aligned}
L_v &= \kappa L_2 \|\text{diag}(R)B\|_2 + \beta^2 + \kappa \|\text{diag}(q)B\|_2, \\
L_m &= H_m M_\varepsilon + \beta_\psi \beta + \kappa \min\{b_* \|q\|_d, \|\text{diag}(q)B\|_2\}.
\end{aligned} \tag{S.40}$$

All of these are functions of disclosed primitives and the action tube. They do not depend on fitted weights, simulated outcomes, or an observed maximum of a derivative.

To record the deterministic error recursion, let $r = 2\varepsilon$ and put

$$\begin{aligned}
D(t) &= r \frac{e^{\beta t} - 1}{\beta}, & I_1(t) &= \int_0^t D(s) ds, & I_2(t) &= \int_0^t D(s)^2 ds, \\
\eta(t) &= h e^{\beta t} \left\{ \frac{Kt}{2} + G \sqrt{\frac{t}{3}} \right\}.
\end{aligned} \tag{S.41}$$

The continuous extension $D(t) = rt$ applies if $\beta = 0$. Set $\xi_0 = 0$ and, successively for $k = 1, \dots, N$, calculate

$$\begin{aligned}
F_k &= \frac{h}{2} \{L_v I_1(t_k) + \beta r t_k\} + h H_\Sigma \sqrt{I_2(t_k)/3} + h H_x \sum_{j < k} D(t_j) \eta(t_j), \\
\xi_k &= F_k + h \beta \sum_{j < k} \xi_j.
\end{aligned} \tag{S.42}$$

This positive recursion retains the forcing accumulated through the current date. It does not propagate a martingale through a future random Jacobian.

The initial distribution assigns probabilities p_ℓ to deterministic profiles y_ℓ . Define

$$S_\ell = \|\Pi y_\ell\|_d + (\kappa + \varepsilon)T + \sigma_I \sqrt{(1 - 1/d)T}, \quad \bar{S} = \sum_\ell p_\ell S_\ell. \tag{S.43}$$

The mean in this expression is an average of conditional bounds. It is not asserted to be the L^2 norm of the state under the mixture of profiles.

PROPOSITION S.6—Paired payoff transfer: *For the two held policies in (S.35), define*

$$\begin{aligned}
\mathfrak{B}_h^{\text{ideal}} &= \sum_{k=0}^{N-1} B_k \{ \beta_\psi \xi_k + H_m D(t_k) \eta(t_k) \} \\
&\quad + \sum_{k=0}^{N-1} J_k \left\{ \frac{L_m}{h} [I_1(t_{k+1}) - I_1(t_k)] + \beta_\psi r \right\}
\end{aligned}$$

$$+ \chi e^{-\rho T} \{ (2\bar{S} + 2D(T))\xi_N + 2D(T)\eta(T) \}. \quad (\text{S.44})$$

Then

$$\left| J(a) - J(b) - \mathbb{E} X_h^{a,b} \right| \leq \mathfrak{B}_h^{\text{ideal}}. \quad (\text{S.45})$$

Suppose δ_{num} bounds, in conditional L^2 at every node, the distance between each auxiliary Euler state and its original rounded, clipped internal state. Suppose ϵ_{stat} bounds the complete arithmetic error of each stored policy-minus-reference statistic, and ϵ_{sub} bounds their final subtraction. Then the original stored direct difference $\hat{X}_h^{a,b}$ satisfies

$$\left| J(a) - J(b) - \mathbb{E} \hat{X}_h^{a,b} \right| \leq \mathfrak{B}_h^{\text{pair}}, \quad (\text{S.46})$$

where

$$\begin{aligned} \mathfrak{B}_h^{\text{pair}} &= \mathfrak{B}_h^{\text{ideal}} + 2\beta_\psi \delta_{\text{num}} \sum_k B_k \\ &\quad + 2\chi e^{-\rho T} \delta_{\text{num}} (2\bar{S} + \delta_{\text{num}}) + 2\epsilon_{\text{stat}} + \epsilon_{\text{sub}}. \end{aligned} \quad (\text{S.47})$$

The conclusion is uniform over both policies satisfying the displayed tube.

E.3. Proof of Proposition S.6

Regularity and integrability. The drift is globally Lipschitz in the state, with constant β , and bounded uniformly over admissible actions. Thus the physical equations have unique strong solutions with finite moments on the finite horizon. The actions are bounded and measurable at the left endpoint of their cell. The first three derivatives of f are bounded. All stochastic integrals used below are consequently square integrable, so their expectations vanish and Itô's isometry applies without an unaccounted localization limit. The quadratic terminal payoff and every product below are integrable.

Derivative and generator bounds. The elementary identities for $\tanh$ give $|\tanh'| \leq 1$, $|\tanh''| \leq L_2$, and $|\tanh'''| \leq 2$. For a held action v , write

$$\mathcal{L}^v u(y) = Du(y) \{ c\mathbf{1} + f(y) - v \} + \frac{1}{2} \text{tr} \{ \Sigma \Sigma^\top D^2 u(y) \}.$$

For vector-valued u , this definition applies to each component. Since $v = m_0(t_k)\mathbf{1} + e$ with $\|e\|_\infty \leq \varepsilon$,

$$\|\mathcal{L}^v f(y)\|_d \leq K, \quad d^{-1/2} \|Df(y)\Sigma\|_{\text{HS}} \leq G.$$

To see the first inequality, bound $\|Df(y)(c\mathbf{1} + f(y) - v)\|_d$ by $\kappa\{C_0\|B\mathbf{1}\|_d + b_*(\kappa + \varepsilon)\}$. The i th second-order term is $\kappa q_i \tanh''(B_i y)/2$, whose normalized norm is at most $\kappa L_2 \|q\|_d/2$.

For two states x, y and two actions v, w in the same tube, splitting the drift part into the change in Df and the change in drift gives

$$\begin{aligned} \|\mathcal{L}^v f(x) - \mathcal{L}^w f(y)\|_d &\leq L_v \|x - y\|_d + \beta \|v - w\|_d, \\ |\mathcal{L}^v \psi(x) - \mathcal{L}^w \psi(y)| &\leq L_m \|x - y\|_d + \beta_\psi \|v - w\|_d. \end{aligned} \quad (\text{S.48})$$

In the first line, the change in Df is bounded by $\kappa L_2 \|\text{diag}(R)B\|_2 \|x - y\|_d$, because $|B_i(c\mathbf{1} + f(x) - v)| \leq R_i$. The remaining first-order terms are bounded by $\beta^2 \|x - y\|_d + \beta \|v - w\|_d$. The change in the second-order term is bounded by $\kappa \|\text{diag}(q)B\|_2 \|x - y\|_d$.

For the second line, the scalar Hessian obeys $\|D^2\psi\|_2 \leq H_m/d$. Its contribution to the drift difference is at most $H_m M_\varepsilon \|x - y\|_d$; the remaining drift terms contribute $\beta_\psi \beta \|x - y\|_d + \beta_\psi \|v - w\|_d$. The change in its second-order term is bounded by either $\kappa b_* \|q\|_d \|x - y\|_d$ or $\kappa \|\text{diag}(q)B\|_2 \|x - y\|_d$. This proves the coefficients in (S.40).

The normalized Hilbert–Schmidt bound for the paired diffusion derivative is

$$\begin{aligned} \frac{1}{d} \|\{Df(x) - Df(y)\}\Sigma\|_{\text{HS}}^2 &= \frac{\kappa^2}{d} \sum_i q_i \{\tanh'(B_i x) - \tanh'(B_i y)\}^2 \\ &\leq \frac{\kappa^2 L_2^2}{d} \|\text{diag}(\sqrt{q})B(x - y)\|_2^2 \leq H_\Sigma^2 \|x - y\|_d^2. \end{aligned} \quad (\text{S.49})$$

This calculation accounts for the dimension normalization explicitly. The scalar production Hessian likewise has no extra $\sqrt{d}$ factor. For the vector four-point bound, an operator-norm estimate does require

$$\|Df(x) - Df(y)\|_2 \leq \kappa L_2 b_* \max_i |B_i(x - y)| \leq H_x \|x - y\|_d.$$

The individual and paired state errors. Fix an initial profile. Let $\Delta Y_t = Y_t^a - Y_t^b$, $\Delta Z_k = Z_k^a - Z_k^b$, and $e_k^v = Y_{t_k}^v - Z_k^v$. The common Brownian increments cancel from each state difference. Since $\|a_k - b_k\|_d \leq r$, pathwise Gronwall gives

$$\|\Delta Y_t\|_d \leq D(t), \quad \|\Delta Z_k\|_d \leq r \frac{(1 + h\beta)^k - 1}{\beta} \leq D(t_k). \quad (\text{S.50})$$

For a single policy the accumulated local drift error through t_k is

$$R_k^v = \sum_{j < k} \int_{t_j}^{t_{j+1}} \{f(Y_s^v) - f(Y_{t_j}^v)\} ds.$$

Apply Itô's formula to $f(Y_s^v)$ within each cell and interchange the integrals. If $\theta_h(s) = t_{j+1} - s$ for $s \in [t_j, t_{j+1})$, Minkowski's inequality and Itô's isometry yield

$$\|R_k^v\|_{L^2(d)} \leq K \int_0^{t_k} \theta_h(s) ds + G \left\{ \int_0^{t_k} \theta_h(s)^2 ds \right\}^{1/2} = h \{Kt_k/2 + G\sqrt{t_k/3}\}.$$

Here $\|U\|_{L^2(d)} = (\mathbb{E}\|U\|_d^2)^{1/2}$, with expectation conditional on the fixed initial profile. The nodal error recursion and discrete Gronwall therefore give

$$\|e_k^v\|_{L^2(d)} \leq \eta(t_k). \quad (\text{S.51})$$

Apply the same representation to $R_k^a - R_k^b$. Equations (S.48) and (S.49) imply

$$\begin{aligned} \|R_k^a - R_k^b\|_{L^2(d)} &\leq \int_0^{t_k} \theta_h(s) \{L_v D(s) + \beta r\} ds \\ &\quad + H_\Sigma \left\{ \int_0^{t_k} \theta_h(s)^2 D(s)^2 ds \right\}^{1/2} \\ &\leq \frac{h}{2} \{L_v I_1(t_k) + \beta r t_k\} + h H_\Sigma \sqrt{I_2(t_k)/3}. \end{aligned}$$

For the last inequality, on each cell θ_h and θ_h^2 decrease while D and D^2 increase. The integral of a decreasing function times an increasing function is no greater than the product of their cell averages times the cell length. Their kernel averages are $h/2$ and $h^2/3$, respectively. The martingale isometry is applied to the single accumulated stochastic integral with predictable integrand. There is no isometry assertion for a future random flow multiplying a past noise increment.

For the difference of the nodal drift errors, the four-point identity and the derivative bound give

$$\begin{aligned} & \|f(Y_{t_j}^a) - f(Y_{t_j}^b) - f(Z_j^a) + f(Z_j^b)\|_d \\ & \leq \beta \|e_j^a - e_j^b\|_d + H_x \|\Delta Z_j\|_d \|e_j^b\|_d. \end{aligned}$$

One obtains this inequality by first changing ΔY_{t_j} to ΔZ_j at the base point $Y_{t_j}^b$, and then changing that base point to Z_j^b with ΔZ_j fixed. Combining it with (S.50) and (S.51) proves inductively that

$$\|e_k^a - e_k^b\|_{L^2(d)} \leq \xi_k. \quad (\text{S.52})$$

All bounds are uniform in the initial profile.

Production at and between decision dates. The corresponding scalar four-point inequality uses H_m in place of H_x and β_ψ in place of β . Thus

$$\mathbb{E}|\psi(Y_{t_k}^a) - \psi(Y_{t_k}^b) - \psi(Z_k^a) + \psi(Z_k^b)| \leq \beta_\psi \xi_k + H_m D(t_k) \eta(t_k).$$

Multiplication by B_k accounts for the first line of (S.44).

For variation inside a cell, take expectations in Itô's formula for $\psi(Y_t^a) - \psi(Y_t^b)$. The martingale has mean zero by the integrability already established. Put $\Psi_k(s) = \int_s^{t_{k+1}} \omega(t) dt$. The expected weighted within-cell error is bounded in absolute value by

$$\int_{t_k}^{t_{k+1}} \Psi_k(s) \{L_m D(s) + \beta_\psi r\} ds \leq J_k \left\{ \frac{L_m}{h} [I_1(t_{k+1}) - I_1(t_k)] + \beta_\psi r \right\}.$$

Here Ψ_k decreases, D increases, and $\int_{t_k}^{t_{k+1}} \Psi_k(s) ds = J_k$. This proves the second line of (S.44).

Terminal dispersion and the initial population. For a fixed profile y_ℓ , projecting the physical state equation removes the scalar schedule, the exogenous scalar drift, and the common shock. Minkowski's inequality yields $\|\Pi Y_T^b\|_{L^2(d)} \leq S_\ell$. The same bound holds for the auxiliary Euler state ΠZ_N^b and for either policy separately. Use

the exact terminal identity

$$\begin{aligned} & \|\Pi Y_T^a\|_d^2 - \|\Pi Y_T^b\|_d^2 - \|\Pi Z_N^a\|_d^2 + \|\Pi Z_N^b\|_d^2 \\ &= 2\langle \Pi Y_T^b, \Pi(e_N^a - e_N^b) \rangle_d + 2\langle \Pi e_N^b, \Pi \Delta Z_N \rangle_d \\ &+ \langle \Pi(\Delta Y_T + \Delta Z_N), \Pi(e_N^a - e_N^b) \rangle_d. \end{aligned}$$

Conditional Cauchy–Schwarz and the pathwise bounds for both differences give the upper bound $(2S_\ell + 2D(T))\xi_N + 2D(T)\eta(T)$ on the absolute expectation. Average this bound over the profiles. Because D, η, ξ are uniform in the initial state, the result contains $\bar{S} = \sum p_\ell S_\ell$. In particular, the calculation uses the mean initial spread after applying Cauchy–Schwarz separately for each profile; it does not replace a mixture second moment by a mean of square roots. Multiplication by $\chi e^{-\rho T}$ proves the remaining line of (S.44), and hence (S.45).

The stored statistic. Let $\tilde{Z}_k^v$ denote the actual rounded, clipped internal state. The inherited forward-arithmetic and normal-clipping bounds, with the actions held fixed as above, provide a uniform value $\delta_{\text{num}} \geq \|Z_k^v - \tilde{Z}_k^v\|_{L^2(d)}$ conditional on each profile. Scalar production then contributes at most $2\beta_\psi \delta_{\text{num}} \sum B_k$ for the two policies. For each policy,

$$\mathbb{E} \left| \|\Pi Z_N^v\|_d^2 - \|\Pi \tilde{Z}_N^v\|_d^2 \right| \leq \delta_{\text{num}}(2S_\ell + \delta_{\text{num}}).$$

Averaging over the profiles and adding the two bounds gives the terminal arithmetic term in (S.47).

Each original stored gain is a policy-minus-reference statistic. Under the verified common-noise and initial-state identities, their exact-real reference terms coincide, and cancel in the direct difference. Charging the complete stored-statistic error for each policy is sufficient even though some numerical terms also cancel. The final floating-point subtraction receives its original additional allowance. This proves (S.46). Q.E.D.

E.4. Outward evaluation and reuse of the original confidence event

The numerical account uses the original $N = 2048$ grid, $T = 1$, and $\varepsilon = 0.1$. Every coefficient is evaluated with the inherited interval arithmetic. The coupling norm is accepted only after positive interval LDL pivots certify a matrix upper bound. For an interval-valued weighted matrix C , the computation uses

$$\|C\|_2 \leq \|C_{\text{mid}}\|_2 + \|C - C_{\text{mid}}\|_F,$$

with a verified norm for the binary64 midpoint and an outward Frobenius bound for its interval radius. Componentwise upper bounds for q and R are legitimate in the weighted norms, since increasing nonnegative diagonal weights can only increase the Euclidean norm of each image.

Positive power series avoid cancellation in D , I_1 , and I_2 near the first node. They are

$$D(t) = r \sum_{m \geq 0} \frac{\beta^m t^{m+1}}{(m+1)!},$$

$$I_1(t) = r \sum_{m \geq 0} \frac{\beta^m t^{m+2}}{(m+2)!},$$

$$I_2(t) = r^2 \sum_{m \geq 0} \frac{(2^{m+2} - 2)\beta^m t^{m+3}}{(m+3)!}.$$

The implementation retains $m = 0, \dots, 28$. Writing $z_* = \beta T < 1$, the successive ratios in each remaining tail are bounded by $z_*/30$, $z_*/31$, and $3z_*/32$, respectively. If u is the last retained term and $r_* < 1$ is its tail-ratio bound, the omitted positive tail is at most $ur_*/(1 - r_*)$. All sums, products, roots, differences of integral enclosures, and the positive recursion (S.42) are rounded outward. The scalar B_k, J_k are their original interval enclosures.

For the implementation terms, the calculation takes the maximum, over the nine initial profiles, of the inherited forward-roundoff plus normal-clipping strong-error bound. It also retains twice the complete original stored-statistic arithmetic allowance and the original 10^{-12} allowance for the direct subtraction. The latter includes no proposed cancellation of scalar quadrature or evaluation errors. The

calculation is conditional on the same binary64 and ideal-iid-normal sampling contract as the original verifier.

This account applies when both compared outputs are held controllers. If both outputs select the analytical schedule, their difference remains exactly zero. If exactly one selects that schedule, the original complete schedule-relative transfer is retained. A continuously varying analytical schedule is not silently treated as a held controller.

Finally, let the original empirical-Bernstein event bound the mean of the same clipped direct-difference observations, with its original clipping threshold and tail allowance. Both the original deterministic transfer and (S.46) hold uniformly for the two held controllers. Their minimum may therefore be used on that same event. This operation spends no additional probability budget, changes no sample or stopping rule, and makes no claim of a smaller empirical variance. The original endpoints remain available together with the refined endpoints and the deterministic accounts that distinguish them.

E.5. Proof of Proposition 10

Condition on the complete executed policy ensemble, including all online checking and stopping decisions. The final confirmation arrays are independent of that information. For $C > 0$, set $Z_{sj} = (Y_{sj} + C)/(2C)$. These variables lie in $[0, 1]$ and are independent; their distributions may differ across streams. Apply Maurer and Pontil (2009, Theorem 11) to the pooled vector and to its complement, allocating $\delta/2$ to each sign. After rescaling, the sampling radius is the first two terms in (49). Equal path counts give

$$\mathbb{E}\bar{Y} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \mathbb{E}Y_{s1}.$$

The triangle inequality and the stream-specific transfer accounts bound the distance between this expression and the desired finite-distribution mean by $\bar{\beta}$. This proves the interval claim. When $C = 0$, the clipped observations vanish identically and the transfer account alone suffices. The conditional bound also holds unconditionally by integration over the executed ensemble.

At an online checkpoint, condition instead on the preceding execution history. The policy at that checkpoint is fixed under this conditioning, and its new checking bank is independent. The same argument applies to that registered checkpoint event. A union bound over the predeclared events covers the selected stopping time. Final method inference uses independent confirmation arrays, so it does not treat the array that triggered a favorable stop as an ordinary fixed-policy sample. Summing the four family allocations establishes the stated 0.05 error bound. On the simultaneous stream event, (50) follows by counting the streams whose lower or upper endpoints cross the target. Q.E.D.

APPENDIX F: COMPLETE ECONOMIC APPLICATIONS

F.1. *Analytical and Nonsmooth Benchmarks*

The time-additive Merton model has wealth dynamics

$$dX = [(r + p(\mu - r))X - c]dt + p\sigma X dW, \quad U(c) = \frac{c^{1-\gamma}}{1-\gamma}.$$

For the stationary infinite-horizon benchmark, set $c = mX$, $V = KX^{1-\gamma}/(1-\gamma)$, and impose the usual admissibility and transversality requirements. Writing

$$p^* = \frac{\mu - r}{\gamma\sigma^2}, \quad R_* = r + p^*(\mu - r) - \frac{1}{2}\gamma(p^*)^2\sigma^2,$$

substitution and maximization give

$$m^* = \frac{\rho + (\gamma - 1)R_*}{\gamma}, \quad K = (m^*)^{-\gamma}. \quad (\text{S.53})$$

With $\rho = 0.04$, $\gamma = 2$, $\mu = 0.08$, $r = 0.02$, and $\sigma = 0.2$, these numbers are $p^* = 0.75$ and $m^* = 0.04125$. These are stationary constants; an arbitrary finite-horizon problem with an unspecified bequest does not have the same constant consumption rule. A finite-horizon verification test can use the matching bequest $KX^{1-\gamma}/(1-\gamma)$, explicitly declared rather than silently omitted.

For $\mu = 0.01$ with no short sales, the unconstrained share is -0.125 and the constrained optimum is zero at every positive wealth. Its effective return is r and $m^* = 0.03$. The maximized HJB remains an equality. The action KKT condition is

TABLE F.I

CORRECTED PORTFOLIO CALIBRATION: ANALYTICAL TARGETS AND EXECUTED BLOCK UPDATES

Model	Target m	Target p	Runs	Maximum policy error
Time-additive Merton	0.04125	0.75000	3	5.6×10^{-16}
No-short-sale corner	0.03000	0.00000	3	2.3×10^{-16}
Recursive matching-bequest test	0.04550	0.30000	3	7.0×10^{-17}

Policy error is the maximum absolute error in the two constant policy outputs, maximized over all three seeds. Calculations use CPU float64 in the declared homothetic subfamily. Machine-level calibration accuracy is not evidence about an unrestricted network, a GPU runtime, or a general recursive infinite-horizon problem.

an inequality at the boundary, but it does not turn this regular-control problem into an obstacle problem or create a wealth-dependent kink. The exit experiment in Section 4, rather than this homothetic corner solution, supplies the nonsmooth selection test.

F.1.1. Executed calibration and interpretation

The replication program trains a homothetic neural subfamily: a positive value coefficient, a sigmoid consumption-share output, and a projected portfolio-share output. These are neural output parameters in a known invariant representation, not an unrestricted multilayer approximation. The two Adam optimizers have learning rates 0.025 and 0.008; each outer iteration contains three critic updates and one actor update. There are 1,600 outer iterations and seeds 0, 1, 2. All attempted runs, including their coefficient and policy traces, are saved. These calibrations optimize the stationary coefficient equation and then check the resulting coefficient against the matching bequest; the coefficient error is a boundary error in that finite-horizon interpretation, not a boundary condition hard-imposed during these particular updates.

Table F.I reports quantities generated by the new code. The normalized policy-evaluation residual is below 4×10^{-16} in these runs. These results establish that the revised graph and formulas agree in the exactly representable laboratory. They do not establish a broad advantage over a direct analytical calculation. In the exit laboratory, grids with 40, 80, 160, and 320 intervals recover the grid value to within

3×10^{-15} ; the wrong smooth sequence has scaled residual between 1.984 and 1.999. All grid nodes, not a selected interior sample, enter that diagnostic.

F.2. Recursive Utility

Let $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $Y = bV > 0$, with $\psi \neq 1$ and $\gamma \neq 1$. We use the difference-form aggregator

$$f(c, V) = \frac{\rho}{a} \left[c^a (bV)^{1-a/b} - bV \right]. \quad (\text{S.54})$$

Its consumption derivative is

$$f_c = \rho c^{a-1} (bV)^{1-a/b} > 0. \quad (\text{S.55})$$

When $\gamma = 1/\psi$, one has $a = b$ and $f = \rho(c^b/b - V)$, a positively normalized time-additive CRRA aggregator. The factor b must not multiply the consumption term outside its power. Doing so reverses consumption monotonicity when $\gamma > 1$. This distinction follows the difference-form convention discussed by [Herdegen et al. \(2021\)](#).

For a homothetic candidate $V = KX^b/b$, the maximization equation becomes

$$0 = \sup_{m,p} \left\{ \frac{\rho}{a} [m^a K^{-a/b} - 1] + r + p(\mu - r) - m - \frac{1}{2} \gamma p^2 \sigma^2 \right\}. \quad (\text{S.56})$$

The consumption first-order condition is $\rho m^{a-1} K^{-a/b} = 1$. Substitution yields

$$p^* = \frac{\mu - r}{\gamma \sigma^2}, \quad m^* = \psi \rho + (1 - \psi) \left[r + \frac{(\mu - r)^2}{2\gamma \sigma^2} \right], \quad K = [\rho (m^*)^{a-1}]^{b/a}. \quad (\text{S.57})$$

For the retained parameters $\gamma = 5$, $\psi = 1.5$, $\rho = 0.04$, $\mu = 0.08$, $r = 0.02$, $\sigma = 0.2$, this gives $(m^*, p^*) = (0.0455, 0.3)$.

F.2.1. A well-defined validation problem at the retained parameters

The numerical calibration uses (S.57) as a finite-horizon matching-bequest test, not as an unqualified infinite-horizon theorem. Choose $T > 0$, stop wealth on leaving $[\underline{x}, \bar{x}]$ with $0 < \underline{x} < \bar{x} < \infty$, and impose the candidate Kx^b/b on the stopping

boundary and at T . Consumption shares are in a compact positive interval containing 0.0455, and portfolio shares are in a compact interval containing 0.3.

For these retained parameters, $a > 0$ and $b < 0$. Since c ranges over a compact positive interval, there exist $0 < \underline{Y} < \bar{Y} < \infty$ such that $f(c, \underline{Y}/b) \leq 0$ and $f(c, \bar{Y}/b) \geq 0$ for every feasible c . Indeed, $c^a Y^{-a/b}$ tends to zero as $Y \downarrow 0$ and to infinity as $Y \uparrow \infty$. Enlarge the interval so that it contains the boundary candidate. The constants $\bar{Y}/b$ and $\underline{Y}/b$ are respectively lower and upper barriers for the backward utility equation. Scalar BSDE comparison therefore keeps every policy utility in this interval. The aggregator is Lipschitz there. The smooth homothetic candidate maximizes the Hamiltonian and matches all stopping payoffs; Theorem 1 identifies it as the value of this specified finite-horizon problem.

The choice of a matching bequest is an analytical validation device. It does not establish that the same candidate is selected by every infinite-horizon recursive formulation. A separate infinite-horizon application must check its utility-process class and transversality, particularly when risk aversion and intertemporal-complementarity parameters straddle one (Herdegen et al., 2021). The paper retains that economic application, but does not use a scalar root to bypass its selection issue.

A critic can enforce $bV > 0$ through $bv_{\xi} = \exp(N_{\xi})$. With a positive terminal target, logarithmic interpolation between bG and the neural output enforces both the terminal condition and the power domain. Additional barriers or bounded transformations enforce a declared invariant interval. Such transformations are stated in the architecture, not imposed retrospectively after a failed evaluation.

F.2.2. *Stability and baseline design*

The recursive experiment in Table F.I tests the corrected sign, coefficient, and block graph. It does not compare independent DDPG training trajectories. The reviewed manuscript's constructed oscillating curve and manually entered coordinates remain in the archival source, but have no evidentiary role in this paper. A matched neural comparison must use the same utility process, domain, bequest, action restrictions, training budget, and held-out evaluation quantities. It must

report all failed seeds and value-domain violations. Neither a zero-centered target nor three successful homothetic calibrations establishes general relative stability.

F.3. *Endogenous Preferences and Portfolio Choice*

The endogenous-preference application retains its internal adjustment margin. Let u be risk aversion, X wealth, and θ adjustment effort. On a declared interval $[\underline{u}, \bar{u}] \subset (1, \infty)$, specify the reflected process

$$du = \theta dt + \sigma_u dW^u + dL^{\underline{u}} - dL^{\bar{u}}, \quad (\text{S.58})$$

$$dX = [(r + p(\mu - r))X - c]dt + p\sigma_X X dW^X, \quad d\langle W^u, W^X \rangle_t = \varrho dt. \quad (\text{S.59})$$

The nondecreasing local-time regulators act only at their respective boundaries. This supplies the state mechanism that a bounded adjustment actor does not supply. The numerical reference uses [1.2, 3], so the utility singularity at $u = 1$ is never reached. In an alternative positive-state model one could specify a multiplicative or transformed diffusion, but that would be a different primitive and must be identified as such.

F.3.1. *Cardinal utility and the objective*

Fix a reference consumption unit $\bar{c} > 0$. The baseline flow utility is

$$U(c, u) = \frac{(c/\bar{c})^{1-u}}{1-u}, \quad r_0(c, u, \theta) = U(c, u) - \frac{1}{2}k\theta^2. \quad (\text{S.60})$$

The reference calculation takes $\bar{c} = 1$, preserving the unnormalized CRRA family of the original application. The agent maximizes

$$\mathbb{E}_{t,u,X} \left[\int_t^\tau e^{-\rho(s-t)} r_0(c_s, u_s, \theta_s) ds + e^{-\rho(\tau-t)} G(u_\tau, X_\tau) \right]. \quad (\text{S.61})$$

Both the payoff and the adjustment cost have the same time-relative discount convention. The terminal/exit payoff and the state domain are stated for each calculation.

A useful family for normalization sensitivity is

$$U_\kappa(c, u) = \frac{(c/\bar{c})^{1-u} - 1}{1-u} + \kappa(u).$$

The baseline corresponds to $\kappa(u) = 1/(1-u)$. The log-continuous normalization corresponds to $\kappa = 0$ and is an economically different preference-formation model. A calibration of κ fixes the cardinal value of a reference consumption stream at each preference state. The choice is not identified by risk-aversion curvature alone.

In particular,

$$U_u(c, u) = \frac{(c/\bar{c})^{1-u} \{1 - (1-u) \log(c/\bar{c})\}}{(1-u)^2}. \quad (\text{S.62})$$

For $c/\bar{c} > 1$ and $u > 1$ this derivative is positive. At $c = 2$, $u = 2$, $\bar{c} = 1$, it is approximately 0.84657. This is a derivative of flow utility, not a sign theorem for the endogenous value derivative V_u . The agent's full incentives also depend on future consumption, boundary behavior, and the adjustment technology.

F.3.2. *Hamiltonian, constrained policies, and economic implications*

The HJB equation is

$$\begin{aligned} -V_t = \sup_{c, p, \theta} \{ & U(c, u) - \frac{1}{2}k\theta^2 - \rho V + \theta V_u + [(r + p(\mu - r))X - c]V_X \\ & + \frac{1}{2}\sigma_u^2 V_{uu} + \frac{1}{2}p^2\sigma_X^2 X^2 V_{XX} + \varrho\sigma_u\sigma_X p X V_{uX} \}. \end{aligned} \quad (\text{S.63})$$

Reflection gives $V_u = 0$ at the two preference boundaries, in the appropriate boundary-value sense. The correlation term is retained with its correct coefficient.

For an interior optimum with $V_X > 0$ and $V_{XX} < 0$, the first-order conditions are

$$c^* = \bar{c} [\bar{c} V_X]^{-1/u}, \quad (\text{S.64})$$

$$\theta^* = V_u/k, \quad (\text{S.65})$$

$$p^* = -\frac{V_X(\mu - r)}{X V_{XX} \sigma_X^2} - \frac{\varrho \sigma_u V_{uX}}{X V_{XX} \sigma_X}. \quad (\text{S.66})$$

For independent box constraints, strict concavity in the corresponding action permits projection of these unconstrained maximizers onto their intervals. If V_{XX} has the wrong sign or constraints couple the actions, a joint feasible maximization is required instead. The formula for p must not be used where its denominator vanishes.

Equation (S.65) gives a conditional shadow-price interpretation of internal adjustment. Holding V_u fixed, a larger k reduces its magnitude. Across economies, however, V_u itself depends on k , so this is not a global policy comparative static. A valid unconditional implication is that the optimized value is weakly decreasing in k : every fixed admissible policy loses $\frac{1}{2}(k_2 - k_1)\mathbb{E} \int e^{-\rho s} \theta_s^2 ds$ when the cost rises. When an envelope derivative exists,

$$\partial_k V^k = -\frac{1}{2}\mathbb{E}^{a^k} \int_t^\tau e^{-\rho(s-t)} \theta_s^2 ds \leq 0. \quad (\text{S.67})$$

The sign of the hedging component in (S.66) remains conditional on V_{uX} and the curvature of wealth value. A negative correlation alone does not determine it.

F.3.3. Independent two-state reference and identification

The executable reference solves the finite-horizon problem with $T = 1$, $u \in [1.2, 3]$, wealth stopped at $[0.5, 2]$, and $G(u, X) = X^{1-u}/(1-u)$ at exit and terminal time. Parameters are $(r, \mu, \sigma_X, \sigma_u, \varrho, \rho) = (0.02, 0.08, 0.2, 0.08, -0.3, 0.04)$. The finite action set is the product of $c/X \in \{0.02, 0.05, 0.1\}$, $p \in \{0, 0.375, 0.75\}$, and $\theta \in \{-0.15, 0, 0.15\}$. A four-point Brownian cubature with positive interpolation weights performs backward induction. The wealth boundary is imposed at every time step; preference overshoots are reflected. This reference defines a finite-action numerical approximation, not continuous-action optimality.

At $(u, X) = (2.1, 1)$, the original 17×25 state grid and 20 time steps give values -11.915211 and -11.956983 for $k = 1$ and $k = 5$. The 25×37 grid and 40 time steps give -11.866768 and -11.908738 . Both select $(c/X, p, \theta) = (0.1, 0.75, -0.15)$ at that point. The approximately 0.048 refinement change is reported separately from training error. The higher-cost value is weakly lower at every recorded node and time. These original references remain useful consistency checks; Section F.3.4 supplies actual neural fits and shows how strongly the finite action domain affects their economic comparison.

The model also specifies what a structural estimation exercise would require. For observations $Y_j = \mathcal{M}(S_{t_j}, a_{t_j}; \vartheta) + \varepsilon_j$, let $m(\vartheta)$ denote selected population moments of the induced observation process. Local identification requires a rank condition on $D_\vartheta m$, after accounting for latent preference states, normalization parameters,

and observation noise. Variation of a simulated policy with k alone is insufficient. Numerical value and policy errors must additionally be propagated into moment errors. The endogenous-preference application is therefore retained as an economic model with an explicit internal margin, not used as evidence that its structural parameters have already been identified from data.

The next comparison is the retained continuous-action study evidence under its original enclosure. The new signed envelope and fresh nested-grid policies are reported separately in Section F.8; historical failure counts are not overwritten.

F.3.4. *Neural preference policies and the role of the action oracle*

We compute the original preference-adjustment economy with the cardinal utility, shock correlation, reflected preference state, and wealth stopping payoff specified above. The main grid has 17 preference points, 25 log-wealth points, and 20 time steps over a unit horizon. Adjustment cost is $k = 1$. The expanded control box is

$$m = c/X \in [0.02, 0.30], \quad p \in [0, 1.5], \quad \theta \in [-0.45, 0.45].$$

The finite-action comparison uses the 175 triples formed by five consumption shares, five portfolio shares, and seven adjustment efforts, including all endpoints. This is the same constrained economy across the matched finite methods, not a change of utility normalization.

Two experiments distinguish action representation from action optimization. First, the finite-policy study finite study trains two-hidden-layer, width-32 tanh critics and categorical actors. At each backward step it computes the complete 175-action payoff vector using the current continuation critic. Its exact argmax provides a label, and the vector also supplies the actor's expected action-loss objective. Thus *the training oracle is exhaustive*, even when the final actor is accurate. It is a useful neural policy-representation experiment, but not evidence that the actor avoided Bellman maximization.

Second, continuous-action study trains a continuous-output actor directly through its own positive-weight payoff. It never receives an all-action value vector or an exact argmax label. The actor receives 160 Adam updates per time level; the continuation critic receives 320 Adam updates and up to 300 L-BFGS iterations, with

failed polishing steps rolled back. The actor-free direct comparator optimizes eight independent bounded controls at every state for the same 160 iterations, followed by the same critic fit. Every restart and rejected action query is charged. Neither procedure invokes a Riccati solution, an analytically invariant policy class, or the independently computed reference during training.

Finite-action neural policies. The raw finite-policy study actor is the primary neural representation result. Its full dynamic policy loss is below the declared 0.1 target for each fixed seed. A top-four shortlist and a separately reported correction map further reduce the loss. The correction map is not part of the raw network. The complete finite policy, not only one-step deviations against a fitted critic, is evaluated by independent backward recursion.

TABLE F.II

FINITE PREFERENCE POLICIES: FINITE-POLICY STUDY EVIDENCE

Method	Seed	Raw loss	Final loss	Value error	Seconds	Guard (%)
NBO	11	0.0445	0.0261	0.0924	30.59	0.294
Direct	11	0.0181	0.0181	0.0790	24.79	0.000
NBO	29	0.0500	0.0260	0.0814	30.49	0.332
Direct	29	0.0263	0.0263	0.0551	25.19	0.000
NBO	47	0.0500	0.0263	0.1325	29.89	0.230
Direct	47	0.0264	0.0264	0.0824	25.17	0.000

Loss bounds cover every stored state and time of the same 175-action economy. Seconds include training and the three reported policy verifications; classical backward induction averages 0.102 seconds. A raw actor, shortlist, and corrected policy are distinct objects. Utility losses are absolute units.

These results repair the earlier initial evaluation study actor failures, which remain in the historical numerical record. They do not reverse the cost comparison: the matched direct finite-Bellman fit is faster and more accurate on average, and classical backward induction is substantially cheaper on this small grid. Tables report those comparisons without selecting a favorable seed. The actor's approx-

imation cost is part of the result, not an argument for discarding the economic application.

Continuous-action optimization without exact labels. The initial continuous actor and direct-gradient pilot fail the 0.1 deviation diagnostic. The actor can saturate at a portfolio bound while a lower portfolio share improves continuation value. Four actor heads with output reinitialization do not cure that failure in the retained pilot. We therefore execute a disclosed local-improvement variant: compare the actor proposal with eight box corners, then perform two coordinate-search sweeps at each of eight decreasing step sizes. The same local search is applied to the direct comparator. An actor-free ablation omits all gradient action updates and starts the identical search from feasible proposals. This ablation measures how much of the hybrid result comes from the search itself.

TABLE F.III
CONTINUOUS-CONTROL TRAINING WITHOUT EXACT MAXIMIZING LABELS

Method	Seed	Raw diagnostic	Hybrid gain	Train (s)	Queries (10^6)
Actor + search	11	0.2596	0.0617	15.01	2.261
Actor + search	29	0.0733	0.0188	28.49	2.261
Actor + search	47	0.2597	0.0475	26.46	2.261
Direct + search	11	0.1554	0.0303	32.02	11.841
Direct + search	29	0.1566	0.0203	32.80	11.841
Direct + search	47	0.1554	0.0251	30.52	11.841
Search only	11	5.5370	0.0617	13.97	0.961
Search only	29	7.2093	0.0616	13.72	0.961
Search only	47	8.0316	0.0616	12.87	0.961

Raw diagnostic is the largest finite-reference payoff minus the raw policy payoff. Hybrid gain is the augmented finite-deviation gain, a lower bound on continuous-action regret, not a certificate. All methods use zero exhaustive training backups and zero exact argmax training labels. Search-only omits action-gradient updates. Training, reference, and full-cover verification costs are separate.

The table distinguishes three quantities. The raw-network loss against the finite reference tests the actor alone. The augmented finite-deviation gain permits all 175

inherited actions and the learned continuous action at every state and time; it is a *lower bound* on missed continuous-action gains. The complete action-box upper bound in Proposition S.2 is a separate verification result. A small augmented gain is not inserted into that proposition as if it were a uniform upper bound. Local changes occur at nearly every active state-time node in the principal hybrid runs, so this experiment does not describe a sparse correction map or an actor-only controller.

TABLE F.IV

FULL CONTINUOUS-ACTION POLICY BOUNDS ON THE TWO-STATE NODAL ECONOMY

Policy	Seed	Uniform bound	Center bound	Eval. (s)	Bound ≤ 0.1
Actor + search	11	0.1294	0.0601	0.22	No
Actor + search	29	0.1227	0.0583	0.22	No
Actor + search	47	0.1301	0.0608	0.25	No
Direct + search	11	0.1153	0.0604	0.23	No
Direct + search	29	0.1156	0.0586	0.22	No
Direct + search	47	0.1203	0.0609	0.25	No
Search only	11	0.1196	0.0647	0.24	No
Search only	29	0.1179	0.0586	0.23	No
Search only	47	0.1209	0.0610	0.22	No

The full action box is covered at all 425 states and 20 times. The common optimal upper envelope costs 1624.59 seconds across the retained implementations and 40.96 million action-box evaluations. Each row adds its own interval policy evaluation. Unresolved leaves retain their upper bounds. No continuous-state/time error is set to zero.

The action cover includes binding constraints and wealth exits. Its unresolved boxes remain in the bound when the declared budget is exhausted. The resulting upper array can audit the actor, the direct optimizer, and the local-search ablation without retraining. This produces an operational error account on a multidimensional state grid with a continuous, nonconcave action objective. It does not establish that all runs meet the tighter 0.1 complete-action target, nor that the actor dominates the direct method at matched economic accuracy.

Alternative representations and economic units. Additional same-model comparisons use a gated neural continuation architecture and tensor Chebyshev projection. The gated method uses DGM-style state-dependent gates (Sirignano and Spiliopoulos, 2018), but here it is fitted to finite Bellman targets, not called a mesh-free differential DGM solver. The projection exploits the same two-state geometry and imposes the same wealth-boundary lifting. Degrees, parameter counts, all-action costs, failures, and actual accuracy appear in the supplement. These are executed post-review comparisons, not results attributed to the earlier unexecuted development protocol. A generic semilinear BSDE comparator is not substituted for this controlled-covariance, stopped problem without reformulating its second-order control term.

All losses are in the specified cardinal utility units. The optimal finite-model value at the center is approximately -4.8410 , and the initial value range is approximately $[-9.1599, -0.1250]$. A loss of 0.1 is about 2.066 percent of the center's *absolute utility magnitude*; this scaling is not a consumption-equivalent welfare percentage. Endogenous curvature and adjustment costs make a common consumption-equivalent conversion a separate economic calculation.

Action bounds and controlled refinements. The sensitivity study retains both adjustment-cost values, the original 27-action truncation, 175 and 1,053 actions in the expanded box, a wider action box, wider wealth bounds, and finer space-time grids. Consumption and portfolio upper bounds remain frequently active. Expanding the box changes the constrained economy; refining actions inside one box does not. continuous-action study additionally evaluates the *same frozen continuous policy*, bilinearly interpolated in state and held on its original time schedule, on three nested state-time levels. This isolates changes in policy evaluation from retraining. The supplemental differences are observed refinement diagnostics, not an estimated order promoted to a continuous-time error bound. The original continuous control problem, its utility normalization, and its comparative-static questions remain the economic objects of interest.

F.4. *Sophisticated Temporal Selves*

To define present bias in continuous time, fix the discount kernel

$$h(s) = \beta e^{-\rho s} + (1 - \beta)e^{-(\rho + \kappa)s}, \quad 0 < \beta \leq 1, \quad \kappa > 0. \quad (\text{S.68})$$

It satisfies $h(0) = 1$ and discounts the immediate future more heavily than the distant future when $\beta < 1$. It is a finite-duration continuous-time specification; it is not the assertion that multiplying current utility by β creates present bias.

For a candidate Markov policy a^* , define the two continuation functions

$$W_j(t, x) = \mathbb{E}_{t,x}^{a^*} \left[\int_t^T e^{-\rho_j(s-t)} u(a_s^*) ds + e^{-\rho_j(T-t)} G(S_T) \right], \quad (\rho_1, \rho_2) = (\rho, \rho + \kappa). \quad (\text{S.69})$$

Write $V = \beta W_1 + (1 - \beta)W_2$. Each component evaluates the *same* future policy:

$$0 = (W_j)_t + \mathcal{L}^{a^*} W_j + u(a^*) - \rho_j W_j, \quad W_j(T) = G. \quad (\text{S.70})$$

The current self's spike-deviation condition is

$$a^*(t, x) \in \arg \max_{a \in \mathcal{A}(t, x)} \{u(a) + \mathcal{L}^a V(t, x)\}. \quad (\text{S.71})$$

The deviations change both current flow utility and the transition of the continuation state. They do not hold all dynamic effects at the equilibrium action. The appendix derives (S.71) from a deviation on $[t, t + \epsilon]$ followed by a^* , thereby fixing the equilibrium concept.

A temporal-self NBO uses two policy-evaluation critics for (S.70) and one actor for (S.71). The current actor does not optimize the sum of future selves' parameter objectives. A target actor can stabilize a numerical iteration, but its agreement with the current actor is not an equilibrium certificate. The relevant diagnostics are component evaluation errors and one-shot deviation gains. The benchmark $\beta = 1$ collapses to the usual exponentially discounted problem.

F.4.1. *An equilibrium calculation with endogenous continuation values*

For deterministic wealth $dX = (rX - c)dt$, CRRA utility with $\gamma > 1$, and an infinite horizon, consider $c = mX$. Set $b = 1 - \gamma$ and $\lambda_j(m) = \rho_j - b(r - m) > 0$.

Then

$$W_j(X) = \frac{m^b}{\lambda_j(m)} \frac{X^b}{b}, \quad 1 = m \left[\frac{\beta}{\lambda_1(m)} + \frac{1-\beta}{\lambda_2(m)} \right]. \quad (\text{S.72})$$

The expression on the right is strictly increasing in m on its admissible interval, starts at zero, and diverges at the first pole for $\beta > 0$. The equilibrium root is unique in this homothetic class. Its derivative with respect to β is negative because $1/\lambda_1 > 1/\lambda_2$. Thus stronger present bias raises consumption in this specified equilibrium, not merely at a fixed shadow price.

With $r = 0.02$, $\rho = 0.04$, $\gamma = 2$, and $\kappa = 0.2$, the roots for $\beta = 1, 0.9, 0.7, 0.5$ are respectively 0.030000, 0.031372, 0.034606, and 0.038760. The code independently optimizes finite-duration deviations and records the gain divided by duration as the duration shrinks. For $\beta = 0.7$, this quantity declines from approximately 4.60×10^{-5} at duration 0.1 to 4.66×10^{-9} at duration 0.001. This is a diagnostic of the derived spike equilibrium. It does not assert that every finite-duration commitment deviation is unprofitable, which would be a different solution concept.

F.5. *Dynamic Games*

The dynamic Cournot application retains firm-specific capital and investment. Firm i has

$$dK^i = (I^i - \delta K^i)dt + \sigma_K K^i dW^i, \quad q_i = AK^i,$$

and payoff rate

$$r_i = P_N(K)q_i - I^i - \frac{1}{2}b_i(I^i)^2, \quad P_N(K) = \max \left\{ 0, P_0 - \frac{1}{M_N} \sum_j q_j \right\}.$$

For each experiment, the market-size normalization M_N , capital domain, investment constraints, covariance, terminal/liquidation values, and horizon are part of the primitives. Keeping $M_N = 1$ is a fixed-market comparison. Taking $M_N = N$ is a growing-market comparison. They do not have the same competitive-limit interpretation.

Given rivals' Markov policies a^{-i} , a player's best-response equation maximizes only its own action. In the stationary additive case,

$$\rho_i V_i = \sup_{a_i \in \mathcal{A}_i} \{r_i(x, a_i, a^{-i}(x)) + \mathcal{L}^{a_i, a^{-i}} V_i\}. \quad (\text{S.73})$$

A Markov-perfect equilibrium is a feasible profile satisfying these simultaneous unilateral conditions, with the same boundary and admissibility conventions. A zero policy-evaluation residual can hold for any consistently valued profile and therefore does not identify this equilibrium.

F.5.1. *Player-specific differentiation and deviation bounds*

Player i 's critic is trained with every policy frozen. Player i 's actor then differentiates only

$$-\mathbb{E}_v \mathcal{H}_i^{a_i(\phi_i), \text{sg}(a^{-i})} [\text{sg}(V_i), \text{sg}(DV_i), \text{sg}(D^2 V_i)]. \quad (\text{S.74})$$

The updated parameter is ϕ_i , not the full vector of all players' parameters. Rival actions are detached, and player j 's loss never contributes to player i 's actor gradient. An explicit regression test exercises this graph.

For each player, define the unilateral Hamiltonian gap by maximizing $\mathcal{H}_i$ over a_i while keeping rivals fixed. Applying Theorem 1 to that best-response problem bounds its finite-horizon exploitability by $2e^{L_i T} B_i + A_{L_i}(T)(2e_i + \eta_i)$. In a discounted finite-state game, Theorem 2 gives $(2\delta_i + \epsilon_i)/(1 - q_i)$. These are player-specific bounds; a sum of critic losses is not a substitute. They remain valid without uniqueness or convergence of the simultaneous policy-update dynamics, provided the reported profile satisfies the stated errors and best-response comparison hypotheses.

The static Cournot diagnostic makes the distinction visible. With payoffs $q_i(1 - q_i - q_j)$, symmetric Nash output is 1/3 per firm. Joint-profit maximization gives a symmetric allocation 1/4. At the latter profile, a firm optimally deviates to 3/8, earning a gain of 1/64. A joint objective can therefore be perfectly optimized and still fail a unilateral equilibrium test. This arithmetic is included as a mandatory regression check.

F.5.2. *Symmetry and market scaling*

Identical initial weights do not prove equilibrium uniqueness or permutation equivariance. An equivariant actor can be specified using own capital and a permutation-invariant aggregation of rival states. Even then, one must check unilateral deviations rather than infer equilibrium from architectural symmetry. Multiple starting profiles and unsuccessful best-response searches belong in a numerical report.

Under fixed market size, positive constant capital per firm cannot be called the competitive limit while total output tends to infinity and the truncated price tends to zero. Maintaining that capital entails investment and adjustment costs. Under $M_N = N$, positive capital per firm may instead be compatible with a growing market, but it describes a different sequence of economies. The original unsupported capital table remains in the archive. Section F.5.3 replaces its evidentiary role with a fully specified finite dynamic game and independent best-response calculations, without asserting convergence of unrestricted neural game training.

F.5.3. *Neural dynamic capital policies and full unilateral deviations*

For two firms, we retain the capital game with $A = 1$, $P_0 = 2$, $\delta = 0.1$, $\sigma_K = 0.15$, $\rho = 0.04$, $T = 3$, investment in $[0, 0.8]$, and capital in $[0.05, 1.5]^2$. Terminal or joint-domain exit pays $0.5K_i$. Baseline adjustment coefficients are $b_1 = b_2 = 1$; the earlier asymmetric experiment has $b_2 = 1.2$. The principal neural grid has 17×17 states, 30 time steps, and 11 actions per player. Positive cubature and interpolation define the finite game. Market normalizations $M_2 = 1$ and $M_2 = 2$ are distinct economies, not evidence for an uncomputed large-population limit.

At each time and state, the finite-policy study calculation trains separate critics and actors. It constructs both full 11×11 continuation-payoff matrices. The training labels are a pure one-step equilibrium when one exists and a minimum-maximum-defect pair otherwise. The actor's own objective is differentiated with the rival action and continuation critic fixed. Thus this study actually trains neural strategic policies, but it also pays for an exhaustive joint-action training oracle. Raw policies, three-by-three shortlist policies, and any corrected policies are distinct stored arrays.

After freezing the entire profile, an independent backward optimization permits every own action at every state and time against the fixed rival policy. Its value minus the profile payoff is the unilateral exploitability map. This is not a one-period test with the approximate continuation held fixed. The raw and final profiles meet the 0.1 absolute-profit target in every principal finite-policy study case; the full-oracle guard is unused in those cases.

TABLE F.V

FINITE-POLICY STUDY NEURAL DYNAMIC-GAME PROFILES AND FULL BEST RESPONSES

Market	Seed	Raw max	Firm 1	Firm 2	Missing	Seconds
1	11	0.0259	0.0164	0.0252	3	52.43
2	11	0.0241	0.0255	0.0163	1	56.13
1	29	0.0300	0.0288	0.0296	0	52.44
2	29	0.0626	0.0624	0.0547	1	57.23
1	47	0.0190	0.0081	0.0183	0	52.53
2	47	0.0101	0.0074	0.0113	0	56.98

All bounds cover every stored finite state, time, and unilateral action, including deviations at the reported missing pure stages. The full-oracle guard is unused in these principal runs. Profit losses are absolute units; market normalizations are different economies.

Five state-time nodes across the six neural runs lack a pure one-step equilibrium for the fitted continuation matrices. Their locations and defects are reported in the supplement. A minimum-defect selection at those nodes is not called an exact equilibrium. Nevertheless, the complete dynamic best-response bound remains valid for the resulting profile: its proof does not assume that every fitted one-step game has a pure solution. Moreover, for fixed Markov rivals, randomized or history-dependent own deviations cannot improve on the finite Markov best-response value, because each conditional action mixture is bounded by its largest pure-action payoff. The calculation therefore checks such unilateral deviations as well; it does not compute a mixed-strategy equilibrium profile or a continuous-action game.

continuous-action study adds actor-free neural continuation fitting and classical backward selection for both market normalizations. The neural comparator allocates 480 critic Adam updates per time level rather than finite-policy study's 320 critic

plus 160 actor updates, with the same L-BFGS cap. Counts and timings reveal the difference between nominal update matching and matched wall time. Both methods receive their own independent full dynamic best responses. The earlier six classical game cases, asymmetric costs, alternative tie rules, boundary frequencies, coarse counterexample, and finer-grid results remain in the supplement and raw record.

TABLE F.VI

CONTINUOUS-ACTION STUDY ACTOR-FREE DYNAMIC-GAME COMPARISONS

Method	Market	Seed	Firm 1	Firm 2	Train (s)	Verify (s)
Classical	1	11	0.0000	0.0000	0.22	6.40
Classical	2	11	0.0000	0.0000	0.23	6.29
Direct	1	11	0.0223	0.0273	37.43	6.09
Direct	1	29	0.0199	0.0282	37.68	6.18
Direct	1	47	0.0223	0.0409	36.03	6.26
Direct	2	11	0.0091	0.0694	40.43	5.75
Direct	2	29	0.0306	0.0185	43.07	6.34
Direct	2	47	0.0206	0.0358	40.47	6.08

The direct method fits two continuation networks with the same class and 480 critic Adam updates, versus finite-policy study's 320 critic and 160 actor updates. Classical selection uses no neural approximation. Every method receives a new full dynamic best response. finite-policy study and continuous-action study timings were measured in their recorded environments; cross-environment ratios are not a hardware-controlled speed experiment.

The finite-game result is a numerically evaluated approximate equilibrium with a specified profit-unit bound. It is not inferred from stationarity of a summed actor loss or the static 1/64 diagnostic. State interpolation, continuous investment, diffusion timing, and continuously monitored exits remain separate terms in a continuous-game error account.

F.6. Scalability and the Diffusion Trace

The original multi-stock consumption-savings specification has additive utility, independent transitions, and no common resource constraint. It decomposes as $V(k) = \sum_j v_j(k_j)$. A separability-aware comparator solves the one-dimensional com-

ponents, and identical components can be reused. An unreduced sparse grid is not an appropriate sole comparator for that economy.

To test a coupled generator, the new laboratory instead uses

$$dX = (AX + a)dt + \Sigma dW, \quad \sup_a \mathbb{E} \int_0^\infty e^{-\rho t} [-X^\top QX - a^\top a] dt. \quad (\text{S.75})$$

The drift is a nonnormal dense matrix $A = -0.7I + 0.06Z/\sqrt{d}$, where Z is generated by the fixed model seed $991 + d$. The payoff matrix is $Q = \text{diag}(1, \dots, 2) + 0.311^\top/d$. The code verifies that $AQ \neq QA$, so the common independent-coordinate reduction of the original benchmark is unavailable. All methods still exploit the declared quadratic structure. A Riccati solver is therefore the appropriate strong reference, rather than a deliberately unreduced general grid.

For $V = -x^\top Px - c_0$ and $a = -Kx$, policy evaluation solves

$$(A - K - \rho I/2)^\top P + P(A - K - \rho I/2) = -(Q + K^\top K), \quad c_0 = \text{tr}(\Sigma \Sigma^\top P)/\rho. \quad (\text{S.76})$$

Hamiltonian improvement gives $K = P$. A positive semidefinite P is representable by a signed sum of squared linear neural units; the actor is linear. The exact block laboratory uses a Lyapunov solve for this critic and exact actor improvement. It is an exactly representable NBO subfamily and a matrix-identity test, not evidence that generic stochastic neural optimization has the same convergence rate.

Across dimensions 2, 5, 10, 20, 50 and three stabilizing actor initializations, the block iteration takes six steps. The maximum error in P , in operator norm, is below 10^{-13} relative to an independent algebraic Riccati solver. The policy-evaluation coefficient residual is below 10^{-12} . On the unit Euclidean ball, the actor gap is bounded by $\|K - P\|_2^2$, and the residual bound follows from the operator norm of the quadratic coefficient error. The program saves matrices, iteration traces, and CPU wall times for both methods. These very short, single-process timings are diagnostics and do not establish a fixed-accuracy runtime ranking.

F.6.1. *Trained, coupled nonquadratic capital economies*

To separate neural scaling from exact quadratic algebra, consider d positive capital stocks and write $Y_i = \log X_i$. Consumption is $c_i = m_i X_i$, with $m_i \in [0.02, 2]$,

and

$$dY_i = \left[0.10 + 0.15 \tanh((B_d Y)_i) - m_i - \frac{1}{2}(0.15^2 + 0.10^2) \right] dt + 0.15 dW_i + 0.10 dW_0, \quad (\text{S.77})$$

$$r_0(Y, m) = d^{-1} \sum_i (\log m_i + Y_i) - 0.10 \bar{m}^2, \quad \bar{m} = d^{-1} \sum_i m_i.$$

The dense row-normalized matrix B_d is generated once from a recorded seed. Horizon is one, discount is 0.04, and terminal utility is $\bar{Y} - 0.03d^{-1} \sum_i (Y_i - \bar{Y})^2$. The actual state domain is $\mathbb{R}^d$; no artificial lateral stopping condition is imposed. Dense nonlinear productivity, shared adjustment cost, common noise, and terminal dispersion couple the coordinates. Neither the solver nor its comparator calls a Riccati or Lyapunov routine, and no exact solution is built into the network.

Strict concavity gives a global Hamiltonian maximizer at each queried state. For $p_i = \partial_i v$, its components satisfy

$$m_i^* = \text{clip} \left((dp_i + 0.2\bar{m}^*)^{-1}, 0.02, 2 \right),$$

with the upper bound selected when the denominator is nonpositive. The scalar consistency equation for $\bar{m}^*$ is monotone and is solved by bisection. This supplies the same feasible maximization routine to the direct-HJB comparator and to action-gap measurement; it does not supply the value function.

Both networks have two hidden tanh layers of width thirty-two. The critic enforces the terminal payoff exactly. The main design uses three seeds at $d = 2, 5, 10, 20, 600$ Adam updates, and batches of 128 independently sampled state-time points. Exact automatic differentiation supplies the trace. The held-out set contains 512 fixed independent points in $[0, 1] \times [-0.5, 0.5]^d$. A common diagnostic target is residual root mean square at most 0.025 and maximum queried action gap at most 0.01. The recorded first crossing is not necessarily sustained; final failures are also reported.

The experiment measures real neural approximation and optimization costs rather than exact quadratic identities. It also shows why a diagnostic target is not a welfare certificate. Simulated policy paths can leave the fitted state box; Euler policy-

TABLE F.VII

ACTUAL TRAINING IN DENSE NONQUADRATIC CAPITAL MODELS

Method	d	Worst RMS	Worst gap	Failures	Median seconds
NBO	2	0.0161	2.42×10^{-4}	0/3	2.6554
Direct HJB	2	0.0161	0	0/3	3.4870
NBO	5	0.0131	0.0020	0/3	4.1518
Direct HJB	5	0.0129	0	0/3	5.0391
NBO	10	0.0188	0.0052	0/3	6.8652
Direct HJB	10	0.0168	0	0/3	7.6988
NBO	20	0.0320	0.0114	2/3	12.8180
Direct HJB	20	0.0170	0	0/3	13.5001

Maxima are across three seeds, each evaluated at 512 independent points in $[0, 1] \times [-0.5, 0.5]^d$. The failure criterion is RMS residual above 0.025 or queried action gap above 0.01 after 600 updates. Both methods use the same critic architecture and global action routine. Times include training diagnostics, not the subsequent Monte Carlo evaluations. These are sampled diagnostics, not uniform economic bounds.

payoff estimates from the initial state need not agree with a critic that has a small sampled residual inside that box. A follow-up expands the fitting and verification box to $[-1.5, 0.5]^d$ and uses 1,000 updates at dimensions ten and twenty. Both designs, paired policy-payoff samples, and forty/eighty-step simulation outputs are retained. Sampling standard errors do not remove time-discretization bias or certify an optimum. Process high-water memory is labeled as such, not interpreted as an isolated per-network allocation. No high-dimensional uniform residual bound is claimed for these sampled quantities.

F.6.2. Method-paired payoff evidence and arithmetic replay

The finite-policy study follow-up evaluates the coupled-capital policies at dimensions ten and twenty, for all three fixed seeds, both fitting domains, and 40, 80, and 160 Euler steps. The same 512 Brownian paths are used by NBO and the direct comparator within every cell. For path j , define $D_j = J_j^{\text{NBO}} - J_j^{\text{direct}}$. We report $\bar{D}$ and the pointwise Student interval $\bar{D} \pm t_{511, 0.975} s_D / \sqrt{512}$. Pairing removes the

irrelevant component of common path variation from the comparison. The complete set of 36 comparisons is reproduced from the pinned raw arrays in the supplement.

TABLE F.VIII

PAIRED PAYOFF DIFFERENCES AT THE FINEST RECORDED EULER LEVEL

Dim.	Seed	Domain	Difference	95% lower	95% upper
10	11	Narrow	-0.03338	-0.03544	-0.03133
10	11	Wide	0.00052	0.00049	0.00054
10	29	Narrow	-0.03094	-0.03250	-0.02938
10	29	Wide	-0.00003	-0.00004	-0.00002
10	47	Narrow	-0.02755	-0.02941	-0.02569
10	47	Wide	0.00000	-0.00001	0.00001
20	11	Narrow	-0.06302	-0.06572	-0.06033
20	11	Wide	0.00084	0.00080	0.00088
20	29	Narrow	-0.01449	-0.01490	-0.01408
20	29	Wide	0.00409	0.00394	0.00424
20	47	Narrow	-0.20161	-0.20211	-0.20111
20	47	Wide	0.00058	0.00053	0.00062

NBO minus direct payoff on the same 512 paths. Full 40/80/160-step comparisons are in the supplement. Confidence intervals quantify Monte Carlo variation only.

The old twenty-dimensional fitting box is exited by essentially all recorded paths. Wider-domain fitting eliminates observed exits in the recorded audits and changes both payoff agreement and method differences. Zero observed exits in 512 paths is not a zero exit-probability theorem. Likewise, these intervals quantify point-wise Monte Carlo uncertainty under a specified Euler scheme. They are neither simultaneous confidence statements across all 36 comparisons nor bounds on time bias, optimal value, or the global differential residual. The experiment identifies a consequential coverage failure and its measured correction without redefining the unbounded economic state space as a stopped box.

The one-state consumption certificate has a different evidentiary status. finite-policy study independently replays its previously generated complete leaf cover with `mpmath.iv` at 40 decimal digits, without importing the original interval

arithmetic implementation. For the stored seed-11 networks, the NBO policy-loss bound is about 0.04961 and the direct-HJB bound about 0.04967. Both meet the stated threshold. This is an *independent interval-arithmetic replay on the previously generated complete cover*, not independent network training, independent cover generation, or a formal machine proof. Together with the new complete three-control cover on the two-state preference grid, it demonstrates two different ways of operationalizing verification. Neither removes the dimensional cost of covering an unrestricted high-dimensional continuous state space.

F.6.3. Exact and stochastic trace computation

Let Σ include any covariance factor and set $B = \Sigma^\top D^2 v \Sigma$. For independent probes with identity covariance,

$$q = \text{tr}(B), \quad \hat{q} = \frac{1}{K} \sum_{k=1}^K z_k^\top B z_k. \quad (\text{S.78})$$

Gaussian probes have variance $2\|B\|_F^2/K$. If $r = d - q/2$, then

$$\mathbb{E}(d - \hat{q}/2)^2 = r^2 + \frac{1}{4} \text{Var}(\hat{q}). \quad (\text{S.79})$$

The extra term depends on the critic. Unbiased trace estimation therefore does not give an unbiased squared-residual objective.

Two conditionally independent probe banks repair this particular bias:

$$\hat{L} = (d - \hat{q}^{(1)}/2)(d - \hat{q}^{(2)}/2), \quad \mathbb{E}[\hat{L} \mid t, x, \xi] = r^2. \quad (\text{S.80})$$

With dominated differentiation, its gradient is also unbiased. The sample product need not be nonnegative and can have substantial variance, so this is an objective identity rather than a guarantee of stable optimization. Exact low-dimensional derivatives remain the independent check. If the actor or deterministic term depends on the same probes, conditional independence must be re-established; the simple identity cannot be reused automatically.

The relevant computational cost depends on the network evaluation and differentiation cost, Brownian rank, covariance factorization, sample count, optimization iterations, and requested accuracy. Forming a dense Hessian involves quadratic

storage; Hessian-vector products can avoid it. Neither operation count by itself is a total-complexity theorem. Active control bounds do not universally create gradient discontinuities, and a noisy trace cannot make a correctly projected box action leave its box. Feasibility and value-derivative accuracy are separate diagnostics.

The independent-bank estimator is also exercised during actual training in the coupled ten-state economy. One implementation uses two independent banks of two Gaussian Hessian-vector probes; the matched single-bank ablation squares a four-probe estimate. Exact traces are used only for the common held-out diagnostics. All three seeds, objective signs, optimization histories, and policy-payoff samples are saved. The two training procedures perform similarly at the retained covariance scale; unbiasedness alone does not establish a finite-budget advantage. A separate Monte Carlo test checks both the product objective and its parameter gradient against exact trace expressions with declared tolerances. These calculations validate the estimator's execution without identifying a random training objective with a deterministic certificate.

F.7. Validated Evaluation and Action Improvement

The neural Bellman map must account for two distinct approximation decisions: which continuation values are passed to the next stage, and how tightly the admissible action supremum is bounded. This section supplies an explicit output rule for the first decision and a signed enclosure for the second. Neither construction presumes that a stationary point of the training loss is economically optimal. Both apply to the original preference economy and leave its controls, state law, and stopping convention unchanged.

F.7.1. An evaluation rule with a prescribed nodal error

Let T_h^a denote a positive-weight, one-step policy operator on a finite state grid, and suppose its continuation weights have sum at most $q_h \leq 1$. At a given date, let w be the continuation vector, a the selected feasible policy, $\hat{v}$ the fitted neural critic, and $\tilde{T}_h^a w$ the computed selected-policy target. We distinguish the computation of this target from maximization over actions. In particular, it uses one action at each

node, not a table of optimizing labels. For a prescribed tolerance $\tau_h > 0$, define

$$C_h(\hat{v}, a, w)_i = \begin{cases} (\tilde{T}_h^a w)_i, & |\hat{v}_i - (\tilde{T}_h^a w)_i| > \tau_h, \\ \hat{v}_i, & \text{otherwise.} \end{cases} \quad (\text{S.81})$$

The comparison in the implementation uses an outward upper bound for the difference. A selected entry is assigned the target directly; it is not reconstructed by adding two cancelling floating-point numbers. The indexed corrections, their masks, and the uncorrected neural outputs are saved. The deployed continuation is therefore a neural representation with a recorded nodal correction, not a raw network.

PROPOSITION S.7—Evaluation guard: *Suppose $\|\tilde{T}_h^a w - T_h^a w\|_\infty \leq \eta_h$. Then*

$$\|C_h(\hat{v}, a, w) - T_h^a w\|_\infty \leq \tau_h + \eta_h.$$

For a frozen sequence of policies over N stages, with exact terminal data, the guarded backward values and the exact values of those policies differ by at most $\sum_{j=0}^{N-1} q_h^j(\tau_h + \eta_h)$. If $Nh = T$, $\tau_h = 0.1h^2$, and $\eta_h = o(h)$, this evaluation perturbation vanishes as $h \downarrow 0$. This statement is a property of the output rule, not a rate for the neural optimizer or an assertion of policy optimality.

The guard gives an implementable meaning to the evaluation component of a vanishing-perturbation condition. It does not supply the action-improvement error, the consistency error of a stopped diffusion approximation, or convergence of an uncorrected network away from the nodes. Its cost can be substantial: in the numerical economy it corrects a large fraction of fitted nodal values. We consequently count the correction memory and compare against an actor-free method with the identical guard. The construction is not evidence of neural compression when almost every node is retained.

F.7.2. Signed enclosures for continuous actions

Fix a state-time node and a continuation vector w . Write $Q_w(a)$ for its one-step payoff. The action set is the full compact box of the preference economy. Let $B = [\ell, r]$

be an action rectangle with centre c . A natural interval calculation supplies an upper bound $U^{\text{nat}}(B)$, while independent feasible evaluations supply a lower incumbent L .

The continuation is bilinear within state cells and continuous across their internal faces. Reflection changes its one-sided slopes. A stopped transition is different: the implementation switches between interpolated continuation and the specified stopping payoff, which need not agree away from boundary grid points. We call B *regular* when every cubature transition stays entirely inside, or entirely outside, each wealth stopping face. Only on such a box do we use derivative information. All intersected interpolation cells and both sides of reflection folds enter the derivative hull.

Suppose $G_j(B) = [g_j^-, g_j^+]$ encloses every one-sided partial derivative with respect to action component j on a regular box. The mean-value upper enclosure is

$$U^{\text{mv}}(B) = \overline{Q}_w(c) + \sum_j \max\{g_j^-(\ell_j - c_j), g_j^-(r_j - c_j), g_j^+(\ell_j - c_j), g_j^+(r_j - c_j)\}, \quad (\text{S.82})$$

where $\overline{Q}_w(c)$ is an outward point upper bound. The algorithm uses the smaller of this bound and $U^{\text{nat}}(B)$. If $g_j^- > 0$, it restricts the supremum to $a_j = r_j$; if $g_j^+ < 0$, it restricts it to $a_j = \ell_j$. These are verified reductions to a maximizing face, not decisions based on a sampled derivative. For boxes crossing a stopping switch, the natural union enclosure is retained without this reduction.

THEOREM S.8—Signed action cover: *Assume that the natural enclosures contain their respective payoffs, that the derivative hulls contain all one-sided derivatives on each regular box, and that feasible point values and arithmetic bounds are outward. Start from a cover of the action box and apply subdivision, reduction to a verified maximizing face, and removal of a box only when its upper bound is at most $L + \varepsilon$. At any interruption, retain the upper bound of every unresolved box. Then*

$$\sup_{a \in \mathcal{A}} Q_w(a) \leq \max\{L + \varepsilon, \max_{B \text{ unresolved}} U(B)\}. \quad (\text{S.83})$$

The bound remains valid when a depth, work, or time budget is exhausted. If r action coordinates have verified strict derivative signs throughout a regular parent box, maximization over that box reduces to a face of dimension at most $d_a - r$.

The last statement describes a source of computational improvement, not a dimension-free worst-case bound. Under a K -Lipschitz gradient, a regular stationary box of diameter s has mean-value excess of order Ks^2 when its point evaluation error is negligible. Boxes that cross stopping switches, have ambiguous derivative signs, or contain many interpolation cells can remain expensive. Our experiment measures their actual cost and keeps unresolved bounds; it does not assume that every box has favorable structure.

The operational NBO output is now explicit: the frozen policy representation, its guarded continuation vector when used, an independently propagated policy interval, and a complete-action optimal upper envelope. A nodal lookup policy is the object certified in the preference experiment. An off-grid interpolation of that policy is a different deployment rule and is evaluated separately. No online search is charged to a stored lookup table; constructing that table includes every actor, critic, and local-search query.

F.8. Cost, Refinement, and Economic Accuracy

The preceding applications define the original economic scope of NBO. We now evaluate the guarded map and signed cover on the same preference economy, and connect the high-dimensional experiment to an absolute economic benchmark. The comparison retains every earlier failure. In particular, a new upper envelope does not turn an actor-plus-search policy into an actor-only policy, and retraining on a finer grid does not repair the off-grid performance of a previously frozen policy.

F.8.1. A tighter certificate for identical frozen policies

The reference specification has 17×25 state nodes, twenty dates, and the continuous action box $[0.02, 0.30] \times [0, 1.5] \times [-0.45, 0.45]$. The ratios constrain consumption, leverage, and preference adjustment in the declared economy. They are economic primitives, not bounds on an unconstrained optimum. Active consumption or port-

folio limits remain meaningful features of this constrained specification; changing them changes the economy.

We freeze the nine previously computed actor-plus-search, direct-plus-search, and search-only policies, including their original seeds. The complete-action upper envelope is recomputed independently of these policies' lower payoff envelopes. The new cover uses a one-step tolerance of 0.0002 and retains every unresolved upper bound. Its output can be shared across policies only because their action set, transition rule, domain, and payoff normalization coincide.

TABLE F.IX

COMPLETE-ACTION BOUNDS FOR THE SAME FROZEN POLICIES

Policy	Seed	Uniform bound	Centre bound	≤ 0.1
Actor + search	11	0.0687	0.0068	Yes
Actor + search	29	0.0553	0.0050	Yes
Actor + search	47	0.0552	0.0075	Yes
Direct + search	11	0.0465	0.0071	Yes
Direct + search	29	0.0536	0.0052	Yes
Direct + search	47	0.0466	0.0075	Yes
Search only	11	0.0633	0.0113	Yes
Search only	29	0.0655	0.0053	Yes
Search only	47	0.0631	0.0076	Yes

Full 17×25 state grid, twenty dates, unchanged three-control box. The common signed cover took 372.53 seconds; 20 of twenty one-step covers closed at the requested tolerance. Other levels retain all unresolved upper bounds. Bounds are conditional on the stated outward-arithmetic execution model; continuous-state/time transfer is not included.

Table F.IX reports the resulting all-date, all-node loss bounds. The improvement relative to the historical table concerns the tightness of the verification calculation, not additional training or a change in preferences. The raw actors remain separately reported in the machine-readable common accounts. Search remains consequential, and these results do not establish an actor advantage over the actor-free comparator. They do show that the earlier failure of a 0.1 upper-bound target need not imply that the frozen hybrid policies themselves are more than 0.1 from the nodal optimum.

For a controlled cost comparison, Table F.X gives one-step corner-only and signed covers at three dates, with identical continuation arrays, initial feasible actions, tolerance, depth limit, and work budget. Wall time includes derivative hull construction for the signed cover. The point of this experiment is not to compare two historical machines. Historical training costs are retained in the supplementary total-cost accounts with that qualification. Shared verification is amortized under explicitly stated populations of one, nine, and one hundred policies, and the welfare conversion has a separately counted cost. No claim of neural speed advantage is inferred from this accounting.

TABLE F.X

MATCHED ONE-STEP ACTION-COVER COMPUTATION

Cover	Date	Seconds	Q boxes, m.	Bracket	Budget hit
Corner	19	18.67	0.414	0.01336	Yes
Signed	19	9.50	0.066	0.00020	No
Corner	10	19.89	0.440	0.02230	Yes
Signed	10	14.60	0.112	0.00020	No
Corner	0	18.32	0.402	0.03430	Yes
Signed	0	25.84	0.446	0.00173	Yes

Continuation, incumbent, tolerance, and budget are identical within each date. Wall time includes gradient work; a query count is not a hardware-independent speed measure.

F.8.2. *The whole-domain refinement failure and its repair*

A frozen policy has to be distinguished from a policy refitted on a new grid. The complete historical refinement arrays reveal large losses away from the centre. Their maximum positive reference advantages, locations, quantiles, boundary-region summaries, and action saturation rates are retained in the complete numerical archive indexed in Appendix N. The largest historical discrepancy is approximately 2.1234 at high risk aversion and wealth close to the lower stopping boundary. Centre values alone do not summarize that experiment. These discrepancies are feasible-reference lower bounds on policy regret, not upper bounds for the continuous economy.

The new refinement comparison uses genuinely nested spatial grids, 17×25 and 33×49 , with twenty and forty dates, respectively. Each declared actor and actor-free search case is trained afresh. The network width, activation, actor budget, critic budget, and seed set are unchanged across the relevant ablation. We compare the unguarded fitted critic with the indexed guard in (S.81). For the forty-date case the target tolerance is 0.0000625. Both the fraction of corrected nodes and their storage cost are reported, rather than being attributed to an uncorrected neural approximation.

TABLE F.XI
FRESH POLICIES ON THE NESTED FINE GRID

Method	Uniform bound range	Ref. advantage	Mean train s.	Passes
Actor	0.0654–0.0963	0.0932	88.32	3/3
Actor + guard	0.0140–0.0356	0.0155	87.75	3/3
Search	0.0655–0.0963	0.0932	51.07	3/3
Search + guard	0.0349–0.0357	0.0156	47.40	3/3

Full 33×49 state grid, forty dates; seeds 11, 29, and 47. Reference advantage is a lower regret diagnostic against the feasible 175-action reference, not an upper bound. Uniform bounds use a separately computed complete-action fine-grid envelope. The guard is a saved indexed correction, not an uncorrected network.

A new complete-action envelope is computed on the fine grid, not interpolated from the coarse certificate. It is applied to all twelve fine-grid policies, including the six unguarded runs. This separates three facts: the failure of a frozen coarse-grid policy; the accuracy achieved by retraining within a declared nodal economy; and the additional error required to transfer a result to a continuously monitored diffusion. The first remains in the record, the second is measured, and the third is not assigned a zero value. Two levels alone do not establish an asymptotic convergence rate.

The deployment specification is a stored state-time action table with its saved neural proposal and continuation correction. Evaluating this table at its nodes requires no online maximizing search. Interpolating the table at off-grid states invokes a different policy rule; the historical experiment shows why this distinction

1 matters. A raw actor without a table or local improvement has a different memory 1
 2 requirement and is not represented by the hybrid loss bounds. 2

3 4 F.8.3. *A model-specific welfare account* 4

5
6 An absolute utility gap is not a consumption-equivalent percentage when the 6
 7 utility exponent is itself a controlled state. We therefore compute a defined com- 7
 8 pensating transfer. Hold the saved policy and its state law fixed, leave preference- 8
 9 adjustment costs and the stopping bequest unchanged, and add externally financed 9
 10 consumption so that flow consumption becomes $(1 + e)c_t X_t$. This intervention 10
 11 changes felicity but does not charge its financing to wealth. It is a compensation 11
 12 experiment, not another admissible self-financed policy in the original box. 12

13 Let J_e^π be the compensated payoff and U the common optimal upper envelope. 13
 14 A verified lower envelope $J_e^\pi \geq U$ at every active starting node and date implies 14
 15 that this transfer compensates the policy's loss everywhere in the declared nodal 15
 16 economy. We search for a common e using monotonicity of CRRA felicity in con- 16
 17 sumption and verify the final inequality by outward propagation. Table F.XII gives 17
 18 the resulting upper percentages. These conservative transfers can be economically 18
 19 substantial even when the absolute utility bound is below 0.1; the threshold is not 19
 20 described as a universally small welfare loss. 20

21 22 F.8.4. *A global benchmark for the coupled capital economy* 22

23
24 A comparison of two frozen policies does not indicate how far either is from 24
 25 the economic optimum. We supplement the paired simulations by a bound on the 25
 26 original, continuous-time, dense nonlinear capital economy. It does not rely on 26
 27 a Riccati equation, separability of the optimal policy, a state mesh, or a sampled 27
 28 residual. 28

29 In the notation of the capital experiment, let Y_i be log capital and $m_i \in [\underline{m}, \bar{m}]$ the 29
 30 consumption ratio. The flow payoff is $\log \bar{m} + \bar{Y} - (\zeta/2)\bar{m}^2$, and the terminal payoff 30
 31 is $\bar{Y} - \chi(Y - \bar{Y})^2$, with $\chi = 0.03$. The drift of Y_i is $a + \kappa \tanh((BY)_i) - m_i - (\sigma_l^2 +$ 31
 32 $\sigma_c^2)/2$. Common noise affects the mean but cancels from cross-sectional dispersion. 32

TABLE F.XII

SUFFICIENT FINANCED FLOW-CONSUMPTION SUPPLEMENTS

Proposal	Seed	Supplement, percent	Verification seconds
Actor	11	20.93	27.42
Actor	29	19.17	26.86
Actor	47	18.87	26.94
Direct	11	15.33	27.06
Direct	29	18.06	28.34
Direct	47	15.33	27.50
Search	11	20.44	31.32
Search	29	22.46	31.35
Search	47	19.42	31.37

Uniform sufficient compensation over every active starting node and date. State dynamics, the original policy, terminal bequest, and adjustment cost stay fixed. Consumption is externally financed. These are not feasible self-financed policy improvements or common homothetic consumption equivalents.

Define

$$b_{\pm} = a \pm \kappa - \frac{1}{2}(\sigma_I^2 + \sigma_C^2), \quad w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)},$$

and let

$$m_0(t) = \text{clip}_{[m, \bar{m}]} \frac{2}{w(t) + \sqrt{w(t)^2 + 4\zeta}}.$$

This common, deterministic schedule is feasible in the unchanged economy.

PROPOSITION S.9—Continuous-economy performance benchmark: *Starting from $Y_i(0) = 0$ for every i , let V^* be the optimal expected payoff of the original d -dimensional capital economy. Define*

$$U = \int_0^T e^{-\rho t} \{ \log m_0(t) - \frac{\zeta}{2} m_0(t)^2 + w(t)[b_+ - m_0(t)] \} dt,$$

$$D_d = \{ \kappa T + \sigma_I \sqrt{(1 - 1/d)T} \}^2,$$

$$L_d = \int_0^T e^{-\rho t} \{ \log m_0(t) - \frac{\zeta}{2} m_0(t)^2 + w(t)[b_- - m_0(t)] \} dt - \chi e^{-\rho T} D_d.$$

Then $L_d \leq J(m_0) \leq V^* \leq U$ for every admissible dense coupling matrix B in the stated model. The inequalities apply to the continuous economy on $\mathbb{R}^d$, not merely to an Euler simulation.

The bound exploits bounded nonlinear drift, concavity of consumption utility, and an explicit feasible policy. Its numerical integration uses outward rectangle sums, interval exponential and logarithm evaluations, and square roots checked by interval squaring. Table F.XIII reports the computed brackets. Their width is economically material and is not presented as a small regret bound for a learned policy. They nevertheless provide an absolute benchmark absent from a paired-policy comparison. Simulations of m_0 use the same Brownian paths as the saved neural policies. The resulting differences and confidence intervals concern their common Euler discretization; an unknown simulation bias is not subtracted from the analytical bound to create a neural-policy certificate.

TABLE F.XIII

GLOBAL BOUNDS IN THE ORIGINAL CONTINUOUS CAPITAL ECONOMY

Dimension	Time panels	Feasible lower	Optimal upper	Width
10	4096	-1.489064	-1.051803	0.437261
10	16384	-1.488883	-1.051998	0.436884
20	4096	-1.489130	-1.051803	0.437327
20	16384	-1.488949	-1.051998	0.436950

Initial log capital is zero in every coordinate; horizon one. The bounds cover every admissible control on $\mathbb{R}^d$. They are not small loss bounds for neural policies. The same feasible schedule is evaluated on the saved Brownian paths in the supplement.

F.9. Benchmark Derivations

F.9.1. Merton consumption and transversality

Write $b = 1 - \gamma$ and $V = KX^b/b$. Then $V_X = KX^{-\gamma}$ and $V_{XX} = -\gamma KX^{-\gamma-1}$. Substituting $c = mX$ into the stationary time-additive equation and dividing by the

positive number KX^b gives

$$0 = \frac{m^b}{bK} - \frac{\rho}{b} + r + p(\mu - r) - m - \frac{1}{2}\gamma p^2 \sigma^2.$$

Consumption maximization gives $m^{-\gamma} = K$; portfolio maximization gives $p = (\mu - r)/(\gamma\sigma^2)$ when interior. With a no-short-sale constraint and negative excess return, the maximizing share is zero. Let $R_* = r + p(\mu - r) - \gamma p^2 \sigma^2 / 2$. After substitution, $\gamma m = \rho + (\gamma - 1)R_*$.

For the optimal constant shares, $e^{-\rho t} \mathbb{E}[X_t^{1-\gamma}]$ decays as e^{-mt} . Thus $m > 0$ gives the candidate's additive transversality condition. Verification against a larger policy class also requires the relevant localization and integrability conditions for the competitors, as specified in the main text. Policies with minus-infinite utility are not preferred to the finite candidate. The numerical laboratory checks the candidate formulas and does not silently apply a stationary constant to an unspecified finite-horizon terminal problem.

F.9.2. Recursive coefficient and the utility interval

For $f(c, V) = \rho[c^a(bV)^{1-a/b} - bV]/a$, the same homothetic substitution gives

$$0 = \frac{\rho}{a}[m^a K^{-a/b} - 1] + R_* - m.$$

Since $f_c = V_X$ implies $\rho m^{a-1} K^{-a/b} = 1$, the first term becomes $(m - \rho)/a$. Consequently, $m = \psi\rho + (1 - \psi)R_*$ and $K = [\rho m^{a-1}]^{b/a}$. The formula assumes the stationary candidate has positive consumption. It is not an infinite-horizon utility-selection theorem.

For the finite-horizon matching-bequest test at $a > 0 > b$, define $Y = bV$. The consumption domain is compact and bounded away from zero. At sufficiently small positive Y , $c^a Y^{-a/b} < 1$ uniformly in consumption, so $f < 0$. At sufficiently large Y , the reverse inequality holds and $f > 0$. Because dividing by negative b reverses the order of the utility endpoints, the large- Y constant is a lower utility barrier and the small- Y constant is an upper utility barrier. The matching bequest lies between them after enlarging the interval. Lipschitz scalar comparison then keeps each policy utility inside the interval, on which the aggregator and its utility derivative

are bounded. This verifies the finite-horizon interpretation without claiming an unrestricted infinite-horizon result at those parameters.

F.10. *Endogenous-Preference Details*

F.10.1. *State restrictions and the reference experiment*

In the baseline application, the risk-aversion interval lies strictly above one and the process is reflected. The reference wealth problem is stopped at positive finite bounds. Consumption shares are bounded away from zero, adjustment efforts are bounded, and the horizon is finite. These choices bound the flow utility, its relevant derivatives away from the terminal-boundary compatibility issue, and the bequest. They provide a well-defined finite reference calculation. The numerical reflection is a folded reflection of Brownian cubature endpoints; the wealth boundary is absorbing with the prescribed payoff. The same terminal and boundary values are imposed for both adjustment-cost parameters.

The grid reference operates in $y = \log X$. Its drift is

$$b_y = r + p(\mu - r) - m - \frac{1}{2}p^2\sigma_X^2,$$

and its covariance with u is $\varrho\sigma_u p\sigma_X$. The two Brownian-factor columns can be written $(\sigma_u, \varrho p\sigma_X)^\top$ and $(0, \sqrt{1 - \varrho^2}p\sigma_X)^\top$. Four symmetric cubature directions reproduce this covariance. The correction $-p^2\sigma_X^2/2$ in log wealth is essential; omitting it would change the economic diffusion.

For each action, the backward update is current utility times the time step plus the exponentially discounted positive-weight interpolation of next-period value. Feasible actions are enumerated on the declared 27-element grid. Because the terminal/exit payoff is fixed, raising k weakly lowers each action's payoff and, by backward induction, lowers value at every state and time. The recorded arrays permit a direct check of this order. The reported grid-refinement difference is not an estimate of unrestricted neural approximation error.

F.10.2. Policy conditions and comparative statics

The consumption marginal utility in reference units is $U_c = \bar{c}^{-1}(c/\bar{c})^{-u}$. Equating it to V_X yields $c = \bar{c}(\bar{c}V_X)^{-1/u}$. Adjustment maximizes $\theta V_u - k\theta^2/2$ and hence uses the projection of V_u/k under a box. Portfolio choice maximizes a quadratic with coefficient $X^2\sigma_X^2 V_{XX}/2$. The projected first-order solution is valid only when this coefficient is negative. These conditions clarify why numerical curvature and feasibility must be checked independently.

The cross derivative V_{uX} is an endogenous object. Its sign cannot be inferred from the label “risk aversion” or from a negative shock correlation. Nor can the sign of $\partial_k \theta^k$ be inferred by differentiating $1/k$ while keeping V_u^k fixed. In contrast, the value monotonicity in k holds policy by policy. At a differentiable optimum the envelope formula follows because k appears only in the quadratic running cost. Changing the normalization function $\kappa(u)$ can change both adjustment and portfolio policies even while leaving utility curvature in consumption unchanged.

An observation model is needed for structural estimation. With latent u , uncertainty in κ , and unknown adjustment or shock parameters, a sensitivity plot for one policy component does not prove identification. A practical estimation exercise would compute moment derivatives jointly, inspect their rank after normalization restrictions, and propagate numerical approximation error into the implied moment map. Those are distinct tasks from solving the controlled economy.

F.11. Temporal-Self Deviations

For the two-exponential kernel, the component value under constant consumption share m is $W_j(X) = m^b X^b / [b\lambda_j(m)]$, where $\lambda_j(m) = \rho_j - b(r - m)$. A deviation to share z for duration ϵ , followed by m , has component value at $X = 1$

$$J_j(z, \epsilon) = \frac{1}{b} \left[z^b \frac{1 - e^{-\lambda_j(z)\epsilon}}{\lambda_j(z)} + e^{-\lambda_j(z)\epsilon} \frac{m^b}{\lambda_j(m)} \right].$$

The continuous extension is used if $\lambda_j(z) = 0$. The deviation gain is $\sum_j w_j J_j(z, \epsilon) - \sum_j w_j W_j(1)$, with weights $(\beta, 1 - \beta)$. This is the quantity independently maximized in the experiment, rather than a restatement of the equilibrium first-order condition.

At an equilibrium share, the first-order gain is nonpositive for every feasible instantaneous share and equals zero at the equilibrium share. A finite-duration self can generally choose a slightly different share, so the maximized gain need not be exactly zero for positive duration. The required diagnostic is its gain divided by duration as duration tends to zero. For $\beta = 1$ the ordinary exponential solution is recovered.

As κ grows, the kernel becomes increasingly concentrated near the present before approaching its long-run component. Taking the instantaneous-deviation limit and taking $\kappa \rightarrow \infty$ are conceptually different operations. The finite- κ model in this paper explicitly specifies the order: the spike condition is derived for fixed κ , and any duration-limit comparative static is taken afterward. This avoids assigning an undefined continuous-time meaning to a discrete present-bias multiplier.

F.12. *Games and Independent Best Responses*

A fixed rival profile turns each player's problem into an ordinary controlled Markov problem, so the main verification results apply player by player. The derivative graph must implement this reduction. During player i 's actor step, every rival policy and every derivative of player i 's critic is detached. During player i 's critic step, all players' actions are detached. A sum of these reporting losses can be logged, but differentiation of a joint-profit objective is not equivalent to those block steps.

The two-firm static diagnostic has best response $q_i = (1 - q_j)/2$ when interior. At the joint-profit allocation $(1/4, 1/4)$, the deviating firm chooses $3/8$ and improves profit from $1/8$ to $9/64$, a gain of $1/64$. The own-action derivative is $1/4$ at the joint optimum. These numbers verify a graph-level distinction; they are not a computation of the dynamic capital game.

For the dynamic problem, a reported equilibrium should include an independent solution of each fixed-rival best-response problem, or an upper bound on the unilateral Hamiltonian gap over the entire verification domain. A finite list of unsuccessful deviations does not supply such an upper bound. Multiple initializations, asymmetric profiles, boundary actions, and the market-size normalization are

part of the same experiment. The specification in the main text retains the dynamic Cournot application but does not invent a dynamic Nash table from the static arithmetic.

F.13. *Proof and Implementation of the Continuous-Action Envelope*

We use the notation of the main text's budget-safe envelope proposition. The nodal economy is fixed before verification. Its terminal and stopping payoffs, primitive coefficients, interpolation abscissae, and compact action box are part of the object being bounded. This is stronger than checking many action samples and weaker than verifying the continuous-state diffusion without a discretization account.

F.13.1. *Complete-cover invariant and backward proof*

At a given state and time, the initial action box covers the entire feasible set. Bisection replaces one closed box by two closed boxes with the same union. Their shared face is harmless. Induction on the number of subdivisions preserves coverage. A pruned box is retained logically through the inequality $u_B \leq c_i + \tau$, where u_B is an outward upper enclosure on that whole box, and c_i is a lower enclosure at a feasible incumbent. If the incumbent later increases, the previous pruning inequality remains true. Consequently every feasible action is either in an unresolved box or in a pruned box with payoff at most the current threshold. Taking the maximum of the threshold and all unresolved upper bounds dominates the action supremum, whether or not the supremum is attained. Termination on a numerical budget does not change this argument.

At N , the terminal inequalities are given. If $V_{n+1}^* \leq U_{n+1}$, positivity gives $T_n V_{n+1}^* \leq T_n U_{n+1} \leq U_n$. If $L_{n+1} \leq V_{n+1}^\pi$, the same argument gives $L_n \leq T_n^{\pi_n} L_{n+1} \leq T_n^{\pi_n} V_{n+1}^\pi = V_n^\pi$. Feasibility implies $V_n^\pi \leq V_n^*$. This proves the main envelope inequality. Substochasticity further gives

$$\|T_n w - T_n z\|_\infty \leq q_n \|w - z\|_\infty.$$

Applying this inequality to $U_n - T_n V_{n+1}^*$ yields the stated excess recursion and its product-sum expansion. The proof uses $q_n > 0$, not a stationary contraction assumption.

tion. Outward addition is used for the pruning threshold and final subtraction. The implementation records a requested tolerance and a retained optimization bracket separately. A budget-exhausted bracket is never replaced by that tolerance. *Q.E.D.*

F.13.2. *Interpolation, reflection, stopping, and arithmetic*

For a cell with coordinates $(u_0, u_1) \times (y_0, y_1)$, the continuation is

$$Iw(u, y) = (1 - s)(1 - t)w_{00} + s(1 - t)w_{10} + (1 - s)tw_{01} + stw_{11},$$

$$s = \frac{u - u_0}{u_1 - u_0}, \quad t = \frac{y - y_0}{y_1 - y_0}.$$

For fixed u , this is affine in y , so its extremes on a subrectangle occur at the two y endpoints. At either endpoint it is affine in u , so the extremes occur at the two u endpoints. Applying this argument to every intersected cell proves the corner enclosure. Nonnegative interpolation weights also permit interval nodal inputs: evaluating the corner expressions outward encloses every admissible nodal realization. A separate valid enclosure evaluates the rectangle center and adds cell-slope bounds over every intersected cell. Slopes are computed from stored nodal differences, not estimated from random derivative samples.

Preference reflection is piecewise affine. Its extrema on an interval occur at the interval endpoints or a crossed folding point. The allowed one-step movement is checked to exclude unhandled multiple folds. For log wealth, boxes fully inside the domain use interpolation, boxes fully outside use the prescribed clipped exit payoff, and boxes crossing an exit face use the union of both. On the stored grid the boundary nodes equal the exact prescribed payoff; at an off-grid exit the analytic payoff need not equal its nodal interpolant. The union construction avoids an unjustified global smoothness or Lipschitz assumption across that switch.

The implementation inherits the archived outward elementary and transcendental routines. Products, quotients, powers through logarithm and exponential, and endpoint rounding are enclosed. The logarithm's input is bounded below by 0.02; the preference denominator is separated from zero because $u \in [1.2, 3]$. The recorded control model treats the stored binary64 primitive coefficients and abscissae as exact real inputs. The evidence also tests agreement with the inherited

training evaluator on the finite action grid. Such a regression check is useful for implementation consistency but is not the proof of global action coverage. That proof is the maintained union invariant and the interval inclusion property. The arithmetic account is conditional on the specified floating-point execution model, not a formal verification of hardware, libraries, or the Python interpreter.

F.13.3. *A common upper envelope for multiple policies*

The optimal upper array depends on the economy, not on which feasible incumbent produced the lower search values. Thus the completed upper array can be reused for any frozen feasible policy on the same state, time, and continuous-action domains. We separately propagate each policy's lower and upper payoff arrays and subtract its lower array from the common optimal upper array. Costs are reported both as shared verification and as the cost of an individual policy evaluation. Sharing a verifier does not make its cost vanish. It also does not justify transferring the upper array to a changed wealth box, action constraint, utility normalization, time step, or interpolation rule.

F.14. *Separating Numerical and Continuous-Economy Error*

The following result specifies the transfer work needed in addition to a nodal certificate. It is not used to set an uncomputed transfer term to zero. Let $\mathcal{T}_n$ be the exact economic one-step optimality operator with its continuously monitored stopping and reflection rule, and let $\mathcal{T}_n^\pi$ be the operator for a declared continuous-state implementation of the numerical policy. Let I_n be a positive, sup-norm-nonexpansive interpolation from grid values to the economic state domain. Suppose the exact operators are q_n -Lipschitz on the comparison class.

PROPOSITION S.10—Transfer with separate optimal and policy defects: *Suppose terminal interpolation error is at most B and, along the relevant nodal continuations,*

$$\begin{aligned} \|\mathcal{T}_n I_{n+1} v_{n+1}^* - I_n \mathcal{T}_n v_{n+1}^*\|_\infty &\leq d_n^*, \\ \|\mathcal{T}_n^\pi I_{n+1} v_{n+1}^\pi - I_n \mathcal{T}_n^\pi v_{n+1}^\pi\|_\infty &\leq d_n^\pi. \end{aligned}$$

Define $D_N^* = D_N^\pi = B$ and $D_n^j = d_n^j + q_n D_{n+1}^j$ for $j \in \{*, \pi\}$. Then the exact economic values satisfy

$$0 \leq \mathcal{V}_n^* - \mathcal{V}_n^\pi \leq I_n(U_n - L_n) + D_n^* + D_n^\pi.$$

The action-domain approximation belongs in d_n^* , while policy implementation, boundary monitoring, state interpolation, and time/cubature errors may affect both defects.

PROOF: Insert $\mathcal{T}_n I_{n+1} v_{n+1}^*$ between the exact and interpolated nodal Bellman values. The Lipschitz property bounds the propagated value difference by $q_n D_{n+1}^*$, and the displayed defect bounds the remaining difference by d_n^* . Backward induction gives $\|\mathcal{V}_n^* - I_n v_n^*\| \leq D_n^*$. The same argument gives the fixed-policy bound. Subtract the two decompositions and use positivity of I_n and $v_n^* - v_n^\pi \leq U_n - L_n$. Feasibility gives the lower inequality. Q.E.D.

The proposition can use local regularity, coupling, barrier, or weak-approximation estimates to bound its defects; it does not claim that those hypotheses hold automatically for a fitted critic near a stopped boundary. In the present two-state computations, action coverage is operationally bounded on the nodal economy. The continuous-state/time defects remain explicitly uncomputed. Nested grids and common frozen policies diagnose sensitivity but do not provide the one-sided inequalities required by this proposition.

TABLE F.XIV: Frozen continuous-policy refinement, common economic domain

Method	Seed	Grid	Steps	Center payoff	Center reference
Actor	11	17×25	20	-4.843442	-4.841008
Actor	11	25×37	40	-4.848086	-4.839462
Actor	11	33×49	80	-4.842578	-4.830165
Actor	29	17×25	20	-4.841652	-4.841008
Actor	29	25×37	40	-4.846754	-4.839462
Actor	29	33×49	80	-4.840763	-4.830165
Actor	47	17×25	20	-4.844135	-4.841008

Method	Seed	Grid	Steps	Center payoff	Center reference
Actor	47	25×37	40	-4.848828	-4.839462
Actor	47	33×49	80	-4.843387	-4.830165
Direct	11	17×25	20	-4.843773	-4.841008
Direct	11	25×37	40	-4.857445	-4.839462
Direct	11	33×49	80	-4.852920	-4.830165
Direct	29	17×25	20	-4.841889	-4.841008
Direct	29	25×37	40	-4.854488	-4.839462
Direct	29	33×49	80	-4.848755	-4.830165
Direct	47	17×25	20	-4.844185	-4.841008
Direct	47	25×37	40	-4.854178	-4.839462
Direct	47	33×49	80	-4.848680	-4.830165

Notes. Policies are frozen on the original time schedule and bilinearly interpolated in state. References still use 175 actions. Differences are sensitivity diagnostics, not continuous-state/time upper bounds.

F.15. *Strategic Deviations and Missing Pure Stages*

For a fixed Markov rival profile in a finite-horizon finite-state/action game, condition on any history. The own player's conditional payoff from a randomized current action is a convex combination of pure-action payoffs with the same continuation best-response values. It cannot exceed the largest such payoff. Backward induction therefore bounds every randomized and history-dependent unilateral policy by the Markov pure-action best-response recursion. Outward propagation then provides a bound over this full deviation class, not merely the particular local deviations observed during actor training.

Existence of a pure equilibrium in a fitted one-step game is a different matter. If none exists, minimum-defect selection produces a feasible profile, not an exact equilibrium. The independent dynamic best response still evaluates that profile correctly. The following records give every missing pure stage in the six principal finite-policy study neural runs, including time index, state, and minimum maximum

stage defect. Indices begin at zero. The six runs have $6 \times 30 \times 225 = 40,500$ active state-time nodes; the five recorded exceptions are not excluded from their final dynamic bounds.

TABLE F.XV

EVERY MISSING PURE FITTED ONE-STEP EQUILIBRIUM IN FINITE-POLICY STUDY

Market	Seed	Time index	K_1	K_2	Minimum defect
1	11	1	0.140625	0.865625	6.492e-06
1	11	1	0.684375	0.140625	4.018e-07
1	11	10	0.321875	0.140625	1.538e-05
2	11	16	0.231250	1.046875	1.534e-05
2	29	13	0.140625	0.956250	1.394e-06

Minimum defect is the smallest maximum unilateral one-step improvement over all joint finite actions. These nodes remain in full dynamic verification; they are not called exact stage equilibria.

The profit targets are absolute units of the displayed per-firm payoff. Raw network policy, shortlist policy, and corrected finite lookup policy are different deployment objects. The principal finite-policy study guard fraction is zero, but the full joint-action oracle was nevertheless constructed during training. A mixed-strategy *profile* was not estimated; the preceding argument explains why mixed unilateral deviations against the fixed pure profile are already covered by the finite best-response bound.

F.16. Paired Payoffs, Costs, and Economic Scaling

Each row below compares the two frozen methods on the same stored Brownian paths, at the indicated dimension, seed, fitting domain, and Euler resolution. Intervals are pointwise Student intervals across 512 pathwise differences. They are not independent across step sizes, because the Brownian increments are nested; no simultaneous-coverage claim is made. The narrow and wide policies were separately fitted, so their comparison also changes the training domain and update budget.

TABLE F.XVI: All method-paired policy-payoff comparisons

Dim.	Seed	Domain	Steps	NBO – direct	Lower	Upper
10	11	Narrow	40	-0.03043	-0.03234	-0.02852
10	11	Narrow	80	-0.03237	-0.03438	-0.03036
10	11	Narrow	160	-0.03338	-0.03544	-0.03133
10	11	Wide	40	0.00061	0.00058	0.00064
10	11	Wide	80	0.00055	0.00052	0.00057
10	11	Wide	160	0.00052	0.00049	0.00054
10	29	Narrow	40	-0.02800	-0.02946	-0.02655
10	29	Narrow	80	-0.02993	-0.03146	-0.02840
10	29	Narrow	160	-0.03094	-0.03250	-0.02938
10	29	Wide	40	-0.00007	-0.00008	-0.00006
10	29	Wide	80	-0.00004	-0.00005	-0.00003
10	29	Wide	160	-0.00003	-0.00004	-0.00002
10	47	Narrow	40	-0.02522	-0.02695	-0.02349
10	47	Narrow	80	-0.02675	-0.02856	-0.02493
10	47	Narrow	160	-0.02755	-0.02941	-0.02569
10	47	Wide	40	0.00004	0.00003	0.00005
10	47	Wide	80	0.00001	0.00001	0.00002
10	47	Wide	160	0.00000	-0.00001	0.00001
20	11	Narrow	40	-0.05798	-0.06054	-0.05543
20	11	Narrow	80	-0.06131	-0.06396	-0.05866
20	11	Narrow	160	-0.06302	-0.06572	-0.06033
20	11	Wide	40	0.00123	0.00118	0.00128
20	11	Wide	80	0.00097	0.00092	0.00101
20	11	Wide	160	0.00084	0.00080	0.00088
20	29	Narrow	40	-0.01308	-0.01346	-0.01269
20	29	Narrow	80	-0.01401	-0.01442	-0.01361
20	29	Narrow	160	-0.01449	-0.01490	-0.01408
20	29	Wide	40	0.00480	0.00463	0.00496
20	29	Wide	80	0.00432	0.00417	0.00448
20	29	Wide	160	0.00409	0.00394	0.00424
20	47	Narrow	40	-0.19715	-0.19765	-0.19666

Dim.	Seed	Domain	Steps	NBO – direct	Lower	Upper
20	47	Narrow	80	-0.20011	-0.20061	-0.19962
20	47	Narrow	160	-0.20161	-0.20211	-0.20111
20	47	Wide	40	0.00084	0.00078	0.00089
20	47	Wide	80	0.00066	0.00061	0.00071
20	47	Wide	160	0.00058	0.00053	0.00062

Notes. Pointwise 95 percent Monte Carlo intervals, 512 paired paths per row. They do not bound optimality, continuous-time bias, or simultaneous coverage.

The additional representation comparisons below use the identical NDU finite model. Gated regression has more parameters and higher per-update work than the width-32 tanh critic. Tensor projection exploits low state dimension. Both facts are explicit rather than hidden by matching only nominal iteration counts.

TABLE F.XVII

ADDITIONAL CONTINUATION REPRESENTATIONS, IDENTICAL FINITE PREFERENCE ECONOMY

Method	Seed	Parameters	Policy bound	Value error	Train (s)	Verify (s)
Chebyshev	11	121	0.0121	0.0775	0.14	4.41
Chebyshev	11	225	0.0039	0.0165	0.15	4.48
Chebyshev	11	49	0.0373	0.2325	0.14	4.36
Gated	11	9345	0.0079	0.0284	41.28	4.12
Gated	29	9345	0.0098	0.0200	42.59	4.49
Gated	47	9345	0.0088	0.0306	38.99	4.30

Gated denotes a DGM-style architecture fitted to finite Bellman targets, not a differential DGM solver. Chebyshev degrees are 6, 10, and 14. All actions are enumerated; construction and reference costs are separately retained in JSON. Deterministic projections are not replicated under fictitious random seeds.

The finite-policy study NDU actor's correction map stores a small number of changed action indices, but its complete policy table occupies 68,000 bytes with the archived int64 representation. A minimal pair of int64 state-time and replacement indices requires 16 bytes per correction, excluding indexing and container

overhead. A stored-table deployment avoids actor evaluation at existing nodes; a raw-network deployment requires its network weights and argmax calculation. Neither observation establishes off-grid policy or guard correctness. The continuous hybrid additionally requires local action queries using the stored continuation representation.

In the continuous-action study, training-query counts exclude verification queries by construction. All-action references are created only after policies and critics have been frozen. The raw actor, direct-gradient, multi-head, and smoke failures are retained with their full configurations. The complete action verifier’s tighter and looser implementations have separate records, including unresolved boxes and declared budgets. A passing regression suite means that arithmetic, scope, and accounting checks passed; it does not change a failed economic accuracy target into a success.

TABLE F.XVIII

RETAINED CONTINUOUS-ACTION DEVELOPMENT FAILURES

Configuration	Seed	Deviation gain	Train (s)	Target 0.1
Actor smoke	11	0.1872	0.53	Fail
Raw actor	11	0.2596	15.77	Fail
Four-head reset	11	1.1033	16.83	Fail
Direct gradient	11	0.1997	16.07	Fail

All executed pilot sources, raw arrays, and histories remain. The smoke grid and budget differ from the principal grid; it is not a principal-seed replicate. Pilot selection and subsequent improvements are explicitly post-review development.

F.16.1. Complete configuration records

The principal application comparisons above retain the whole-domain failure, fresh-grid comparison, policy correction costs, and consumption compensation. The complete configuration tables additionally report every boundary-region summary, action-saturation rate, training seed, correction record, and capital payoff pair. They remain in the exact numerical archive identified in Appendix N; the editorial map

records the original table labels and source lines. A policy refitted on a nested grid remains a different object from a frozen policy evaluated off grid. None of those negative results is replaced by the successful comparison at the center of a state domain.

APPENDIX G: STOCHASTIC HESSIAN OBJECTIVES AND COMPLEXITY

Let $B = \Sigma^\top D^2 v \Sigma$, a symmetric matrix in Brownian-factor space. For Gaussian probes, Isserlis' identity gives $\mathbb{E}(z^\top B z) = \text{tr } B$ and $\text{Var}(z^\top B z) = 2 \text{tr}(B^2) = 2 \|B\|_F^2$. Averaging K independent probes divides the variance by K . Rademacher probes instead have variance $2 \sum_{i \neq j} B_{ij}^2 / K$. The variance depends on the chosen covariance factor and probe law, both of which must be specified.

Write $\hat{q}_j = q + \epsilon_j$, with conditionally mean-zero independent errors. Then

$$\mathbb{E}[(d - \hat{q}_1/2)(d - \hat{q}_2/2)] = (d - q/2)^2 + \frac{1}{4} \mathbb{E}[\epsilon_1 \epsilon_2] = (d - q/2)^2.$$

Using one bank twice would retain its variance term. In the scalar Gaussian example, a single-bank squared residual has expectation $1 - h + 3h^2/4$, minimized at $h = 2/3$, whereas the exact residual $(1 - h/2)^2$ is minimized at $h = 2$. The independent product restores the latter population objective. The saved Monte Carlo diagnostic uses 400,000 pairs and records both empirical estimates; it does not claim that the sampling error is identically zero.

The core implementation uses exact Hessians to make the covariance contraction independently testable. A Hessian-vector implementation can replace it without changing the conditional-independence requirement. The cost of a Hessian-vector product depends on the network graph, not just the state dimension. Total cost also includes sampling, optimization, domain coverage, action maximization, and stopping at an agreed error level. A quadratic-neural representation with a Lyapunov solve has different approximation and optimization costs from a generic multilayer network. The quadratic laboratory is explicitly the former; the new dense nonlinear experiments actually train generic networks and Hessian-vector estimators.

G.1. *Implemented Error Accounts and Arithmetic*

The implemented map returns a candidate and a verification record, rather than assigning authority to an optimizer’s exit code. In the differential consumption economy, the record identifies the binary64 weights, both stopping faces, all dyadic leaves, global consumption maximization, and the error bounds for the continuous diffusion. In NDU it instead identifies the finite state–action–time model, its positive transition weights, and time-indexed evaluation and improvement errors. The dynamic-game record fixes the entire rival policy before an independent backward best-response solve. An unchecked continuous-action or time-discretization term is represented as unavailable in the machine-readable output, not as zero.

For finite time levels, let $\delta_t = \|v_t - T_t^a v_{t+1}\|_\infty$ and $\epsilon_t = \|T_t v_{t+1} - T_t^a v_{t+1}\|_\infty$. If both maps are β_t -Lipschitz, the triangle inequality gives

$$\|v_t - V_t^a\| \leq \delta_t + \beta_t \|v_{t+1} - V_{t+1}^a\|, \quad \|v_t - V_t^*\| \leq \delta_t + \epsilon_t + \beta_t \|v_{t+1} - V_{t+1}^*\|.$$

Summing the two inequalities yields the regret account. The same recurrence with a player’s unilateral gap bounds the gain from a dynamic deviation when that player’s policy evaluation is exact. Accounts must be checked at each time level: a time-zero bound need not dominate a later-time bound when discounting and errors differ. The coarse-game regression exercises this distinction and verifies that a missing pure equilibrium is reported rather than silently treated as exact.

G.1.1. *Full-domain interval implementation*

An interval $[l, u]$ encloses an exact real quantity. Each elementary operation expands its rounded endpoints with the adjacent binary64 number. Affine layers split positive and negative weights and include a conservative dot-product forward-error allowance. The arithmetic contract requires binary64 round-to-nearest arithmetic, the stated dot-product error bound, and absence of exceptional overflow or flush-to-zero behavior. Stored parameters denote their exact binary64 real values; the verifier does not assert an unrecorded decimal model identity.

For an exponential argument x , range reduction gives $r = x/2^s$ with $|r| \leq 1/8$. A degree-fourteen Taylor polynomial has remainder bounded by

$$e^{|r|}|r|^{15}/15! < 2(1/8)^{15}/15! < 2^{-80}.$$

Interval Horner evaluation and repeated squaring enclose e^x . The identity $\tanh x = 1 - 2/(e^{2x} + 1)$ then supplies $\tanh$ values and derivative channels $1 - \tanh^2 x$ and $-2 \tanh x(1 - \tanh^2 x)$. Powers of two and the range-reduction domain are checked in the code.

For logarithms, exact binary decomposition gives $x = 2^k m$, with $1 \leq m < 2$. Writing $z = (m - 1)/(m + 1)$,

$$\log m = 2 \sum_{j=0}^{24} \frac{z^{2j+1}}{2j+1} + R, \quad |R| \leq \frac{2|z|^{51}}{51(1-z^2)} < 2^{-72}$$

for the outward-enclosed range $|z| < 0.334$. The same expansion encloses $\log 2$. Neither construction presumes correctly rounded library $\tanh$ or logarithm functions. The saved test suite checks transcendental endpoints against eighty-digit arithmetic, differentiates the trained network independently, and checks all certificate hashes and leaf adjacencies. These tests supplement, but do not replace, the arithmetic contract.

The scalar action objective $h(c) = \log c - pc$ has second derivative $-1/c^2$. If C encloses both the actor and its maximizing alternative, strong concavity on their joining segment gives

$$h(c^*) - h(c) \leq h'(c)(c^* - c) - (c^* - c)^2/(2C^2) \leq C^2 h'(c)^2/2.$$

The argument includes constrained endpoints. Taking the minimum with the natural interval action-gap enclosure markedly reduces verification cost, without using a sampled gradient as a bound. Residual targets are decided on every leaf; reaching a maximum subdivision depth does not change a failed target into a pass.

G.1.2. *Application-specific hypotheses*

The stopped consumption diffusion is uniformly nondegenerate on its compact domain; controls and coefficients are bounded and its endpoint data are continuous.

The smooth networks and strictly positive consumption map meet the regularity and log-domain requirements of the verification argument. Both stopping-face errors and the full action interval are checked. The upwind reference error is not a premise of its continuous certificate.

The NDU computation is a positive-weight, finite-action, finite-horizon economy with prescribed discrete stopping and folded preference transitions. Positivity, all terminal and wealth-face values, and every finite action are evaluated. The continuous reflected PDE's corner compatibility, continuous-action coverage error, and diffusion approximation error are not inferred from these tests. Expanding a feasible box changes the constrained economy; densifying a fixed box does not. The sensitivity record keeps these operations separate.

The dense capital computation uses bounded nonlinear drift in log coordinates, bounded positive consumption fractions, constant diffusion, and finite horizon. Its flow has at most linear log-state growth and its terminal payoff at most quadratic growth; these are integrable under the declared bounded-drift diffusion. The fitted box is a sampling domain, not a state constraint. Its sampled differential residuals do not check comparison barriers or uniform errors on $\mathbb{R}^d$. Euler payoff estimates carry both sampling error and unbounded-state/time-discretization issues not covered by the reported Monte Carlo standard error.

The dynamic capital game is a fully enumerated finite approximation with stopped joint boundaries. Stage equilibria and dynamic fixed-rival best responses are numerical statements about that model. The continuous-control best-response comparison assumptions in the main theorem remain logically distinct. The retained homothetic recursive-utility and temporal-self examples use their stated utility domains, bequests, and integrability conditions; initial evaluation study does not relabel them as unrestricted neural experiments.

APPENDIX H: OBSERVATION AND IMPLEMENTATION

H.1. *Observation and the Economic Implementation*

The capital state and the innovation-driven numerical controller are distinct. We now specify an observation model under which the controller uses the history of

observed capital rather than latent economic shocks. The model assumes continuous noiseless observation of every capital coordinate, known diffusion and drift coefficients, knowledge of the controller's own actions, and independent private randomization. It does not assume that a finite set of noisy observations reveals Brownian increments.

Write $\Sigma = [\sigma_I I_d, \sigma_C \mathbf{1}]$, $Q = \Sigma \Sigma^\top$, and let n be a unit vector spanning the one-dimensional null space of Σ . Since $\sigma_I > 0$, Q is invertible. The observed state path and known actions determine

$$M_t = Y_t - Y_0 - \int_0^t \{c\mathbf{1} + f(Y_s) - a_s\} ds.$$

Let B^0 be a scalar Brownian motion independent of the economic shocks and initial state, used only for private randomization.

PROPOSITION S.11—Implementation from observed state history: *Define*

$$\tilde{W}_t = \Sigma^\top Q^{-1} M_t + n B_t^0. \quad (\text{S.84})$$

Then $\tilde{W}$ is a $(d + 1)$ -dimensional Brownian motion in the enlarged observation filtration and $\Sigma \tilde{W}_t = M_t$. Feeding its cell increments into the prescribed numerical controller, including the componentwise clipping rule, yields the same closed-loop law and payoff as the innovation-driven controller. Thus its existing transfer and policy-specific performance statements apply under this observation model.

Private randomization in (S.84) is essential for this exact equivalence: the right-inverse term alone has rank d , whereas the implemented clipping acts separately on $d + 1$ drivers. The construction does not recover the original unobservable null component path by path. It replaces that component by an independent one with the correct law. Nor is the resulting controller a deterministic function of the state at a single decision date; its memory and independent randomization are part of its definition. The integrated drift in the proposition is an observation-model primitive. A sensor grid, drift quadrature or measurement error would need a further transfer allowance and is not silently treated as exact.

1 H.1.1. Information and the comparison class 1

2 The comparison class is defined on a canonical product space. Let $W =$ 2
 3 $(W^1, \dots, W^{d+1})$ be the economic Brownian motion and let R be an independent 3
 4 random element on a standard Borel space. The element R may contain a private 4
 5 Brownian path and a sequence of independent sensor errors. Write $\mathbb{F}^W$ for the usual 5
 6 augmentation of the economic Brownian filtration and $\mathbb{F}^{W,R}$ for the corresponding 6
 7 product filtration; the latter reveals only the portion of R specified by its observa- 7
 8 tion clock. An admissible control is a progressively measurable, nonanticipative 8
 9 functional of the available paths, taking values in the primitive action box. Bounded 9
 10 controls and the globally Lipschitz drift give a unique strong state process and an 10
 11 integrable payoff. These conventions apply to the comparison class in Theorems S.4 11
 12 and 4. 12

13
 14 PROPOSITION S.12—Independent randomization and the optimum: *Let $V_W^*(y)$ 14*
 15 *and $V_{W,R}^*(y)$ be the expected-payoff suprema over the preceding economic and product 15*
 16 *filtrations. Then 16*

$$17 \quad V_{W,R}^*(y) = V_W^*(y). \quad 17$$

18 *The anchor envelope G_d and the policy-specific regret conclusions therefore compare the 18*
 19 *observed-history and finite-observation policies with the same economic optimum. The 19*
 20 *equality concerns the supremum; it neither requires an optimal policy nor asserts that a 20*
 21 *single fixed randomizer realization attains that supremum. 21*

22
 23 PROOF: The inequality $V_{W,R}^* \geq V_W^*$ follows by ignoring R . For the reverse in- 23
 24 equality, represent an admissible randomized control by its nonanticipative mea- 24
 25 surable functional $a(t, W_{[0,t]}, R_{[0,t]})$ before completion of the filtrations. Fixing a 25
 26 full realization r of R leaves a progressively measurable functional of W . For al- 26
 27 most every r , Fubini's theorem preserves feasibility and the state equation; strong 27
 28 uniqueness identifies the resulting state with the controlled state for that section. 28
 29 Independence leaves the law of W equal to Wiener measure after conditioning on 29
 30 $R = r$. Thus the conditional payoff is at most $V_W^*(y)$ for almost every r . Integrat- 30
 31 ing over r proves the reverse inequality. Null-set modifications in the augmented 31
 32 filtration do not change the state law or payoff. Alternatively, the anchor proof 32

may be repeated directly on the product filtration: its synchronous comparison and martingale identities use only adaptedness, bounded actions, and the Brownian property, all of which are preserved. *Q.E.D.*

Exact continuous observation. For precision in Proposition S.11, work first in the product filtration generated by W and an independent scalar Brownian motion B^0 . Let $\mathbb{G}$ be the usual augmentation of $\sigma(Y_s, B_s^0 : 0 \leq s \leq t)$, with the initial profile included. The stored numerical state and held actions are functions of this history. Consequently

$$M_t = Y_t - Y_0 - \int_0^t \{c1 + f(Y_s) - a_s\} ds$$

is $\mathbb{G}$ -adapted. Since it equals ΣW_t , it is a $\mathbb{G}$ -martingale: conditional expectation first in the product filtration and then in its subfiltration $\mathbb{G}$ proves this claim. With $A = \Sigma^\top (\Sigma \Sigma^\top)^{-1}$ and a unit null vector n , the process $AM + nB^0$ is a continuous $\mathbb{G}$ -local martingale starting at zero. Its bracket is $I_{d+1}t$, because $A\Sigma$ and $nn^\top$ are complementary orthogonal projections and $[M, B^0] = 0$. The multivariate Lévy characterization gives a Brownian motion in the stated observation filtration. The identity $\Sigma(AM + nB^0) = M$ holds pathwise.

This argument also establishes causality. At a decision date, the already observed drift integral determines the preceding innovation increment and hence the next internal state. The next action is then fixed before the next Brownian increment. The held-action state equation is uniquely solvable on that cell. Induction identifies the joint law of physical state, internal state, actions, and recovered innovations with the prescribed innovation controller. The independent null component supplies the correct law when clipping is applied separately to $d + 1$ coordinates; it does not recover the original latent null component. The exact observation result is separate from the finite measurement implementation below.

H.1.2. Finite observations with a continuous-economy transfer

At dates $t_k = kh$, a finite-observation controller receives $O_k = \bar{Y}_{t_k} + \eta_k$ and holds $\bar{a}_k = \hat{\pi}_k(O_k)$ until the next date. The sensor error is measurable at t_k in the product filtration just specified and satisfies $\|\eta_k\|_{2,d} \leq \nu_{\text{obs}}$, where $\|X\|_{2,d} = (\mathbb{E}\|X\|_d^2)^{1/2}$.

Both the sensor state $\bar{Y}$ and the comparison parent's physical state Y obey the original continuous diffusion. No Brownian coordinate or continuous drift integral is supplied to the sensor controller.

The protected parent has internal state $Z_0 = y$ and selects $a_k = \hat{\pi}_k(Z_k)$. Its update is the specified clipped-innovation Euler update, with an implementation error of normalized norm at most ζ_h per cell. This includes innovation conversion and recurrence arithmetic. Both action maps use the same inward guard and lie within ϵ of the common schedule $m_0(t_k)$ coordinatewise. For a real reference map π_k , suppose

$$\|\pi_k(x) - \pi_k(z)\|_d \leq L\|x - z\|_d, \quad \sup_x \|\hat{\pi}_k(x) - \pi_k(x)\|_d \leq \delta_\pi.$$

Global matrix-norm products and forward error propagation through the protected activation supply L and δ_π for the saved weights. Measurement error is measured before the protected map; input representation within that map is included in δ_π .

Retain the production coefficient κ of the economic model and let $\beta \geq \kappa\|B\|_2$. Put $\bar{\sigma} = (\sigma_I^2 + \sigma_C^2)^{1/2}$ and let M bound $\|c1 + f(y) - m\|_d$ for all feasible held actions. Define the generator constants H and $K = \beta M + Q_f$ as in the diffusion-transfer argument of Theorem 4, where $H^2 \geq d^{-1} \sup_y \|Df(y)\Sigma\|_F^2$, $q_i = (B\Sigma\Sigma^\top B^\top)_{ii}$, and $Q_f = (\kappa/2)(d^{-1} \sum_i q_i^2)^{1/2}$ is the generator bound, not a diffusion covariance. Let $z > 0$ be the clipping threshold and let $v_{\text{clip}} \geq \mathbb{E}(G - \text{clip}(G, -z, z))^2$ for a standard normal G . The bound $2\varphi(z)(z + z^{-1})$ is valid. Set

$$b_k = h\{Kt_k/2 + H\sqrt{t_k/3}\} + \bar{\sigma}\sqrt{t_k v_{\text{clip}}} + k\zeta_h, \quad b_0 = 0, \quad (\text{S.85})$$

$$e_0 = 0, \quad a_k^\Delta = \min\{2\epsilon, L(e_k + v_{\text{obs}}) + 2\delta_\pi\},$$

$$e_{k+1} = (1 + \beta h)e_k + ha_k^\Delta + (b_{k+1} - b_k), \quad 0 \leq k < N. \quad (\text{S.86})$$

The forcing is nondecreasing. Each term retains the accumulated martingale cancellation; in particular, the clipping term is a finite-horizon second-moment account rather than a sum of cellwise absolute clipping errors.

Define the nonnegative function

$$D(t) = \int_0^t e^{\beta(t-s)} a_{[s/h]}^\Delta ds,$$

$$Q_T = s_0 + (\kappa + \epsilon)T + \sigma_I \sqrt{(1 - 1/d)T}.$$

Here $s_0 \geq \|\Pi Y_0\|_{2,d}$, with Π the projection away from the common-state direction. For $t \in [t_k, t_{k+1})$, choose $m_{-,k} > 0$ below every action in the common tube and put

$$r_k(t) = \epsilon + |m_0(t) - m_0(t_k)|, \quad C_k(t) = r_k(t) \{[m_0(t)m_{-,k}]^{-1} + \zeta\}.$$

With $W(t) = e^{-\rho t} w(t)$ as before, define

$$\begin{aligned} B_{h,\nu_{\text{obs}}}^{\text{sens}} &= \beta \int_0^T W(t) D(t) dt + \sum_{k=0}^{N-1} a_k^\Delta \int_{t_k}^{t_{k+1}} e^{-\rho t} C_k(t) dt \\ &\quad + 2\chi e^{-\rho T} Q_T D(T). \end{aligned} \tag{S.87}$$

PROPOSITION S.13—Finite-observation payoff transfer: *Under the stated assumptions,*

$$\|\bar{Y}_{t_k} - Z_k\|_{2,d} \leq e_k,$$

$$\|\bar{a}_k - a_k\|_{2,d} \leq a_k^\Delta,$$

$$\|\bar{Y}_t - Y_t\|_{2,d} \leq D(t),$$

and

$$|J_y(\bar{a}) - J_y(a)| \leq B_{h,\nu_{\text{obs}}}^{\text{sens}}.$$

Consequently, a valid improvement interval $[L_a, U_a]$ for the protected parent at the same time grid transfers to $[L_a - B_{h,\nu_{\text{obs}}}^{\text{sens}}, U_a + B_{h,\nu_{\text{obs}}}^{\text{sens}}]$ for the finite-observation controller. This deterministic operation uses no additional statistical confidence allocation. It requires a certificate for that exact protected parent and grid.

The allowance concerns the original physical diffusion and uses only the stated sensor moment bound. A shared-noise fine-Euler comparison checks the finite-observation implementation and reports its observed payoff difference. Such a diagnostic does not replace the allowance or certify the Euler bias. For fixed weights and horizon, uniformly bounded L, β, ϵ and generator constants, and $m_{-,k} \geq \underline{m} > 0$, the forcing vanishes if $h \rightarrow 0$, $v_{\text{clip}}(h) \rightarrow 0$, $\xi_h/h \rightarrow 0$, $\nu_{\text{obs}} \rightarrow 0$, and $\delta_\pi \rightarrow 0$.

Discrete Gronwall then makes the transfer allowance vanish. Fixed clipping and fixed numerical or measurement error leave the corresponding terms in the finite-grid bound.

H.2. Finite-Observation Transfer Proof

H.2.1. Proof of Proposition S.13

All processes are coupled using the same economic Brownian motion and initial profile; the sensor noise uses the independent randomizer. The real-map Lipschitz bound, uniform implementation error, and nonexpansiveness of the common guard give

$$\|\hat{\pi}_k(\bar{Y}_{t_k} + \eta_k) - \hat{\pi}_k(Z_k)\|_{2,d} \leq L\{\|\bar{Y}_{t_k} - Z_k\|_{2,d} + \nu_{\text{obs}}\} + 2\delta_\pi.$$

The common action tube also bounds this difference by 2ϵ .

Apply Itô's formula to $f(\bar{Y})$ on each cell, where the sensor action has already been chosen. The generator and derivative bounds used in the policy-specific study proof remain valid for this adapted held action. Stochastic Fubini yields

$$\begin{aligned} \int_{t_j}^{t_{j+1}} \{f(\bar{Y}_s) - f(\bar{Y}_{t_j})\} ds &= \int_{t_j}^{t_{j+1}} (t_{j+1} - v) \mathcal{A}^{\bar{a}_j} f(\bar{Y}_v) dv \\ &\quad + \int_{t_j}^{t_{j+1}} (t_{j+1} - v) Df(\bar{Y}_v) \Sigma dW_v. \end{aligned}$$

The L^2 norm of the accumulated first term through t_k is at most $Kht_k/2$. The stochastic integrals are martingale increments over disjoint cells. Itô's isometry bounds their accumulated normalized L^2 norm by $Hh\sqrt{t_k/3}$.

Before representation error, every component of $\Delta W_j - \sqrt{h} \text{clip}(\Delta W_j / \sqrt{h}, -z, z)$ has zero mean and variance at most $h\nu_{\text{clip}}$. Components and cells are independent. Therefore the normalized second moment of the accumulated projected clipping error is at most

$$\frac{1}{d} \text{tr}(\Sigma \Sigma^\top) t_k \nu_{\text{clip}} = \bar{\sigma}^2 t_k \nu_{\text{clip}}.$$

The clipping error and the preceding local-drift martingale need not be independent: the triangle inequality between their already accumulated norms is sufficient. The implementation error contributes at most $k\tilde{\zeta}_h$.

Let $E_k = \|\bar{Y}_{t_k} - Z_k\|_{2,d}$. Subtracting the internal recursion from the integrated sensor state equation thus gives

$$E_k \leq \beta h \sum_{j < k} E_j + h \sum_{j < k} \|\bar{a}_j - a_j\|_{2,d} + b_k.$$

The recursion in (S.86) is equivalent to $e_k = \beta h \sum_{j < k} e_j + h \sum_{j < k} a_j^\Delta + b_k$. Induction, using the action bound already proved, establishes $E_k \leq e_k$ and the asserted action inequalities.

The two physical diffusions have the same Brownian term. Subtracting their integral equations and applying the β -Lipschitz production bound gives

$$\|\bar{Y}_t - Y_t\|_{2,d} \leq \beta \int_0^t \|\bar{Y}_s - Y_s\|_{2,d} ds + \int_0^t a_{[s/h]}^\Delta ds.$$

The integral form of Gronwall's inequality proves the bound by $D(t)$. Projection away from the common-state direction eliminates common drift, the common schedule, and common diffusion. Bounded production, the action tube, and Itô's isometry give, for both physical states, $\sup_{t \leq T} \|\Pi Y_t\|_{2,d} \leq Q_T$.

It remains to compare payoffs. Subtract the two exact identities (25); the common schedule terms cancel. Production contributes at most $\beta \int_0^T W(t) D(t) dt$. For the consumption deficit R_t in (24), its gradient in the normalized inner product has components

$$-\frac{1}{m_i} + w(t) + \zeta \bar{m} = \frac{1}{m_0(t)} - \frac{1}{m_i} + \zeta \{\bar{m} - m_0(t)\}.$$

Every point on the segment between the two held actions remains in their common coordinate tube at t_k . Its distance from $m_0(t)$ is at most $r_k(t)$, and its coordinates are at least $m_{-,k}$. Thus the normalized gradient norm is bounded by $C_k(t)$. The mean-value inequality and Cauchy-Schwarz bound the absolute expected deficit difference by $C_k(t) a_k^\Delta$.

Finally,

$$\left| \mathbb{E} \{ \|\Pi \bar{Y}_T\|_d^2 - \|\Pi Y_T\|_d^2 \} \right| \leq \|\bar{Y}_T - Y_T\|_{2,d} \{ \|\Pi \bar{Y}_T\|_{2,d} + \|\Pi Y_T\|_{2,d} \} \leq 2Q_T D(T).$$

Adding these three contributions proves (S.87). Adding and subtracting the parent's payoff on the existing confidence event proves the interval transfer.

An outward finite computation. One need not infer a scalar integral error from a refinement plot. The function D is nondecreasing and obeys the exact cell recursion

$$D(t_{k+1}) = e^{\beta h} D(t_k) + a_k^\Delta \int_0^h e^{\beta s} ds.$$

An interval enclosure of m_0 on each cell gives an upper bound C_k^+ for $C_k(t)$. With the already enclosed weights A_k, B_k , a computable upper allowance is

$$\beta \sum_k B_k D(t_{k+1}) + \sum_k A_k C_k^+ a_k^\Delta + 2\chi e^{-\rho T} Q_T D(T),$$

evaluated outward. In the forcing recurrence, use nondecreasing saved upper endpoints $b_k^+ \geq b_k$ and the differences of those exact saved scalars. This preserves the telescoping majorant. The network and recurrence arithmetic assumptions are the same conditional binary64 assumptions used for the protected parent.

APPENDIX I: INDEPENDENT NUMERICAL DIAGNOSTICS

I.1. A Trained Neural Boundary-Value Problem

Consider a household with labor income, risky wealth, and a prescribed retirement payoff on exit. Before exit from $D = (0.5, 2.5)$,

$$dX_s = (0.12 + 0.02X_s - c_s) ds + 0.15X_s dW_s, \quad c_s \in [0.02, 0.6]. \quad (\text{S.88})$$

Discount is $\rho = 0.08$, flow utility is $\log c$, and exit pays $g(x) = \log(0.12 + 0.08x)/0.08$. This is a specified stopped economy, not a claim about an unbounded-wealth problem. Income and the exit option destroy the homothetic reduction used in the portfolio calibration. Bounded coefficients, a compact positive consumption interval, positive diffusion variance on the closed domain, and continuous endpoint

data yield the usual stopped-domain comparison problem. A neural approximation must solve its nonlinear differential equation and both boundary conditions.

The critic has two hidden tanh layers of width sixteen and the lifting

$$v_{\xi}(x) = g(\ell) + \frac{g(u) - g(\ell)}{u - \ell}(x - \ell) + (x - \ell)(u - x)N_{\xi}\left(2\frac{x - \ell}{u - \ell} - 1\right).$$

The actor uses the same hidden architecture and a sigmoid into the consumption interval. Here the optimum is not forced to an endpoint by that representation. Independent verification nevertheless maximizes over the *closed* interval: for $p = v'(x)$,

$$c^*(p) = \begin{cases} \text{clip}(1/p, 0.02, 0.6), & p > 0, \\ 0.6, & p \leq 0. \end{cases} \quad (\text{S.89})$$

Training imports the general differential core used for the earlier routing tests. It does not fit reference values, derivatives, or policy labels.

PROPOSITION S.14—Continuous-domain economic certificate: *For a stopped discounted additive economy obeying comparison, let $v \in C^2(D) \cap C(\overline{D})$ and let a be admissible. Suppose throughout D that*

$$|\rho v - r_0^a - \mathcal{L}^a v| \leq e, \quad 0 \leq \sup_b (r_0^b + \mathcal{L}^b v) - (r_0^a + \mathcal{L}^a v) \leq \eta, \quad \|v - g\|_{\partial D} \leq B.$$

Then

$$\|v - V^a\|_{\infty} \leq B + \frac{e}{\rho}, \quad \|v - V^*\|_{\infty} \leq B + \frac{e + \eta}{\rho}, \quad 0 \leq V^* - V^a \leq 2B + \frac{2e + \eta}{\rho}. \quad (\text{S.90})$$

Indeed, constant upper and lower shifts of v dominate the evaluation or optimality residual and the stopping data; comparison gives the first two inequalities and their sum gives the third. No reference-grid error is used in this argument.

The executable verifier covers the entire state interval by dyadic subintervals. Outward-rounded propagation through the stored network encloses v , v' , v'' , and the actor on every leaf. Equation (S.89) supplies a global action enclosure. We also

use the concavity bound

$$\log c^* - pc^* - (\log c - pc) \leq \frac{1}{2}C^2(1/c - p)^2, \quad C \geq \max\{c, c^*\}, \quad (\text{S.91})$$

which follows by completing the square in the strong-concavity inequality on the segment joining the two actions. It does not assume an interior maximizer. Taking the better of this bound and a direct interval enclosure avoids dependency inflation. Terminal-free boundary errors are evaluated separately at both endpoints, including rounding in the lifting constants.

The certificate saves every leaf, verifies that their union is exactly $[0.5, 2.5]$, and binds the parameter and arithmetic files by hashes. Exponential and logarithmic enclosures use range reduction and polynomial remainder bounds rather than an assumed accuracy of a library transcendental. Its mathematical conclusion is conditional on the stated binary64 arithmetic contract; high-precision and derivative tests check the implementation but are not a formal machine proof. The construction does not estimate a global Lipschitz constant from samples.

I.1.1. *Controlled comparison and failures*

We use seeds 11, 29, and 47, each with 256 prespecified jittered collocation points. Eight outer rounds have a common cap of 800 L-BFGS inner iterations. NBO allocates each round sixty iterations to evaluation and forty to improvement; direct neural HJB uses the same critic and the global action rule in all one hundred iterations. A joint-loss ablation differentiates the sum of squared residual and negative Hamiltonian through both blocks. Line searches use different numbers of closures, so this is a matched iteration cap, not an assertion of equal floating-point work. Closure counts and elapsed times are saved.

An independent positive-rate upwind policy-iteration solver uses continuous-action maximization within each drift region. Reference grids have 320, 640, 1,280, and 2,560 intervals. Value and derivative comparisons use the finest grid; its refinement differences are not substituted for a rigorous discretization bound. The continuous certificate instead checks the neural functions directly.

All three NBO and direct-HJB runs attain the declared interval-verification thresholds. The joint-loss ablation does not: a residual near one half is accompanied by

TABLE I.I

MULTILAYER CONSUMPTION: APPROXIMATION AND CONTINUOUS-DOMAIN VERIFICATION

Method	Seed	$\ v - V_h^*\ _\infty$	Sampled e	Uniform η	Regret bound
NBO	11	3.30×10^{-4}	6.18×10^{-4}	4.24×10^{-6}	0.0501
NBO	29	5.75×10^{-4}	0.0015	2.66×10^{-6}	0.0500
NBO	47	5.46×10^{-4}	0.0014	4.36×10^{-6}	0.0501
Direct HJB	11	4.14×10^{-4}	9.72×10^{-4}	0	0.0500
Direct HJB	29	3.55×10^{-4}	7.13×10^{-4}	0	0.0500
Direct HJB	47	6.00×10^{-4}	0.0017	0	0.0500
Joint loss	11	5.4142	0.5002	—	—
Joint loss	29	5.4435	0.5003	—	—
Joint loss	47	5.5677	0.5003	—	—

The comparison grid has 2,560 intervals on $[0.5, 2.5]$. The last two columns cover the entire continuous interval and consumption set $[0.02, 0.6]$. All six certificates have uniform evaluation error at most 0.002 and endpoint error below 10^{-12} . Dashes indicate no accepted certificate, not zero error. Reference error and the sampled differential residual are not used as uniform bounds.

large value and action errors. Width, depth, and update-allocation variations are retained in the supplement, including unfavorable outcomes. The direct method is a strong comparator: these results do not establish that carrying an actor is uniformly faster or more accurate. They establish an end-to-end neural solve with an independently computable welfare bound, and identify a concrete failure of joint differentiation. Fitting and full-domain verification are separately timed because their computational burdens are different.

I.2. Independent Scalar Reference: Equations and Complete Diagnostics

I.2.1. The scalar reference and its residual account

The scalar reference isolates policy evaluation and policy improvement in a nonlinear member of the coupled-capital family. It uses the three stored scalar actors selected in the original experiment and the analytical anchor, without retraining. The four state–time–domain specifications are $(401, 256, 6)$, $(801, 512, 6)$, $(1601, 1024, 6)$ and $(1601, 1024, 8)$, where the last coordinate is the half-width L of the log-capital

interval. All model coefficients, action limits, terminal values and boundary data remain those of the original reference. The diagnostic program checks the source and input hashes before computation and verifies the input hashes again afterwards.

Write $\eta = (\sigma_{\text{id}}^2 + \sigma_{\text{com}}^2)/2$, $c = \alpha - \eta$, and a for the adjustment-cost coefficient. The realized scalar coupling entry is $B_{11} = 1$. For scalar log capital y and consumption rate $m \in [\underline{m}, \bar{m}] = [0.02, 2]$, the running payoff and drift are

$$u(y, m) = y + \log m - \frac{a}{2}m^2, \quad \mu(y, m) = c + \beta \tanh y - m.$$

The terminal payoff is y ; the cross-sectional dispersion term vanishes in one dimension. Put $t_k = kh$, $h = T/n_t$ and let δ be the uniform state spacing. The original upwind generator is

$$(L_\delta^m v)_i = q_i^-(m)(v_{i-1} - v_i) + q_i^+(m)(v_{i+1} - v_i),$$

$$q_i^- = \frac{\eta}{\delta^2} + \frac{(-\mu_i)^+}{\delta},$$

$$q_i^+ = \frac{\eta}{\delta^2} + \frac{\mu_i^+}{\delta}.$$

Both coefficients are nonnegative. A fixed-policy backward step solves

$$\frac{v_i^k - v_i^{k+1}}{h} + \rho v_i^k - (L_\delta^{m_i^k} v^k)_i - u(y_i, m_i^k) = 0$$

at every interior node. Boundary values enter neighboring differences exactly once. In particular, residuals are evaluated in this original equation, not by subtracting a boundary-adjusted right-hand side after reintroducing the boundary terms.

For Howard improvement, the Hamiltonian depending on m is

$$\log m - \frac{a}{2}m^2 + (d_i - m)^+ p_i^f + (d_i - m)^- p_i^b,$$

where

$$d_i = c + \beta \tanh y_i, \quad p_i^f = \frac{v_{i+1} - v_i}{\delta}, \quad p_i^b = \frac{v_i - v_{i-1}}{\delta},$$

where $z^- = \min(z, 0)$. On each of the intervals cut by $m = d_i$, the objective is strictly concave. Its unconstrained maximizer has the form $2/(p + \sqrt{p^2 + 4a})$, with the

algebraically equivalent stable expression used when $p < 0$. Clipping this root to each nonempty interval and comparing the two attained values gives the global scalar action maximum for the upwind Hamiltonian. Thus the reported HJB residual uses the full economic action interval, without action-grid enumeration or a neural critic.

The imposed boundary functions have the form $b_{\pm}(t) = \pm Lw(t) + Q_{\pm}(t)$, where

$$w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)}, \quad \bar{m}(t) = \frac{2}{w(t) + \sqrt{w(t)^2 + 4a}},$$

and

$$Q_{\pm}(t) = \int_t^T e^{-\rho(s-t)} \left[\log \bar{m}(s) - \frac{a}{2} \bar{m}(s)^2 + w(s) \{c \pm \beta - \bar{m}(s)\} \right] ds.$$

The original cumulative-trapezoid rule and time interpolation evaluate these functions. They represent asymptotic boundary data shared by all policies; they are not independently verified exact boundary values for each learned policy in the unbounded economy. Their satisfaction at grid endpoints is an algebraic check, not a bound on economic boundary bias.

For the optimized grid value, define the original-equation HJB defect

$$r_i^k = \frac{v_i^k - v_i^{k+1}}{h} + \rho v_i^k - \max_m \{u(y_i, m) + (L_{\delta}^m v^k)_i\}.$$

For a fixed policy, omit the maximization and use its stored feedback action. The diagnostic files retain the maximum residual at every time step, the linear residual of every Howard evaluation, every iterate difference, and the iteration count. The original stopping rule is an iterate difference below 10^{-11} , with a cap of 100 iterations. A separate Howard check evaluates $\|\mathcal{T}_k(v^k) - v^k\|_{\infty}/h$, where $\mathcal{T}_k$ solves the linear system for the greedy policy selected from v^k .

The significance of these residuals can be stated directly. Let $\tilde{v}$ be the exact solution of the same finite-grid backward equations, with the same frozen policy (or the same Bellman operator), terminal data and boundary data, and let v be the approximate sequence whose defect is r . In exact arithmetic, the fixed-policy matrix has positive diagonal, nonpositive off-diagonal entries, and row sums at least $h^{-1} + \rho$ after

boundary elimination. The discrete maximum principle therefore yields

$$\|\tilde{v}^k - v^k\|_\infty \leq \frac{\|\tilde{v}^{k+1} - v^{k+1}\|_\infty + hR_k}{1 + \rho h}$$

whenever the boundary data agree and $\|r^k\|_\infty \leq R_k$. The same estimate holds for the Bellman step: at a positive maximum of the difference, select an action maximizing the Hamiltonian of the upper candidate and apply the fixed-action maximum principle; interchange the candidates for the negative part. With exact common terminal data this gives

$$\|\tilde{v}^0 - v^0\|_\infty \leq \sum_{k=0}^{n_t-1} \frac{hR_k}{(1 + \rho h)^{k+1}} \leq \frac{\max_k R_k}{\rho} \{1 - (1 + \rho h)^{-n_t}\}.$$

The archived propagation diagnostic substitutes the observed floating-point residuals in this formula. These residuals are not outward-rounded enclosures, so that substitution is a numerical diagnostic rather than a certified bound. The estimate concerns the finite-grid equations with the prescribed boundary data; it does not include consistency, state interpolation or boundary bias.

The tables report all four grids, the five initial states $y_0 \in \{-1, -0.5, 0, 0.5, 1\}$, the policies' action-bound frequencies, and changes in values, optimal actions and policy losses. Initial-state values are linearly interpolated from nodal time-zero values. Bound frequencies use an absolute tolerance of 10^{-12} and count equally weighted interior state-time nodes; they are not occupation probabilities. The first two grid comparisons halve both h and δ . Increasing L from 6 to 8 at fixed $(n_y, n_t) = (1601, 1024)$ changes δ from 0.0075 to 0.01. This comparison measures joint domain and spacing sensitivity, not boundary truncation error in isolation. Full common-domain changes and their locations accompany the central-region summaries.

The scalar calculation supplies an independent finite-grid benchmark for non-linear feedback, with a reproducible algebraic error account. In the 10-, 20- and 50-dimensional economies, cross-sectional dispersion, the coupling matrix and the covariance structure remain economically active. Moreover, this scalar benchmark evaluates Markov feedback, whereas the continuous-time deployment theorem

specifies a randomized controller with innovation history. Scalar reference losses therefore have their stated one-dimensional meaning; multidimensional conclusions rely on the separate multidimensional policy comparisons and certification arguments.

TABLE I.II

SCALAR FINITE-GRID POLICY LOSSES AT FIVE INITIAL STATES

Grid (n_y, n_t, L)	y_0	NBO	DPO	Affine	Anchor
(401, 256, 6)	-1	1.02×10^{-5}	6.78×10^{-5}	9.63×10^{-5}	1.16×10^{-4}
	-0.5	7.17×10^{-6}	8.61×10^{-5}	5.67×10^{-5}	5.64×10^{-4}
	0	2.55×10^{-6}	1.09×10^{-4}	1.93×10^{-4}	1.54×10^{-3}
	0.5	3.23×10^{-6}	1.04×10^{-4}	1.00×10^{-4}	1.68×10^{-3}
	1	9.79×10^{-5}	5.30×10^{-5}	1.69×10^{-4}	7.17×10^{-4}
(801, 512, 6)	-1	1.08×10^{-5}	6.93×10^{-5}	9.51×10^{-5}	1.17×10^{-4}
	-0.5	8.23×10^{-6}	8.96×10^{-5}	5.73×10^{-5}	5.70×10^{-4}
	0	3.12×10^{-6}	1.14×10^{-4}	2.00×10^{-4}	1.56×10^{-3}
	0.5	3.08×10^{-6}	1.04×10^{-4}	9.95×10^{-5}	1.67×10^{-3}
	1	1.02×10^{-4}	5.11×10^{-5}	1.72×10^{-4}	6.99×10^{-4}
(1601, 1024, 6)	-1	1.11×10^{-5}	7.00×10^{-5}	9.45×10^{-5}	1.17×10^{-4}
	-0.5	8.81×10^{-6}	9.13×10^{-5}	5.77×10^{-5}	5.74×10^{-4}
	0	3.44×10^{-6}	1.16×10^{-4}	2.03×10^{-4}	1.57×10^{-3}
	0.5	2.99×10^{-6}	1.03×10^{-4}	9.89×10^{-5}	1.67×10^{-3}
	1	1.05×10^{-4}	5.02×10^{-5}	1.74×10^{-4}	6.90×10^{-4}
(1601, 1024, 8)	-1	1.10×10^{-5}	6.98×10^{-5}	9.46×10^{-5}	1.17×10^{-4}
	-0.5	8.62×10^{-6}	9.08×10^{-5}	5.76×10^{-5}	5.72×10^{-4}
	0	3.32×10^{-6}	1.15×10^{-4}	2.02×10^{-4}	1.57×10^{-3}
	0.5	3.02×10^{-6}	1.04×10^{-4}	9.90×10^{-5}	1.67×10^{-3}
	1	1.04×10^{-4}	5.04×10^{-5}	1.73×10^{-4}	6.92×10^{-4}

Entries are the Howard grid value minus the value of the unchanged stored feedback policy, with common asymptotic Dirichlet data. Values at the five stated initial log-capital levels use linear interpolation of time-zero nodal values when necessary. No policy is refitted or chosen using this table. These are scalar finite-grid losses; they do not estimate the optimum in dimensions 10, 20 or 50.

TABLE I.III

SCALAR VALUE AND POLICY CHANGES UNDER REFINEMENT

Comparison	Policy	Core value	Core action	Core loss	Common value
401 $\rightarrow$ 801, $L = 6$	Howard	6.59×10^{-4}	2.87×10^{-4}	0	6.59×10^{-4}
	NBO	6.57×10^{-4}	0	4.28×10^{-6}	6.57×10^{-4}
	DPO	6.61×10^{-4}	0	4.64×10^{-6}	6.61×10^{-4}
	Affine	6.60×10^{-4}	0	7.20×10^{-6}	6.60×10^{-4}
	Anchor	6.78×10^{-4}	0	1.91×10^{-5}	6.78×10^{-4}
801 $\rightarrow$ 1601, $L = 6$	Howard	3.31×10^{-4}	1.45×10^{-4}	0	3.31×10^{-4}
	NBO	3.30×10^{-4}	0	2.22×10^{-6}	3.30×10^{-4}
	DPO	3.32×10^{-4}	0	2.40×10^{-6}	3.32×10^{-4}
	Affine	3.32×10^{-4}	0	3.72×10^{-6}	3.32×10^{-4}
	Anchor	3.40×10^{-4}	0	9.74×10^{-6}	3.40×10^{-4}
$L = 6 \rightarrow 8$, $n_y = 1601$	Howard	5.21×10^{-5}	4.72×10^{-5}	0	2.72×10^{-4}
	NBO	5.23×10^{-5}	7.59×10^{-7}	6.88×10^{-7}	5.03×10^{-4}
	DPO	5.28×10^{-5}	7.14×10^{-7}	8.13×10^{-7}	2.89×10^{-4}
	Affine	5.25×10^{-5}	1.02×10^{-8}	1.29×10^{-6}	7.37×10^{-3}
	Anchor	5.39×10^{-5}	0	3.63×10^{-6}	2.72×10^{-4}

Entries are maximum absolute changes in values, actions or policy losses. The first three numerical columns use time-zero nodes with $|y| \leq 1$; the final column uses the full common state interval. The second grid is linearly interpolated at nodes of the first. Policy loss is $\ell = V^H - V^m$. The first two comparisons halve both state spacing and time step. The last expands the domain at fixed n_y, n_t , changing state spacing from 0.0075 to 0.01; it is joint domain/spacing sensitivity and does not isolate boundary truncation error. Frozen-policy action changes arise only from interpolation and floating-point evaluation. Full common-interior action changes and their locations are archived.

I.3. *Finite-Observation Diagnostic Accounts*

The experiment fixes the same saved NBO checkpoint before drawing the diagnostic bank. The sensor uses finite state measurements and independent private noise; the parent uses a clipped-innovation internal state. The deterministic allowance and the finite-Euler measurement answer different questions. Neither the observed mean nor its simulation standard error supplies the fine-Euler bias.

TABLE I.IV: Components of the finite-observation allowance

d	Cells	ν_{obs}	Production	Consumption	Terminal
10	64	0	0.000270	0.000833	0.000169
10	64	0.001	0.000687	0.001876	0.000361
10	64	0.01	0.004433	0.011263	0.002095
10	128	0	0.000137	0.000412	0.000087
10	128	0.001	0.000553	0.001432	0.000282
10	128	0.01	0.004295	0.010610	0.002037
20	64	0	0.000627	0.001756	0.000369
20	64	0.001	0.001535	0.003886	0.000787
20	64	0.01	0.009706	0.023047	0.004553
20	128	0	0.000320	0.000876	0.000191
20	128	0.001	0.001232	0.002973	0.000619
20	128	0.01	0.009442	0.021853	0.004471
50	64	0	0.001754	0.005424	0.001192
50	64	0.001	0.004613	0.013341	0.002877
50	64	0.01	0.022028	0.044889	0.008488
50	128	0	0.000912	0.002769	0.000634
50	128	0.001	0.003840	0.010758	0.002406
50	128	0.01	0.021850	0.043484	0.008503

Each component is displayed upward. The machine-readable records give the full recurrence, network Lipschitz and roundoff bounds, initial-profile bounds, seeds, pathwise diagnostic arrays, and the separate Monte Carlo standard errors.

APPENDIX J: COMPLETE COMPUTATIONAL OUTCOMES

The following tables retain every declared stream and saved checkpoint.

TABLE J.I: Every Declared Stream Payoff and Direct Contrast

d	Stream	Endpoint	Mean	Lower	Upper
10	2095568532	NBO	0.001273	0.000537	0.002008
10	2095568532	Raw	0.001273	0.000537	0.002009
10	2095568532	Direct policy	0.001267	0.000531	0.002002
10	2095568532	Neural HJB	0.001120	0.000386	0.001855
10	2095568532	NBO–Raw	0.000000	-0.001152	0.001152
10	2095568532	NBO–Direct policy	0.000006	-0.001146	0.001159
10	2095568532	NBO–Neural HJB	0.000152	-0.001007	0.001311
10	2133755523	NBO	0.001257	0.000521	0.001992
10	2133755523	Raw	0.001257	0.000521	0.001993
10	2133755523	Direct policy	0.001249	0.000513	0.001984
10	2133755523	Neural HJB	0.001071	0.000337	0.001805
10	2133755523	NBO–Raw	0.000000	-0.001153	0.001152
10	2133755523	NBO–Direct policy	0.000008	-0.001145	0.001161
10	2133755523	NBO–Neural HJB	0.000186	-0.000974	0.001346
10	153722668	NBO	0.001262	0.000527	0.001998
10	153722668	Raw	0.001261	0.000526	0.001997
10	153722668	Direct policy	0.001256	0.000520	0.001991
10	153722668	Neural HJB	0.001135	0.000401	0.001869
10	153722668	NBO–Raw	0.000001	-0.001151	0.001153
10	153722668	NBO–Direct policy	0.000007	-0.001146	0.001159
10	153722668	NBO–Neural HJB	0.000127	-0.001030	0.001285
10	603655094	NBO	0.001272	0.000536	0.002007
10	603655094	Raw	0.001269	0.000533	0.002004
10	603655094	Direct policy	0.001267	0.000531	0.002002
10	603655094	Neural HJB	0.001079	0.000343	0.001815
10	603655094	NBO–Raw	0.000003	-0.001149	0.001155
10	603655094	NBO–Direct policy	0.000005	-0.001147	0.001158
10	603655094	NBO–Neural HJB	0.000193	-0.000968	0.001354

	<i>d</i>	Stream	Endpoint	Mean	Lower	Upper	
1							1
2	10	216592699	NBO	0.001266	0.000531	0.002002	2
3	10	216592699	Raw	0.001264	0.000529	0.002000	3
4	10	216592699	Direct policy	0.001260	0.000525	0.001995	4
5	10	216592699	Neural HJB	0.001156	0.000420	0.001892	5
6	10	216592699	NBO–Raw	0.000002	-0.001150	0.001154	6
7	10	216592699	NBO–Direct policy	0.000006	-0.001146	0.001159	7
8	10	216592699	NBO–Neural HJB	0.000110	-0.001046	0.001266	8
9	10	1950831757	NBO	0.001266	0.000530	0.002001	9
10	10	1950831757	Raw	0.001264	0.000528	0.002000	10
11	10	1950831757	Direct policy	0.001260	0.000524	0.001995	11
12	10	1950831757	Neural HJB	0.001055	0.000322	0.001788	12
13	10	1950831757	NBO–Raw	0.000001	-0.001151	0.001153	13
14	10	1950831757	NBO–Direct policy	0.000006	-0.001147	0.001158	14
15	10	1950831757	NBO–Neural HJB	0.000211	-0.000952	0.001373	15
16	10	107486742	NBO	0.001271	0.000536	0.002007	16
17	10	107486742	Raw	0.001272	0.000536	0.002007	17
18	10	107486742	Direct policy	0.001266	0.000530	0.002001	18
19	10	107486742	Neural HJB	0.000876	0.000143	0.001609	19
20	10	107486742	NBO–Raw	0.000000	-0.001153	0.001152	20
21	10	107486742	NBO–Direct policy	0.000006	-0.001147	0.001158	21
22	10	107486742	NBO–Neural HJB	0.000395	-0.000770	0.001561	22
23	10	1142235589	NBO	0.001269	0.000534	0.002005	23
24	10	1142235589	Raw	0.001268	0.000532	0.002004	24
25	10	1142235589	Direct policy	0.001263	0.000528	0.001999	25
26	10	1142235589	Neural HJB	0.000034	-0.000700	0.000768	26
27	10	1142235589	NBO–Raw	0.000001	-0.001151	0.001153	27
28	10	1142235589	NBO–Direct policy	0.000006	-0.001147	0.001159	28
29	10	1142235589	NBO–Neural HJB	0.001235	0.000054	0.002417	29
30	10	1824517164	NBO	0.001269	0.000533	0.002005	30
31	10	1824517164	Raw	0.001269	0.000534	0.002005	31
32	10	1824517164	Direct policy	0.001267	0.000531	0.002002	32
	10	1824517164	Neural HJB	0.001130	0.000395	0.001866	
	10	1824517164	NBO–Raw	0.000000	-0.001152	0.001152	

	<i>d</i>	Stream	Endpoint	Mean	Lower	Upper	
1							1
2							2
3	10	1824517164	NBO-Direct policy	0.000002	-0.001151	0.001155	3
4	10	1824517164	NBO-Neural HJB	0.000139	-0.001018	0.001295	4
5	10	1965147083	NBO	0.001267	0.000532	0.002003	5
6	10	1965147083	Raw	0.001267	0.000532	0.002003	6
7	10	1965147083	Direct policy	0.001263	0.000528	0.001999	7
8	10	1965147083	Neural HJB	0.001063	0.000326	0.001799	8
9	10	1965147083	NBO-Raw	0.000000	-0.001152	0.001152	9
10	10	1965147083	NBO-Direct policy	0.000004	-0.001149	0.001156	10
11	10	1965147083	NBO-Neural HJB	0.000204	-0.000957	0.001365	11
12	10	1860620983	NBO	0.001270	0.000535	0.002006	12
13	10	1860620983	Raw	0.001270	0.000534	0.002005	13
14	10	1860620983	Direct policy	0.001268	0.000532	0.002003	14
15	10	1860620983	Neural HJB	0.001149	0.000414	0.001884	15
16	10	1860620983	NBO-Raw	0.000001	-0.001151	0.001153	16
17	10	1860620983	NBO-Direct policy	0.000003	-0.001150	0.001155	17
18	10	1860620983	NBO-Neural HJB	0.000121	-0.001036	0.001279	18
19	10	1618144748	NBO	0.001275	0.000539	0.002010	19
20	10	1618144748	Raw	0.001275	0.000540	0.002010	20
21	10	1618144748	Direct policy	0.001268	0.000532	0.002003	21
22	10	1618144748	Neural HJB	0.001119	0.000385	0.001854	22
23	10	1618144748	NBO-Raw	0.000000	-0.001152	0.001152	23
24	10	1618144748	NBO-Direct policy	0.000007	-0.001145	0.001160	24
25	10	1618144748	NBO-Neural HJB	0.000156	-0.001004	0.001315	25
26	10	592131000	NBO	0.001265	0.000530	0.002001	26
27	10	592131000	Raw	0.001264	0.000528	0.001999	27
28	10	592131000	Direct policy	0.001259	0.000523	0.001994	28
29	10	592131000	Neural HJB	0.001076	0.000342	0.001811	29
30	10	592131000	NBO-Raw	0.000001	-0.001151	0.001153	30
31	10	592131000	NBO-Direct policy	0.000007	-0.001146	0.001159	31
32	10	592131000	NBO-Neural HJB	0.000189	-0.000972	0.001349	32
	10	964082955	NBO	0.001275	0.000540	0.002011	
	10	964082955	Raw	0.001273	0.000538	0.002009	
	10	964082955	Direct policy	0.001271	0.000535	0.002006	

	<i>d</i>	Stream	Endpoint	Mean	Lower	Upper	
1							1
2	10	964082955	Neural HJB	0.001106	0.000372	0.001840	2
3	10	964082955	NBO–Raw	0.000002	-0.001150	0.001154	3
4	10	964082955	NBO–Direct policy	0.000005	-0.001148	0.001157	4
5	10	964082955	NBO–Neural HJB	0.000169	-0.000990	0.001328	5
6	10	887230079	NBO	0.001265	0.000530	0.002000	6
7	10	887230079	Raw	0.001265	0.000530	0.002001	7
8	10	887230079	Direct policy	0.001260	0.000525	0.001995	8
9	10	887230079	Neural HJB	0.001081	0.000343	0.001818	9
10	10	887230079	NBO–Raw	0.000000	-0.001152	0.001152	10
11	10	887230079	NBO–Direct policy	0.000005	-0.001148	0.001157	11
12	10	887230079	NBO–Neural HJB	0.000184	-0.000974	0.001343	12
13	10	1818435806	NBO	0.001264	0.000528	0.001999	13
14	10	1818435806	Raw	0.001262	0.000526	0.001997	14
15	10	1818435806	Direct policy	0.001258	0.000523	0.001994	15
16	10	1818435806	Neural HJB	0.000320	-0.000410	0.001050	16
17	10	1818435806	NBO–Raw	0.000002	-0.001150	0.001154	17
18	10	1818435806	NBO–Direct policy	0.000006	-0.001147	0.001158	18
19	10	1818435806	NBO–Neural HJB	0.000944	-0.000230	0.002118	19
20	50	2095568532	NBO	0.001341	0.000558	0.002124	20
21	50	2095568532	Raw	0.001343	0.000560	0.002125	21
22	50	2095568532	Direct policy	0.001343	0.000561	0.002126	22
23	50	2095568532	Neural HJB	0.001127	0.000343	0.001911	23
24	50	2095568532	NBO–Raw	-0.000002	-0.001204	0.001201	24
25	50	2095568532	NBO–Direct policy	-0.000002	-0.001206	0.001201	25
26	50	2095568532	NBO–Neural HJB	0.000214	-0.000995	0.001423	26
27	50	2133755523	NBO	0.001345	0.000562	0.002128	27
28	50	2133755523	Raw	0.001347	0.000564	0.002130	28
29	50	2133755523	Direct policy	0.001346	0.000563	0.002129	29
30	50	2133755523	Neural HJB	-0.001572	-0.002492	-0.000652	30
31	50	2133755523	NBO–Raw	-0.000002	-0.001204	0.001201	31
32	50	2133755523	NBO–Direct policy	-0.000001	-0.001204	0.001202	32
	50	2133755523	NBO–Neural HJB	0.002917	0.001566	0.004269	
	50	153722668	NBO	0.001353	0.000570	0.002135	

	<i>d</i>	Stream	Endpoint	Mean	Lower	Upper	
1							1
2	50	153722668	Raw	0.001354	0.000571	0.002137	2
3	50	153722668	Direct policy	0.001351	0.000568	0.002134	3
4	50	153722668	Neural HJB	-0.001096	-0.001986	-0.000205	4
5	50	153722668	NBO–Raw	-0.000001	-0.001204	0.001201	5
6	50	153722668	NBO–Direct policy	0.000002	-0.001201	0.001205	6
7	50	153722668	NBO–Neural HJB	0.002449	0.001128	0.003769	7
8	50	603655094	NBO	0.001346	0.000563	0.002129	8
9	50	603655094	Raw	0.001349	0.000566	0.002132	9
10	50	603655094	Direct policy	0.001349	0.000566	0.002132	10
11	50	603655094	Neural HJB	0.000703	-0.000091	0.001497	11
12	50	603655094	NBO–Raw	-0.000002	-0.001205	0.001200	12
13	50	603655094	NBO–Direct policy	-0.000002	-0.001206	0.001201	13
14	50	603655094	NBO–Neural HJB	0.000643	-0.000580	0.001866	14
15	50	216592699	NBO	0.001345	0.000562	0.002129	15
16	50	216592699	Raw	0.001346	0.000563	0.002130	16
17	50	216592699	Direct policy	0.001344	0.000561	0.002127	17
18	50	216592699	Neural HJB	-0.000866	-0.001738	0.000006	18
19	50	216592699	NBO–Raw	-0.000001	-0.001204	0.001201	19
20	50	216592699	NBO–Direct policy	0.000002	-0.001202	0.001205	20
21	50	216592699	NBO–Neural HJB	0.002211	0.000909	0.003513	21
22	50	1950831757	NBO	0.001344	0.000561	0.002127	22
23	50	1950831757	Raw	0.001347	0.000564	0.002129	23
24	50	1950831757	Direct policy	0.001348	0.000565	0.002131	24
25	50	1950831757	Neural HJB	0.000737	-0.000051	0.001526	25
26	50	1950831757	NBO–Raw	-0.000003	-0.001205	0.001200	26
27	50	1950831757	NBO–Direct policy	-0.000004	-0.001207	0.001199	27
28	50	1950831757	NBO–Neural HJB	0.000607	-0.000612	0.001825	28
29	50	107486742	NBO	0.001328	0.000545	0.002111	29
30	50	107486742	Raw	0.001330	0.000547	0.002113	30
31	50	107486742	Direct policy	0.001350	0.000567	0.002133	31
32	50	107486742	Neural HJB	-0.000616	-0.001469	0.000237	32
	50	107486742	NBO–Raw	-0.000002	-0.001204	0.001201	
	50	107486742	NBO–Direct policy	-0.000022	-0.001225	0.001182	

	<i>d</i>	Stream	Endpoint	Mean	Lower	Upper	
1							1
2							2
3	50	107486742	NBO–Neural HJB	0.001944	0.000659	0.003230	3
4	50	1142235589	NBO	0.001344	0.000561	0.002127	4
5	50	1142235589	Raw	0.001345	0.000562	0.002128	5
6	50	1142235589	Direct policy	0.001348	0.000566	0.002131	6
7	50	1142235589	Neural HJB	-0.000003	-0.000825	0.000819	7
8	50	1142235589	NBO–Raw	-0.000001	-0.001203	0.001202	8
9	50	1142235589	NBO–Direct policy	-0.000004	-0.001208	0.001199	9
10	50	1142235589	NBO–Neural HJB	0.001347	0.000096	0.002598	10
11	50	1824517164	NBO	0.001348	0.000565	0.002131	11
12	50	1824517164	Raw	0.001349	0.000566	0.002132	12
13	50	1824517164	Direct policy	0.001346	0.000564	0.002129	13
14	50	1824517164	Neural HJB	-0.000298	-0.001130	0.000534	14
15	50	1824517164	NBO–Raw	-0.000001	-0.001204	0.001201	15
16	50	1824517164	NBO–Direct policy	0.000001	-0.001202	0.001205	16
17	50	1824517164	NBO–Neural HJB	0.001646	0.000384	0.002907	17
18	50	1965147083	NBO	0.001323	0.000540	0.002107	18
19	50	1965147083	Raw	0.001324	0.000540	0.002107	19
20	50	1965147083	Direct policy	0.001345	0.000562	0.002128	20
21	50	1965147083	Neural HJB	-0.000361	-0.001197	0.000475	21
22	50	1965147083	NBO–Raw	-0.000001	-0.001203	0.001202	22
23	50	1965147083	NBO–Direct policy	-0.000022	-0.001225	0.001181	23
24	50	1965147083	NBO–Neural HJB	0.001684	0.000418	0.002950	24
25	50	1860620983	NBO	0.001346	0.000563	0.002129	25
26	50	1860620983	Raw	0.001347	0.000564	0.002130	26
27	50	1860620983	Direct policy	0.001349	0.000566	0.002131	27
28	50	1860620983	Neural HJB	0.000779	-0.000005	0.001564	28
29	50	1860620983	NBO–Raw	-0.000001	-0.001204	0.001202	29
30	50	1860620983	NBO–Direct policy	-0.000002	-0.001205	0.001201	30
31	50	1860620983	NBO–Neural HJB	0.000567	-0.000648	0.001782	31
32	50	1618144748	NBO	0.001330	0.000547	0.002113	32
	50	1618144748	Raw	0.001331	0.000548	0.002114	
	50	1618144748	Direct policy	0.001350	0.000567	0.002132	
	50	1618144748	Neural HJB	-0.000491	-0.001334	0.000353	

	<i>d</i>	Stream	Endpoint	Mean	Lower	Upper	
1							1
2							2
3	50	1618144748	NBO–Raw	-0.000001	-0.001203	0.001201	3
4	50	1618144748	NBO–Direct policy	-0.000019	-0.001223	0.001184	4
5	50	1618144748	NBO–Neural HJB	0.001821	0.000545	0.003096	5
6	50	592131000	NBO	0.001346	0.000563	0.002129	6
7	50	592131000	Raw	0.001346	0.000563	0.002129	7
8	50	592131000	Direct policy	0.001346	0.000563	0.002129	8
9	50	592131000	Neural HJB	-0.000211	-0.001040	0.000617	9
10	50	592131000	NBO–Raw	0.000000	-0.001203	0.001202	10
11	50	592131000	NBO–Direct policy	0.000000	-0.001203	0.001203	11
12	50	592131000	NBO–Neural HJB	0.001557	0.000298	0.002816	12
13	50	964082955	NBO	0.001346	0.000563	0.002129	13
14	50	964082955	Raw	0.001348	0.000565	0.002131	14
15	50	964082955	Direct policy	0.001347	0.000564	0.002129	15
16	50	964082955	Neural HJB	-0.000332	-0.001165	0.000502	16
17	50	964082955	NBO–Raw	-0.000002	-0.001204	0.001201	17
18	50	964082955	NBO–Direct policy	-0.000001	-0.001204	0.001203	18
19	50	964082955	NBO–Neural HJB	0.001678	0.000414	0.002942	19
20	50	887230079	NBO	0.001343	0.000560	0.002126	20
21	50	887230079	Raw	0.001345	0.000562	0.002128	21
22	50	887230079	Direct policy	0.001345	0.000562	0.002128	22
23	50	887230079	Neural HJB	-0.000282	-0.001110	0.000547	23
24	50	887230079	NBO–Raw	-0.000002	-0.001204	0.001201	24
25	50	887230079	NBO–Direct policy	-0.000002	-0.001205	0.001202	25
26	50	887230079	NBO–Neural HJB	0.001625	0.000366	0.002884	26
27	50	1818435806	NBO	0.001344	0.000562	0.002127	27
28	50	1818435806	Raw	0.001345	0.000562	0.002127	28
29	50	1818435806	Direct policy	0.001346	0.000563	0.002128	29
30	50	1818435806	Neural HJB	0.000280	-0.000525	0.001085	30
31	50	1818435806	NBO–Raw	-0.000001	-0.001203	0.001202	31
32	50	1818435806	NBO–Direct policy	-0.000001	-0.001205	0.001202	32
	50	1818435806	NBO–Neural HJB	0.001064	-0.000172	0.002301	

Notes: Every declared stream and endpoint is retained. These conditional-policy intervals and the method averages belong to the single registered confirmation family; their simulation banks are independent of stopping. Every lower endpoint is rounded down and every upper endpoint up to six decimal places; displayed intervals therefore contain the original protected intervals. Means are descriptive and rounded to nearest.

TABLE J.II: Actual online certification by work checkpoint

d	Method	Checkpoint	Attempted	Completed	Certified by then
10	NBO	1	16	16	0/16
10	NBO	2	16	16	0/16
10	NBO	3	16	16	0/16
10	Raw	1	16	16	0/16
10	Raw	2	16	16	0/16
10	Raw	3	16	16	0/16
10	Direct policy	1	16	16	0/16
10	Direct policy	2	16	16	0/16
10	Direct policy	3	16	16	0/16
10	Neural HJB	1	16	16	0/16
10	Neural HJB	2	16	16	0/16
10	Neural HJB	3	16	16	0/16
50	NBO	1	16	16	0/16
50	NBO	2	16	16	3/16
50	NBO	3	13	13	16/16
50	Raw	1	16	16	0/16
50	Raw	2	16	16	3/16
50	Raw	3	13	13	16/16
50	Direct policy	1	16	16	0/16
50	Direct policy	2	16	16	0/16
50	Direct policy	3	16	16	16/16
50	Neural HJB	1	16	16	0/16
50	Neural HJB	2	16	16	0/16
50	Neural HJB	3	16	16	0/16

Every listed check was executed before the continuation or stopping decision. The last column counts all streams that had attained the primary 0.0005 lower-gain target by the checkpoint, retaining earlier stopped streams. Missing later checks represent actual stopping or recorded failures. This table does not infer an unexecuted fixed-budget payoff frontier.

APPENDIX K: COMPLETE MECHANISM ASSESSMENT

The following records retain every candidate and the work used in its independent mechanism assessment.

TABLE K.I

COMPLETE PROTECTED BELLMAN-BRIDGE STATISTICS

d	Statistic	Raw mean	Lower	Upper	Clip bound	Arithmetic
10	M	0.0012636	-0.0008484	0.0033757	0.21174	0.00000019
10	A	0.0008803	0.0007515	0.0010091	0.01003	0.00000001
10	E	0.0000045	-0.0313992	0.0314082	3.25082	0.00000001
10	D	0.0000272	-0.0040585	0.0041129	0.42285	0.00000001
10	E^{raw}	0.0000302	-0.0029862	0.0030465	0.31182	0.00000001
10	$E - E^{\text{raw}}$	-0.0000257	-0.0344447	0.0343934	3.56264	0.00000001
50	M	0.0013307	-0.0001528	0.0028143	0.14696	0.00000012
50	A	0.0009390	0.0008149	0.0010631	0.01003	0.00000001
50	E	0.0000398	-0.0171787	0.0172583	1.78228	0.00000001
50	D	0.0001221	-0.0027173	0.0029614	0.29347	0.00000001
50	E^{raw}	0.0000259	-0.0053392	0.0053911	0.55513	0.00000001
50	$E - E^{\text{raw}}$	0.0000139	-0.0225684	0.0225961	2.33741	0.00000001

Notes: Every registered statistic and dimension appears. These intervals concern the exact represented-query sample expectations before the additional state-representation conversion used in the welfare bound. The arithmetic column is the outward sample-mean and sample-standard-deviation perturbation allowance, not an empirical population range. The source records each stream's scalar weights, verified network norms, curvature, clipping tail and state/adjoint error constants. There is no per-stream inferential claim.

TABLE K.II

COMPLETE WORK FOR THE INDEPENDENT MECHANISM ASSESSMENT

d	Process seconds	CPU seconds	Peak MiB	Future updates	Padded rows
10	339.05	334.24	461.1	33,646,832	66,803,712
50	561.96	558.12	810.1	33,517,664	66,828,288

Notes: Process and CPU seconds sum all 16 fresh stream processes in each dimension; peak memory is their maximum. The assessment includes loading the already selected policy, rebuilding constants, candidate occupation paths, both continuation banks, reverse derivatives, interval arithmetic and output. Future updates count mathematically active continuation transitions; padded rows additionally disclose vectorized inactive arithmetic. This work is reported in addition to the primary algorithms' complete stopping and independent payoff-assessment costs.

APPENDIX L: AN OUTWARD ALGEBRAIC ERROR ACCOUNT FOR THE SCALAR COMPARATOR

Fix the declared spatial nodes, common spacing Δy , time step h , action endpoints, and terminal and boundary data. Each stored binary64 input denotes an exact real in this finite-grid problem. Let $\eta = (\sigma_I^2 + \sigma_C^2)/2$, $d_0(y) = a - \eta + \kappa \tanh y$, and define the forward and backward slopes $p_f = (v_{i+1} - v_i)/\Delta y$ and $p_b = (v_i - v_{i-1})/\Delta y$. The action-dependent upwind Hamiltonian is

$$\mathcal{H}_i(m; v) = \log m - \frac{\zeta}{2} m^2 + \max\{d_0(y_i) - m, 0\} p_f + \min\{d_0(y_i) - m, 0\} p_b. \quad (\text{S.92})$$

The positive-drift and negative-drift branches are each concave in m . Their feasible intervals are, respectively, $[\ell, \min\{u, d_0(y_i)\}]$ and $[\max\{\ell, d_0(y_i)\}, u]$, when nonempty. On either branch, the unconstrained positive critical point solves $1/m - \zeta m = p_j$. The Howard update clips this point to the branch interval and compares the two resulting objective values. Thus the sign of the drift selects the appropriate difference operator throughout maximization.

The floating-point Howard solve proposes values and actions. A separate outward calculation encloses the Bellman residual, including the error in the proposed maximizer. For a concave branch H on $[a, b]$ and any $m_\circ \in [a, b]$,

$$\max_{m \in [a, b]} H(m) \leq H(m_\circ) + \max\{0, H'(m_\circ)(a - m_\circ), H'(m_\circ)(b - m_\circ)\}. \quad (\text{S.93})$$

Interval evaluations of log, tanh, the slopes, and the arithmetic operations enclose the right-hand side. Interval uncertainty in $d_0(y_i)$ is handled by enlarging each branch interval before constructing its upper bound. An arbitrary globally feasible action supplies a lower bound for the maximized Hamiltonian. Combining the two bounds encloses

$$r_{k,i} = \frac{v_{k,i} - v_{k+1,i}}{h} + \rho v_{k,i} - \eta \frac{v_{k,i+1} - 2v_{k,i} + v_{k,i-1}}{(\Delta y)^2} - y_i - \max_m \mathcal{H}_i(m; v_k). \quad (\text{S.94})$$

The artifact stores every proposed value and action, the boundary arrays, the action endpoints, and the outward residual bound at every time step.

PROPOSITION S.15—Algebraic error in the declared scalar boundary problem: Fix finitely many ordered spatial coordinate labels and a common positive difference spacing Δy , $h > 0$, $\rho \geq 0$, $\eta \geq 0$, compact action intervals $[\ell_k, u_k]$ with $0 < \ell_k \leq u_k < \infty$, and finite terminal and boundary data. Let $\tilde{v}$ denote the exact solution of the backward Bellman equations (S.94) with zero residual. Let v have identical terminal and boundary data and residuals $r_{k,i}$. For any $R_k \geq \max_i |r_{k,i}|$, define $E_N = 0$ and

$$E_k = \frac{E_{k+1} + hR_k}{1 + \rho h}. \quad (\text{S.95})$$

Then the exact finite-grid solution exists uniquely, and $\|v_k - \tilde{v}_k\|_\infty \leq E_k$ at every time step.

PROOF: For a fixed action m , write the discrete generator at an interior node as

$$(L_m w)_i = \lambda_i^-(m)(w_{i-1} - w_i) + \lambda_i^+(m)(w_{i+1} - w_i),$$

$$\lambda_i^-(m) = \frac{\eta}{(\Delta y)^2} + \frac{\max\{m - d_0(y_i), 0\}}{\Delta y},$$

$$\lambda_i^+(m) = \frac{\eta}{(\Delta y)^2} + \frac{\max\{d_0(y_i) - m, 0\}}{\Delta y}.$$

Both rates are nonnegative. Each frozen-policy matrix therefore has nonpositive off-diagonal entries and diagonal dominance at least $h^{-1} + \rho$ after the boundary values are moved to the right-hand side. This is the M -matrix comparison property used below.

For completeness, existence can be established at each backward step by uniformizing these finitely many bounded rates. Choose $\Lambda \geq \max_{i,m} \{\lambda_i^-(m) + \lambda_i^+(m)\}$. When $\Lambda > 0$, $L_m = \Lambda(P_m - I)$, where P_m has nonnegative entries and row sums one before the fixed boundary values are substituted. The Bellman equation is a fixed-point equation whose dependence on the unknown interior values has supremum-norm Lipschitz constant at most $\Lambda/(h^{-1} + \rho + \Lambda) < 1$. The fixed-point theorem gives a unique solution. If $\Lambda = 0$, the equation is pointwise and its coefficient $h^{-1} + \rho$ is strictly positive. Backward recursion starts from the specified terminal values.

To establish the error bound, suppose $\|v_{k+1} - \tilde{v}_{k+1}\|_\infty \leq E_{k+1}$ and set $e_i = v_{k,i} - \tilde{v}_{k,i}$. If $M = \max_i e_i > 0$, the maximum occurs at an interior node i , because the boundary error is zero. Nonnegative rates give $(L_m e)_i \leq 0$ for every feasible m . Writing $f_i(m) = y_i + \log m - \zeta m^2/2$, it follows that

$$\sup_m \{(L_m v_k)_i + f_i(m)\} - \sup_m \{(L_m \tilde{v}_k)_i + f_i(m)\} \leq 0.$$

Subtract the exact and approximate Bellman equations at this node. The preceding inequality implies

$$(h^{-1} + \rho)M \leq h^{-1}(v_{k+1,i} - \tilde{v}_{k+1,i}) + r_{k,i} \leq h^{-1}E_{k+1} + R_k.$$

Interchanging v_k and $\tilde{v}_k$ gives the same bound for the largest negative error. Equation (S.95) follows, and induction from $E_N = 0$ completes the proof. *Q.E.D.*

The implementation evaluates this recurrence outward under the inherited arithmetic contract. It controls algebraic solution error for a specified finite-grid boundary problem. The separate grid, domain, and deployed-policy mesh comparisons measure the remaining numerical sensitivities.

COROLLARY S.16—Loss from a nested scalar action restriction: *Consider two problems in Proposition S.15 with identical spatial labels, difference and time steps, primitives, and terminal and boundary data. Suppose the action intervals of the second problem are contained in those of the first at every date. Denote their exact values by $\tilde{v}^F$ and $\tilde{v}^T$, their computed values by v^F and v^T , and their algebraic error bounds by E^F and E^T . Then, at*

every initial-date node,

$$0 \leq \tilde{v}_{0,i}^F - \tilde{v}_{0,i}^T \leq \|v_0^F - v_0^T\|_\infty + E_0^F + E_0^T.$$

In particular, identical stored initial-date values reduce the upper bound to $E_0^F + E_0^T$.

PROOF: At each backward step, use the common uniformization constant from the preceding proof for both action sets. The resulting Bellman maps are monotone in continuation and current interior values. Enlarging the maximization set increases the map pointwise. The contraction fixed points therefore inherit this order; backward induction from the common terminal data gives $\tilde{v}^F \geq \tilde{v}^T$. The upper inequality follows by adding and subtracting $v_{0,i}^F$ and $v_{0,i}^T$ and applying the two algebraic error bounds. Q.E.D.

APPENDIX M: COMPLETE SCALAR COMPARISON TABLES

TABLE M.I: Deployed scalar policy gains on common diffusion paths

Policy	Steps	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.
Schedule	1024	0.0000	0.0000	0.0000	0.0000	0.0000
F1	1024	0.1103	1.5441	0.6666	0.7725	0.0068
F2	1024	0.1103	1.5441	0.6670	0.7726	0.0068
F3	1024	0.1103	1.5440	0.6672	0.7726	0.0068
F4	1024	0.1103	1.5440	0.6672	0.7726	0.0068
T1	1024	0.1103	1.5441	0.6666	0.7725	0.0068
T2	1024	0.1103	1.5441	0.6670	0.7726	0.0068
T3	1024	0.1103	1.5440	0.6672	0.7726	0.0068
T4	1024	0.1103	1.5440	0.6672	0.7726	0.0068
NBO (R12)	1024	0.1017	1.5418	0.5541	0.7312	0.0070
DPO (R12)	1024	0.0471	1.4365	0.6231	0.7010	0.0066
Schedule	2048	0.0000	0.0000	0.0000	0.0000	0.0000
F1	2048	0.1137	1.5586	0.6768	0.7818	0.0069
F2	2048	0.1137	1.5587	0.6771	0.7819	0.0069

Policy	Steps	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.
F3	2048	0.1136	1.5586	0.6772	0.7819	0.0069
F4	2048	0.1136	1.5586	0.6772	0.7819	0.0069
T1	2048	0.1137	1.5586	0.6768	0.7818	0.0069
T2	2048	0.1137	1.5587	0.6771	0.7819	0.0069
T3	2048	0.1136	1.5586	0.6772	0.7819	0.0069
T4	2048	0.1136	1.5586	0.6772	0.7819	0.0069
NBO (R12)	2048	0.1036	1.5557	0.5683	0.7412	0.0071
DPO (R12)	2048	0.0468	1.4460	0.6303	0.7065	0.0066

All payoff gains and standard errors are multiplied by 10^3 . The comparator is the analytical schedule evaluated on the same numerical mesh and Brownian path. Population denotes the mean over independent uniform initial-state draws; the last column is its paired Monte Carlo standard error. Initial-state columns retain their separate rankings. NBO and DPO are the prespecified frozen R12 scalar policies. These are conditional Euler-deployment diagnostics; Monte Carlo standard errors do not include continuous-time or boundary approximation error.

TABLE M.II

SCALAR TRAINING-GRID AND DOMAIN SENSITIVITY

Comparison	Change	$y = -1$	$y = 0$	$y = 1$
F1–F2	Mesh	3.454e-04	3.424e-04	6.555e-04
F2–F3	Mesh	1.742e-04	1.724e-04	3.250e-04
F3–F4	Domain	4.441e-16	-4.441e-16	-2.220e-16
T1–T2	Mesh	3.454e-04	3.424e-04	6.555e-04
T2–T3	Mesh	1.742e-04	1.724e-04	3.250e-04
T3–T4	Domain	4.441e-16	-4.441e-16	-2.220e-16

Entries are later-grid minus earlier-grid values, retaining their signs. Domain changes keep the declared difference spacing and time step fixed. The differences are numerical sensitivities, separate from the algebraic residual bound.

TABLE M.III

SCALAR DEPLOYMENT-MESH SENSITIVITY

Policy	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.
Schedule	-1.564e-04	-1.702e-04	-1.323e-04	-1.529e-04	7.858e-07
F1	-1.531e-04	-1.557e-04	-1.221e-04	-1.436e-04	7.856e-07
F2	-1.530e-04	-1.556e-04	-1.222e-04	-1.436e-04	7.854e-07
F3	-1.530e-04	-1.556e-04	-1.222e-04	-1.436e-04	7.853e-07
F4	-1.530e-04	-1.556e-04	-1.222e-04	-1.436e-04	7.853e-07
T1	-1.531e-04	-1.557e-04	-1.221e-04	-1.436e-04	7.856e-07
T2	-1.530e-04	-1.556e-04	-1.222e-04	-1.436e-04	7.854e-07
T3	-1.530e-04	-1.556e-04	-1.222e-04	-1.436e-04	7.853e-07
T4	-1.530e-04	-1.556e-04	-1.222e-04	-1.436e-04	7.853e-07
NBO (R12)	-1.544e-04	-1.563e-04	-1.181e-04	-1.429e-04	7.934e-07
DPO (R12)	-1.567e-04	-1.608e-04	-1.250e-04	-1.475e-04	7.884e-07

Entries are payoff at 2,048 steps minus payoff at 1,024 steps using the same fine Brownian increments and initial draws. The frozen policy table or neural weights are unchanged between deployments. These paired numerical changes are not a time-discretization error bound.

TABLE M.IV: Every paired scalar policy comparison

Left minus right	Steps	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.
Schedule-F1	1024	-0.1103	-1.5441	-0.6666	-0.7725	0.0068
Schedule-F2	1024	-0.1103	-1.5441	-0.6670	-0.7726	0.0068
Schedule-F3	1024	-0.1103	-1.5440	-0.6672	-0.7726	0.0068
Schedule-F4	1024	-0.1103	-1.5440	-0.6672	-0.7726	0.0068
Schedule-T1	1024	-0.1103	-1.5441	-0.6666	-0.7725	0.0068
Schedule-T2	1024	-0.1103	-1.5441	-0.6670	-0.7726	0.0068
Schedule-T3	1024	-0.1103	-1.5440	-0.6672	-0.7726	0.0068
Schedule-T4	1024	-0.1103	-1.5440	-0.6672	-0.7726	0.0068
Schedule-NBO (R12)	1024	-0.1017	-1.5418	-0.5541	-0.7312	0.0070
Schedule-DPO (R12)	1024	-0.0471	-1.4365	-0.6231	-0.7010	0.0066
F1-F2	1024	0.0000	0.0000	-0.0004	-0.0001	0.0000
F1-F3	1024	0.0001	0.0001	-0.0005	-0.0001	0.0000

	Left minus right	Steps	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.	
1								1
2								2
3	F1-F4	1024	0.0001	0.0001	-0.0005	-0.0001	0.0000	3
4	F1-T1	1024	0.0000	0.0000	0.0000	0.0000	0.0000	4
5	F1-T2	1024	0.0000	0.0000	-0.0004	-0.0001	0.0000	5
6	F1-T3	1024	0.0001	0.0001	-0.0005	-0.0001	0.0000	6
7	F1-T4	1024	0.0001	0.0001	-0.0005	-0.0001	0.0000	7
8	F1-NBO (R12)	1024	0.0087	0.0023	0.1125	0.0412	0.0007	8
9	F1-DPO (R12)	1024	0.0633	0.1076	0.0436	0.0714	0.0004	9
10	F2-F3	1024	0.0000	0.0001	-0.0001	0.0000	0.0000	10
11	F2-F4	1024	0.0000	0.0001	-0.0001	0.0000	0.0000	11
12	F2-T1	1024	0.0000	0.0000	0.0004	0.0001	0.0000	12
13	F2-T2	1024	0.0000	0.0000	0.0000	0.0000	0.0000	13
14	F2-T3	1024	0.0000	0.0001	-0.0001	0.0000	0.0000	14
15	F2-T4	1024	0.0000	0.0001	-0.0001	0.0000	0.0000	15
16	F2-NBO (R12)	1024	0.0086	0.0023	0.1129	0.0414	0.0007	16
17	F2-DPO (R12)	1024	0.0632	0.1076	0.0440	0.0715	0.0004	17
18	F3-F4	1024	0.0000	0.0000	0.0000	0.0000	0.0000	18
19	F3-T1	1024	-0.0001	-0.0001	0.0005	0.0001	0.0000	19
20	F3-T2	1024	0.0000	-0.0001	0.0001	0.0000	0.0000	20
21	F3-T3	1024	0.0000	0.0000	0.0000	0.0000	0.0000	21
22	F3-T4	1024	0.0000	0.0000	0.0000	0.0000	0.0000	22
23	F3-NBO (R12)	1024	0.0086	0.0022	0.1131	0.0414	0.0007	23
24	F3-DPO (R12)	1024	0.0632	0.1075	0.0441	0.0716	0.0004	24
25	F4-T1	1024	-0.0001	-0.0001	0.0005	0.0001	0.0000	25
26	F4-T2	1024	0.0000	-0.0001	0.0001	0.0000	0.0000	26
27	F4-T3	1024	0.0000	0.0000	0.0000	0.0000	0.0000	27
28	F4-T4	1024	0.0000	0.0000	0.0000	0.0000	0.0000	28
29	F4-NBO (R12)	1024	0.0086	0.0022	0.1131	0.0414	0.0007	29
30	F4-DPO (R12)	1024	0.0632	0.1075	0.0441	0.0716	0.0004	30
31	T1-T2	1024	0.0000	0.0000	-0.0004	-0.0001	0.0000	31
32	T1-T3	1024	0.0001	0.0001	-0.0005	-0.0001	0.0000	32
	T1-T4	1024	0.0001	0.0001	-0.0005	-0.0001	0.0000	
	T1-NBO (R12)	1024	0.0087	0.0023	0.1125	0.0412	0.0007	
	T1-DPO (R12)	1024	0.0633	0.1076	0.0436	0.0714	0.0004	

	Left minus right	Steps	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.	
1								1
2								2
3	T2-T3	1024	0.0000	0.0001	-0.0001	0.0000	0.0000	3
4	T2-T4	1024	0.0000	0.0001	-0.0001	0.0000	0.0000	4
5	T2-NBO (R12)	1024	0.0086	0.0023	0.1129	0.0414	0.0007	5
6	T2-DPO (R12)	1024	0.0632	0.1076	0.0440	0.0715	0.0004	6
7	T3-T4	1024	0.0000	0.0000	0.0000	0.0000	0.0000	7
8	T3-NBO (R12)	1024	0.0086	0.0022	0.1131	0.0414	0.0007	8
9	T3-DPO (R12)	1024	0.0632	0.1075	0.0441	0.0716	0.0004	9
10	T4-NBO (R12)	1024	0.0086	0.0022	0.1131	0.0414	0.0007	10
11	T4-DPO (R12)	1024	0.0632	0.1075	0.0441	0.0716	0.0004	11
12	NBO (R12)-DPO (R12)	1024	0.0546	0.1053	-0.0690	0.0302	0.0010	12
13	Schedule-F1	2048	-0.1137	-1.5586	-0.6768	-0.7818	0.0069	13
14	Schedule-F2	2048	-0.1137	-1.5587	-0.6771	-0.7819	0.0069	14
15	Schedule-F3	2048	-0.1136	-1.5586	-0.6772	-0.7819	0.0069	15
16	Schedule-F4	2048	-0.1136	-1.5586	-0.6772	-0.7819	0.0069	16
17	Schedule-T1	2048	-0.1137	-1.5586	-0.6768	-0.7818	0.0069	17
18	Schedule-T2	2048	-0.1137	-1.5587	-0.6771	-0.7819	0.0069	18
19	Schedule-T3	2048	-0.1136	-1.5586	-0.6772	-0.7819	0.0069	19
20	Schedule-T4	2048	-0.1136	-1.5586	-0.6772	-0.7819	0.0069	20
21	Schedule-NBO (R12)	2048	-0.1036	-1.5557	-0.5683	-0.7412	0.0071	21
22	Schedule-DPO (R12)	2048	-0.0468	-1.4460	-0.6303	-0.7065	0.0066	22
23	F1-F2	2048	0.0000	-0.0001	-0.0003	-0.0001	0.0000	23
24	F1-F3	2048	0.0000	-0.0001	-0.0004	-0.0001	0.0000	24
25	F1-F4	2048	0.0000	-0.0001	-0.0004	-0.0001	0.0000	25
26	F1-T1	2048	0.0000	0.0000	0.0000	0.0000	0.0000	26
27	F1-T2	2048	0.0000	-0.0001	-0.0003	-0.0001	0.0000	27
28	F1-T3	2048	0.0000	-0.0001	-0.0004	-0.0001	0.0000	28
29	F1-T4	2048	0.0000	-0.0001	-0.0004	-0.0001	0.0000	29
30	F1-NBO (R12)	2048	0.0100	0.0029	0.1085	0.0406	0.0007	30
31	F1-DPO (R12)	2048	0.0669	0.1126	0.0465	0.0753	0.0004	31
32	F2-F3	2048	0.0000	0.0000	-0.0001	0.0000	0.0000	32
	F2-F4	2048	0.0000	0.0000	-0.0001	0.0000	0.0000	
	F2-T1	2048	0.0000	0.0001	0.0003	0.0001	0.0000	
	F2-T2	2048	0.0000	0.0000	0.0000	0.0000	0.0000	

	Left minus right	Steps	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.	
1								1
2								2
3	F2-T3	2048	0.0000	0.0000	-0.0001	0.0000	0.0000	3
4	F2-T4	2048	0.0000	0.0000	-0.0001	0.0000	0.0000	4
5	F2-NBO (R12)	2048	0.0100	0.0030	0.1088	0.0407	0.0007	5
6	F2-DPO (R12)	2048	0.0669	0.1127	0.0468	0.0754	0.0004	6
7	F3-F4	2048	0.0000	0.0000	0.0000	0.0000	0.0000	7
8	F3-T1	2048	0.0000	0.0001	0.0004	0.0001	0.0000	8
9	F3-T2	2048	0.0000	0.0000	0.0001	0.0000	0.0000	9
10	F3-T3	2048	0.0000	0.0000	0.0000	0.0000	0.0000	10
11	F3-T4	2048	0.0000	0.0000	0.0000	0.0000	0.0000	11
12	F3-NBO (R12)	2048	0.0100	0.0030	0.1089	0.0407	0.0007	12
13	F3-DPO (R12)	2048	0.0669	0.1127	0.0469	0.0754	0.0004	13
14	F4-T1	2048	0.0000	0.0001	0.0004	0.0001	0.0000	14
15	F4-T2	2048	0.0000	0.0000	0.0001	0.0000	0.0000	15
16	F4-T3	2048	0.0000	0.0000	0.0000	0.0000	0.0000	16
17	F4-T4	2048	0.0000	0.0000	0.0000	0.0000	0.0000	17
18	F4-NBO (R12)	2048	0.0100	0.0030	0.1089	0.0407	0.0007	18
19	F4-DPO (R12)	2048	0.0669	0.1127	0.0469	0.0754	0.0004	19
20	T1-T2	2048	0.0000	-0.0001	-0.0003	-0.0001	0.0000	20
21	T1-T3	2048	0.0000	-0.0001	-0.0004	-0.0001	0.0000	21
22	T1-T4	2048	0.0000	-0.0001	-0.0004	-0.0001	0.0000	22
23	T1-NBO (R12)	2048	0.0100	0.0029	0.1085	0.0406	0.0007	23
24	T1-DPO (R12)	2048	0.0669	0.1126	0.0465	0.0753	0.0004	24
25	T2-T3	2048	0.0000	0.0000	-0.0001	0.0000	0.0000	25
26	T2-T4	2048	0.0000	0.0000	-0.0001	0.0000	0.0000	26
27	T2-NBO (R12)	2048	0.0100	0.0030	0.1088	0.0407	0.0007	27
28	T2-DPO (R12)	2048	0.0669	0.1127	0.0468	0.0754	0.0004	28
29	T3-T4	2048	0.0000	0.0000	0.0000	0.0000	0.0000	29
30	T3-NBO (R12)	2048	0.0100	0.0030	0.1089	0.0407	0.0007	30
31	T3-DPO (R12)	2048	0.0669	0.1127	0.0469	0.0754	0.0004	31
32	T4-NBO (R12)	2048	0.0100	0.0030	0.1089	0.0407	0.0007	32
	T4-DPO (R12)	2048	0.0669	0.1127	0.0469	0.0754	0.0004	
	NBO (R12)-DPO (R12)	2048	0.0569	0.1097	-0.0621	0.0347	0.0010	

All 55 pairs on both meshes are retained; means and standard errors are multiplied by 10^3 . The full machine-readable account additionally retains every initial-state standard error and raw payoff array. These conditional numerical comparisons spend no confidence allocation and do not claim continuous-time coverage.

M.1. Complete protected-parent and finite-observation results

The source package fixes the two saved policies, the four parent grids, the population, all observation precisions, and the confidence allocation before simulation. Each cell uses an independent process with pinned numerical dependencies. The archive retains all raw paired payoffs, final states, per-cell transfer recurrences, and the source and output digests.

TABLE M.V

COMPLETE FINITE-OBSERVATION PAYOFF TRANSFERS

d	N	ν	Allowance	Lower	Upper	Allowance/parent lower
10	1024	0.0000	0.000080	0.000687	0.002205	0.103
10	1024	0.0001	0.000241	0.000526	0.002366	0.314
10	1024	0.0010	0.001692	-0.000925	0.003817	2.206
10	4096	0.0000	0.000020	0.001050	0.001822	0.019
10	4096	0.0001	0.000181	0.000889	0.001983	0.169
10	4096	0.0010	0.001630	-0.000559	0.003432	1.522
50	1024	0.0000	0.000555	0.000252	0.002920	0.687
50	1024	0.0001	0.001845	-0.001038	0.004211	2.285
50	1024	0.0010	0.013458	-0.012651	0.015824	16.673
50	4096	0.0000	0.000139	0.001054	0.002122	0.116
50	4096	0.0001	0.001432	-0.000239	0.003415	1.200
50	4096	0.0010	0.013066	-0.011873	0.015049	10.949

Notes: Every declared dimension, grid, and observation precision appears. Lower endpoints are printed downward; upper endpoints and allowances are printed upward to six decimal places. Ratios are descriptive rounded values. The allowance is subtracted from the parent lower endpoint and added to its upper endpoint with outward arithmetic. The final column compares implementation error with the positive parent guarantee; a ratio above one removes that guarantee. The candidate's physical state remains the original unbounded diffusion. Its measurements arrive only at the N decision dates and have normalized root mean square error at most ν .

TABLE M.VI

MEASURED WORK FOR PROTECTED-PARENT VERIFICATION AND SENSOR TRANSFERS

d	N	Complete seconds	Peak MiB	State transitions
10	1024	37.334	383.4	16,777,216
10	4096	126.355	434.6	67,108,864
50	1024	55.194	405.0	16,777,216
50	4096	263.063	451.6	67,108,864

Notes: Each row is a fresh CPU process. Its elapsed time includes imports, weight loading, interval constants, protected simulation, all three sensor allowances, raw-file writes, and the final result record. Environment provisioning is reported separately. No policy is refitted in this experiment.

TABLE M.VII: Every registered direct stream comparison before and after deterministic refinement

d	Stream	NBO minus	Mean	Original interval	Refined interval	Case
10	2095568532	Raw	-0.000000	[-0.001152, 0.001152]	[-0.000901, 0.000900]	Normal
10	2133755523	Raw	-0.000000	[-0.001153, 0.001152]	[-0.000901, 0.000900]	Normal
10	153722668	Raw	0.000001	[-0.001151, 0.001153]	[-0.000899, 0.000902]	Normal
10	603655094	Raw	0.000003	[-0.001149, 0.001155]	[-0.000898, 0.000903]	Normal
10	216592699	Raw	0.000002	[-0.001150, 0.001154]	[-0.000899, 0.000902]	Normal
10	1950831757	Raw	0.000001	[-0.001151, 0.001153]	[-0.000899, 0.000902]	Normal
10	107486742	Raw	-0.000000	[-0.001153, 0.001152]	[-0.000901, 0.000900]	Normal
10	1142235589	Raw	0.000001	[-0.001151, 0.001153]	[-0.000899, 0.000902]	Normal
10	1824517164	Raw	-0.000000	[-0.001152, 0.001152]	[-0.000901, 0.000900]	Normal
10	1965147083	Raw	-0.000000	[-0.001152, 0.001152]	[-0.000900, 0.000900]	Normal
10	1860620983	Raw	0.000001	[-0.001151, 0.001153]	[-0.000900, 0.000901]	Normal
10	1618144748	Raw	-0.000000	[-0.001152, 0.001152]	[-0.000901, 0.000900]	Normal
10	592131000	Raw	0.000001	[-0.001151, 0.001153]	[-0.000899, 0.000902]	Normal
10	964082955	Raw	0.000002	[-0.001150, 0.001154]	[-0.000898, 0.000903]	Normal
10	887230079	Raw	-0.000000	[-0.001152, 0.001152]	[-0.000901, 0.000900]	Normal
10	1818435806	Raw	0.000002	[-0.001150, 0.001154]	[-0.000898, 0.000903]	Normal
10	2095568532	Direct policy	0.000006	[-0.001146, 0.001159]	[-0.000895, 0.000907]	Normal
10	2133755523	Direct policy	0.000008	[-0.001145, 0.001161]	[-0.000893, 0.000909]	Normal
10	153722668	Direct policy	0.000007	[-0.001146, 0.001159]	[-0.000894, 0.000907]	Normal

<i>d</i>	Stream	NBO minus	Mean	Original interval	Refined interval	Case
10	603655094	Direct policy	0.000005	[-0.001147, 0.001158]	[-0.000896, 0.000906]	Normal
10	216592699	Direct policy	0.000006	[-0.001146, 0.001159]	[-0.000895, 0.000907]	Normal
10	1950831757	Direct policy	0.000006	[-0.001147, 0.001158]	[-0.000895, 0.000907]	Normal
10	107486742	Direct policy	0.000006	[-0.001147, 0.001158]	[-0.000895, 0.000906]	Normal
10	1142235589	Direct policy	0.000006	[-0.001147, 0.001159]	[-0.000895, 0.000907]	Normal
10	1824517164	Direct policy	0.000002	[-0.001151, 0.001155]	[-0.000899, 0.000903]	Normal
10	1965147083	Direct policy	0.000004	[-0.001149, 0.001156]	[-0.000897, 0.000905]	Normal
10	1860620983	Direct policy	0.000003	[-0.001150, 0.001155]	[-0.000898, 0.000904]	Normal
10	1618144748	Direct policy	0.000007	[-0.001145, 0.001160]	[-0.000894, 0.000908]	Normal
10	592131000	Direct policy	0.000007	[-0.001146, 0.001159]	[-0.000894, 0.000907]	Normal
10	964082955	Direct policy	0.000005	[-0.001148, 0.001157]	[-0.000896, 0.000905]	Normal
10	887230079	Direct policy	0.000005	[-0.001148, 0.001157]	[-0.000896, 0.000906]	Normal
10	1818435806	Direct policy	0.000006	[-0.001147, 0.001158]	[-0.000895, 0.000906]	Normal
10	2095568532	Neural HJB	0.000152	[-0.001007, 0.001311]	[-0.000755, 0.001060]	Normal
10	2133755523	Neural HJB	0.000186	[-0.000974, 0.001346]	[-0.000723, 0.001094]	Normal
10	153722668	Neural HJB	0.000127	[-0.001030, 0.001285]	[-0.000779, 0.001034]	Normal
10	603655094	Neural HJB	0.000193	[-0.000968, 0.001354]	[-0.000717, 0.001102]	Normal
10	216592699	Neural HJB	0.000110	[-0.001046, 0.001266]	[-0.000795, 0.001015]	Normal
10	1950831757	Neural HJB	0.000211	[-0.000952, 0.001373]	[-0.000700, 0.001122]	Normal
10	107486742	Neural HJB	0.000395	[-0.000770, 0.001561]	[-0.000519, 0.001309]	Normal
10	1142235589	Neural HJB	0.001235	[0.000054, 0.002417]	[0.000306, 0.002165]	Normal
10	1824517164	Neural HJB	0.000139	[-0.001018, 0.001295]	[-0.000767, 0.001044]	Normal
10	1965147083	Neural HJB	0.000204	[-0.000957, 0.001365]	[-0.000705, 0.001114]	Normal
10	1860620983	Neural HJB	0.000121	[-0.001036, 0.001279]	[-0.000785, 0.001027]	Normal
10	1618144748	Neural HJB	0.000156	[-0.001004, 0.001315]	[-0.000752, 0.001063]	Normal
10	592131000	Neural HJB	0.000189	[-0.000972, 0.001349]	[-0.000720, 0.001098]	Normal
10	964082955	Neural HJB	0.000169	[-0.000990, 0.001328]	[-0.000738, 0.001077]	Normal
10	887230079	Neural HJB	0.000184	[-0.000974, 0.001343]	[-0.000722, 0.001091]	Normal
10	1818435806	Neural HJB	0.000944	[-0.000230, 0.002118]	[0.000022, 0.001867]	Normal
50	2095568532	Raw	-0.000002	[-0.001204, 0.001201]	[-0.000908, 0.000905]	Normal
50	2133755523	Raw	-0.000002	[-0.001204, 0.001201]	[-0.000909, 0.000905]	Normal
50	153722668	Raw	-0.000001	[-0.001204, 0.001201]	[-0.000908, 0.000906]	Normal
50	603655094	Raw	-0.000002	[-0.001205, 0.001200]	[-0.000909, 0.000905]	Normal

<i>d</i>	Stream	NBO minus	Mean	Original interval	Refined interval	Case
50	216592699	Raw	-0.000001	[-0.001204, 0.001201]	[-0.000908, 0.000906]	Normal
50	1950831757	Raw	-0.000003	[-0.001205, 0.001200]	[-0.000909, 0.000904]	Normal
50	107486742	Raw	-0.000002	[-0.001204, 0.001201]	[-0.000908, 0.000905]	Normal
50	1142235589	Raw	-0.000001	[-0.001203, 0.001202]	[-0.000908, 0.000906]	Normal
50	1824517164	Raw	-0.000001	[-0.001204, 0.001201]	[-0.000908, 0.000905]	Normal
50	1965147083	Raw	-0.000001	[-0.001203, 0.001202]	[-0.000907, 0.000906]	Normal
50	1860620983	Raw	-0.000001	[-0.001204, 0.001202]	[-0.000908, 0.000906]	Normal
50	1618144748	Raw	-0.000001	[-0.001203, 0.001201]	[-0.000908, 0.000906]	Normal
50	592131000	Raw	-0.000000	[-0.001203, 0.001202]	[-0.000907, 0.000907]	Normal
50	964082955	Raw	-0.000002	[-0.001204, 0.001201]	[-0.000909, 0.000905]	Normal
50	887230079	Raw	-0.000002	[-0.001204, 0.001201]	[-0.000908, 0.000905]	Normal
50	1818435806	Raw	-0.000001	[-0.001203, 0.001202]	[-0.000907, 0.000906]	Normal
50	2095568532	Direct policy	-0.000002	[-0.001206, 0.001201]	[-0.000910, 0.000905]	Normal
50	2133755523	Direct policy	-0.000001	[-0.001204, 0.001202]	[-0.000909, 0.000906]	Normal
50	153722668	Direct policy	0.000002	[-0.001201, 0.001205]	[-0.000906, 0.000909]	Normal
50	603655094	Direct policy	-0.000002	[-0.001206, 0.001201]	[-0.000910, 0.000905]	Normal
50	216592699	Direct policy	0.000002	[-0.001202, 0.001205]	[-0.000906, 0.000909]	Normal
50	1950831757	Direct policy	-0.000004	[-0.001207, 0.001199]	[-0.000911, 0.000904]	Normal
50	107486742	Direct policy	-0.000022	[-0.001225, 0.001182]	[-0.000929, 0.000886]	Normal
50	1142235589	Direct policy	-0.000004	[-0.001208, 0.001199]	[-0.000912, 0.000903]	Normal
50	1824517164	Direct policy	0.000001	[-0.001202, 0.001205]	[-0.000906, 0.000909]	Normal
50	1965147083	Direct policy	-0.000022	[-0.001225, 0.001181]	[-0.000930, 0.000886]	Normal
50	1860620983	Direct policy	-0.000002	[-0.001205, 0.001201]	[-0.000910, 0.000905]	Normal
50	1618144748	Direct policy	-0.000019	[-0.001223, 0.001184]	[-0.000927, 0.000888]	Normal
50	592131000	Direct policy	-0.000000	[-0.001203, 0.001203]	[-0.000908, 0.000907]	Normal
50	964082955	Direct policy	-0.000001	[-0.001204, 0.001203]	[-0.000908, 0.000907]	Normal
50	887230079	Direct policy	-0.000002	[-0.001205, 0.001202]	[-0.000909, 0.000906]	Normal
50	1818435806	Direct policy	-0.000001	[-0.001205, 0.001202]	[-0.000909, 0.000906]	Normal
50	2095568532	Neural HJB	0.000214	[-0.000995, 0.001423]	[-0.000699, 0.001127]	Normal
50	2133755523	Neural HJB	0.002917	[0.001566, 0.004269]	[0.001862, 0.003973]	Normal
50	153722668	Neural HJB	0.002449	[0.001128, 0.003769]	[0.001424, 0.003474]	Normal
50	603655094	Neural HJB	0.000643	[-0.000580, 0.001866]	[-0.000284, 0.001570]	Normal
50	216592699	Neural HJB	0.002211	[0.000909, 0.003513]	[0.001205, 0.003217]	Normal

<i>d</i>	Stream	NBO minus	Mean	Original interval	Refined interval	Case
50	1950831757	Neural HJB	0.000607	[-0.000612, 0.001825]	[-0.000316, 0.001529]	Normal
50	107486742	Neural HJB	0.001944	[0.000659, 0.003230]	[0.000955, 0.002934]	Normal
50	1142235589	Neural HJB	0.001347	[0.000096, 0.002598]	[0.000392, 0.002303]	Normal
50	1824517164	Neural HJB	0.001646	[0.000384, 0.002907]	[0.000680, 0.002611]	Normal
50	1965147083	Neural HJB	0.001684	[0.000418, 0.002950]	[0.000714, 0.002654]	Normal
50	1860620983	Neural HJB	0.000567	[-0.000648, 0.001782]	[-0.000352, 0.001486]	Normal
50	1618144748	Neural HJB	0.001821	[0.000545, 0.003096]	[0.000841, 0.002800]	Normal
50	592131000	Neural HJB	0.001557	[0.000298, 0.002816]	[0.000594, 0.002521]	Normal
50	964082955	Neural HJB	0.001678	[0.000414, 0.002942]	[0.000710, 0.002647]	Normal
50	887230079	Neural HJB	0.001625	[0.000366, 0.002884]	[0.000662, 0.002588]	Normal
50	1818435806	Neural HJB	0.001064	[-0.000172, 0.002301]	[0.000123, 0.002005]	Normal

All declared streams are retained. Normal means two simulated total-held policies; Mixed means one analytical fallback; Both means two exact analytical references. Only Normal rows are eligible for the new transfer. No payoff vector is regenerated and no skipped stopping event is recycled.

APPENDIX N: SOURCE AND EVIDENCE INDEX

The current article has one evaluation–improvement algorithm and one set of theorem statements. Complete application models and their proofs remain in this supplement. The preceding reading copy and complete numerical record are preserved at repository commit `f5021cefa71492babfcbfa580e0c984f59a9de26`. The exact main and supplementary roots have Git blob identities, respectively,

`c13f5e951b47577f0cd4bcbbb8b3d003e92e6a33e`
`9ae1d87ddc6da2e0cd0475522c269da8ee55a1be`.

The revision’s source inventory records every source file and its digest in that reading copy. Its editorial map assigns every preceding equation, result, section, and table label to the current main text, current supplement, or the exact archived record. The latter holds complete seed-level tables, checkpoint histories, earlier fitting protocols, and the successive revision narratives. Their raw arrays, unsuccessful runs, and compiled records remain available at their original repository paths.

The existing unfavorable direct and costate comparisons are summarized in the main article; reorganizing their source does not alter those results or convert an inconclusive interval into equivalence.

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